Some documentation for

Minos (V2.2.0)

Contest Logging Software

(V2.2.0.0)

This document is based on,

and specifically relevant to,

V2.2.0

The **Minos Contest Logger** is supplied under Version 3 of the GPL or any later version.

The Development Team are Mike Goodey (G0GJV), David Balharrie (M0DGB/G8FKH), Neil Yorke (M0NKE), Dave Sergeant (G3YMC), Peter Burton (G3ZPB), and Ken Punshon (G4APJ)

The **Minos Contest Logger** is **not** a supported product; no commitment or liability is or will be accepted to address any problems that may be encountered in using it.

However, any bugs or deficiencies should be notified to assist in the continuous development and improvement. If you use Minos and want to discuss the program or have any problems with it you are encouraged to join the project discussion mailing list on Groups.io.

Either visit https://minos.groups.io/g/main or send a blank email to main+subscribe@minos.groups.io

This document is intended to provide information ranging from simple installation and usage through to detailed internals, so don't worry if some of it is of no use/interest to you. There have been extensive changes since Minos V1. With V2 the software is (usually) installed (see section 1.1), and the Applications (see section 0) are a major new facility.

As of V2.0.7 the user has the ability to configure the layout of the Logger main panel (see section 3.5). Unless otherwise stated, all descriptions and illustrations in this document use the default configuration.

This is a 'live' document that **may** be revised on the basis of any comments, feedback, or additional information received (via the project discussion mailing list). No commitment or liability is or will be accepted to address any problems that may be encountered in using it.

The software it describes is also under continuing development, so may not exactly match what is described here.

This is the latest release of the document for software that is currently under development. As with software, expect bugs; some of these will be known, others will not. Please report any errors that you find (after checking that you have the latest version of the document).

There is an implicit assumption that readers and users of the software are 'PC literate' and know about windows, panels, scroll bars etc.

The Logger runs on Windows 7 or later, but the document is written from the perspective of running the programme on Windows 10 Home (various versions, latterly 1803). At this time there are no known problems with running it on other versions, indeed it has been extensively tested on a number of other platforms, but no guarantees are given.

The Logger also builds and runs native on Linux. The intention is to also release binaries for Ubuntu (32 and 64 bit), and Raspberry Pi. Further platforms (other varieties of Linux, Android, Mac) may *eventually* be added. It is yet to be decided what will happen with documentation for these other platforms.

The general usage should be the same, or very similar; any significant differences will be noted as and when discovered/reported to the author.

Please consider any trademarks, copyrights, etc. that have not been acknowledged as acknowledged.

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1 Introduction

The Minos Contest Logger is a logging program originally written by Mike Goodey G0GJV, now being developed by G0GJV and others. All RSGB and Region 1 VHF and UHF amateur radio contests are supported. The Logger supports real time or post event data entry and production of an entry log file for submission to the adjudicators.

Operation of the program is intended to reflect the normal practice of filling in a log book. The fields appear in that order on the screen, and are below a similarly laid out list of previous contacts - similar to a log book.

It is mainly intended for **on-line** log entry, but can be used to enter logs post event. The less done after the event the better; all relevant data should be captured during the contest.

A single operator can't keep both computer and paper up to date together. With only one operator, keep the computer log up to date - it is your duplicate check system as well as the log that will eventually be used to submit your contest entry.

Where possible a second operator/logger should use the computer, and the main operator should maintain a traditional paper log. This way, the main operator always has a piece of paper to scribble notes on, as well as a providing a backup log.

The user is ALWAYS right - even if insisting that a W9W/MM worked him from ZZ99ZZ square. Although errors may be reported, in the end the program can be made to log it like that anyway. However, a report of 6 and 0, or invalid dates and times will **not** be allowed.

Error messages should never get in the way of entering more data. So, except in unusual circumstances, the program will not display error messages that must be acknowledged.

For more on the history of Minos, and its predecessor GJVLog, see
http://minos.sourceforge.net/history.html

1.1 Installation

The Minos installation programme (MinosInstall_2_2_0.exe) is downloaded (and saved to some convenient location) from https://sourceforge.net/projects/minos/files/

It is recommended that Minos V2 is installed to a different location than that used for Minos V1, even after release. Since V2.0 there have been, and continue to be, significant changes in the way configuration is handled. It is prudent to save a copy of the \Configuration folder prior to installing the new version in case you want to revert back.

Updates to the Beta can be installed to the same location as an existing V2 or later Beta. ***However***, be aware that the format of the Rig and Rotator configuration files has changed during development so earlier versions should be deleted and the details set up again. If you fail to do this an error message will inform you of the problem when you invoke the application(s).

Administrator privileges are needed to invoke the installation programme. The usual 'changes to system' dialogue may be seen, depending on system version and configuration:

- Open File – Security Warning (click Run)
- User Account Control 'changes to device' (click Yes)

The standard installation dialogue then follows, as shown below.

Accept the licence agreement:

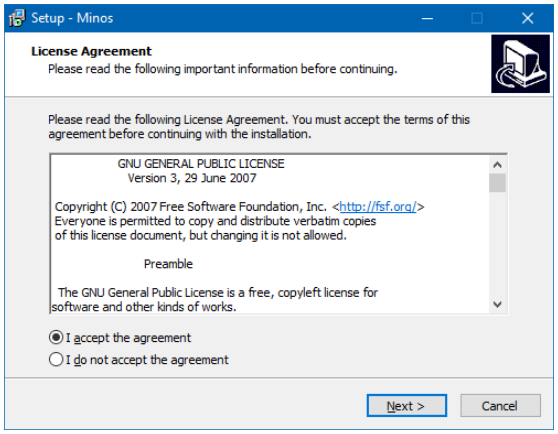

Figure 1 – Licence agreement

The default location for the installation is C:\Minos. This can be changed, but if this is done via the 'Browse' button be sure that it ends up where you want (by default a sub-folder is created).

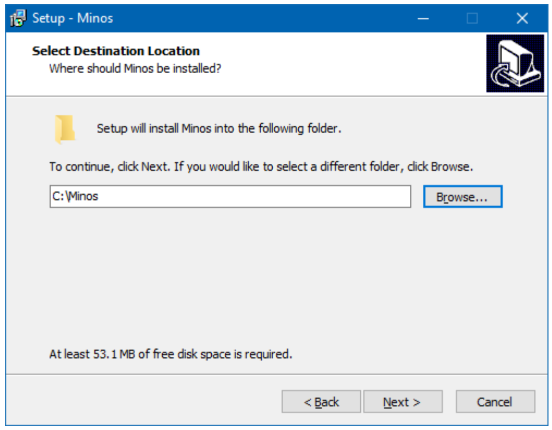

Figure 2 – Target location

You may of course get a message that the folder already exists, or needs to be created.

The log (.minos) files are by default placed in a sub-folder to the programme ‘root’ directory (see a description of the folders in Section 5.1).

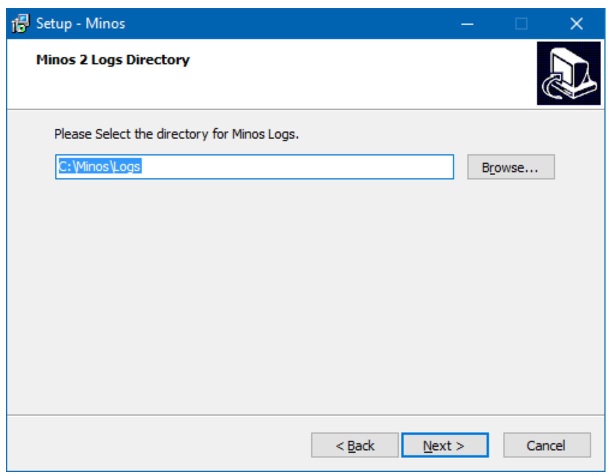

Figure 3 – Location of logs

By default a start menu entry is created, but this can be suppressed if so desired.

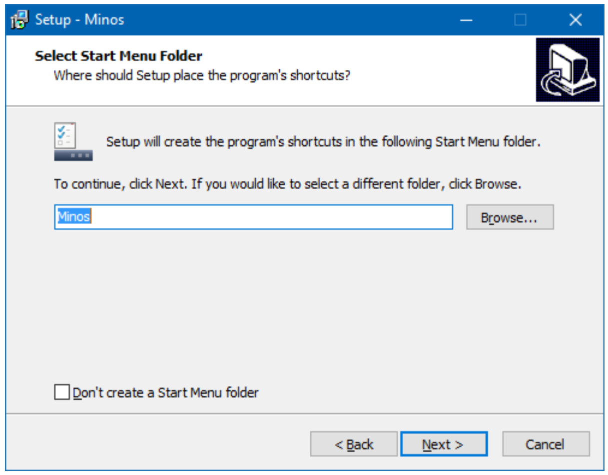

Figure 4 – Start menu

The "**Additional Tasks**" panel (Figure 5) gives the option of creating a desktop shortcut for invoking the Logger in a convenient fashion.

There is also an option to associate the **".minos**" extension with the Logger so that double-clicking a log file will open the Logger.

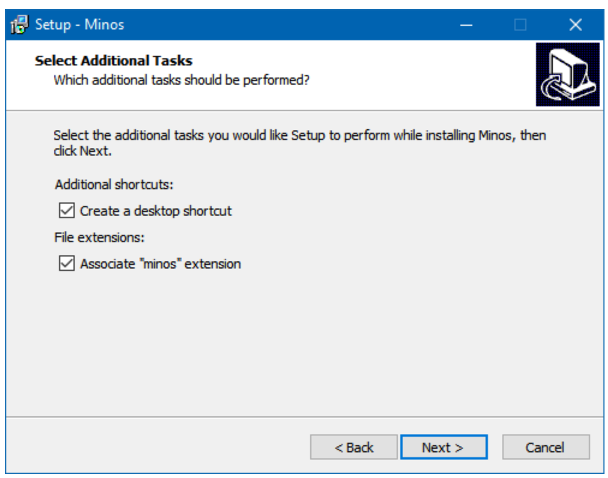

Figure 5 – Additional Tasks

Then there is a final check before installation starts:

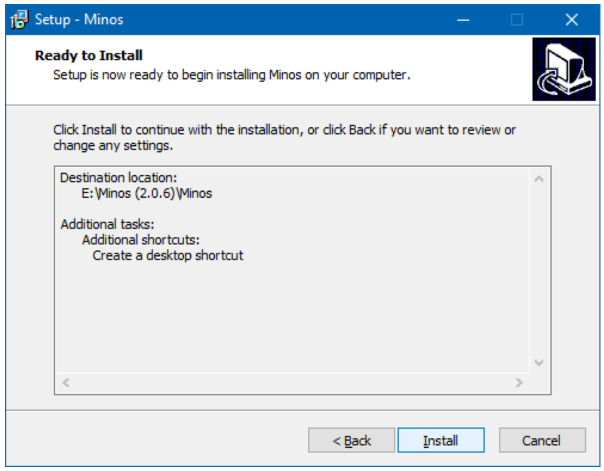

Figure 6 – Ready to install

After a short period of activity the completion notification (Figure 7) gives you the option to run the programme immediately. If this is not your first installation you may want to copy over configuration files before starting the programme.

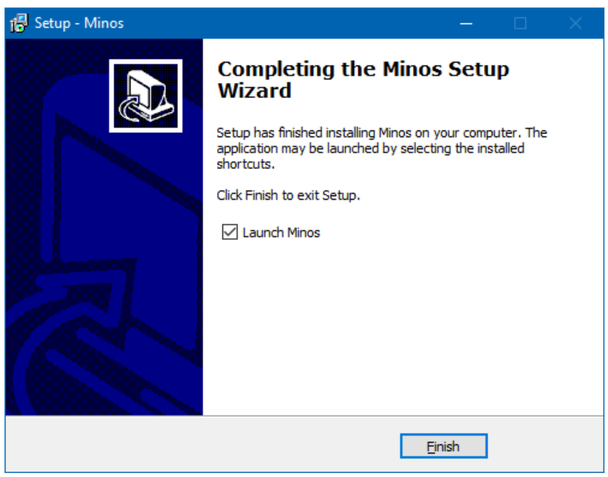

Figure 7 – Installation complete

1.1.1 Uninstallation

The Logger is uninstalled using the standard "**Control Panel/Programs and Features**" mechanism.

This removes all the binaries (\Bin), and the general Configuration information (that is not specific to the particular station installation), leaving (under the root folder):

- \Configuration: Station specific (The Display, Entry, ListPreload, LogsPreload, MinosConfig, MinosLogger, MinosRotatorConfig, QTH, and Station INI files
- All application configuration such as defined rigs and rotators is also saved, including screen layouts in ScreenConfigs.json.
- \Lists: (CSL files).

- \Logs: (Minos files).
- \TraceLog: (diagnostics).
- Changes to fonts and panel layout for the Logger will be retained in the Registry[1].
- Any personal files placed in the \Bin folder will be retained (and hence the folder).
- Any other folders that the user has created under the main Minos folder

1.2 Basic Configuration

Ensure that the computer clock is reasonably accurate (within 1 or 2 seconds or better, particularly for MGM contests), and set to BST if applicable (as can be seen at the bottom right of Figure 12, Minos calculates and displays UTC[2] - in the Auxiliary panel as well).

In order to perform the initial set-up, invoke the program and the Splash Screen will be displayed.

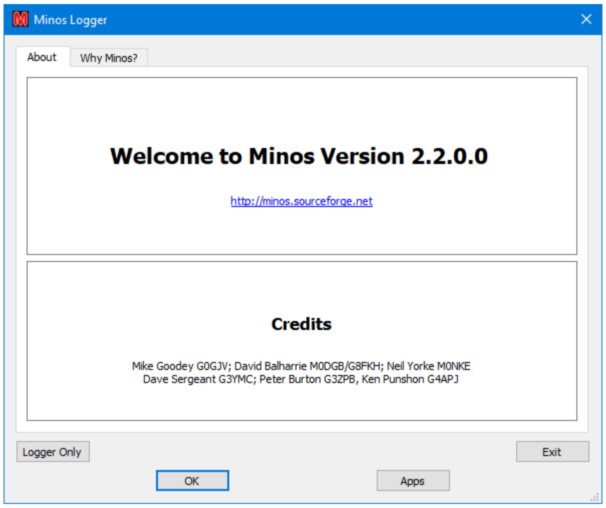

Figure 8 – Welcome screen

Click on "**OK**"[3] (or "**Logger Only**") to dismiss the splash screen. **Ignore the Apps button for now**.

You will certainly want to expand the Logger main panel. Depending on your screen size and resolution you may want to expand to full screen, particularly if you have a wide screen.

Click on "**File/New Contest**" to be presented with "**Details of Contest Entry**" (see Figure 9).

[1] These can be reset using "**File/Exit MinosQT and Clear registry**".
[2] See section 3.4.4 for information about time adjustments that can be made.
[3] You may be interested to read "**Why Minos**" on the "**About**" panel for background on the origin of the name.

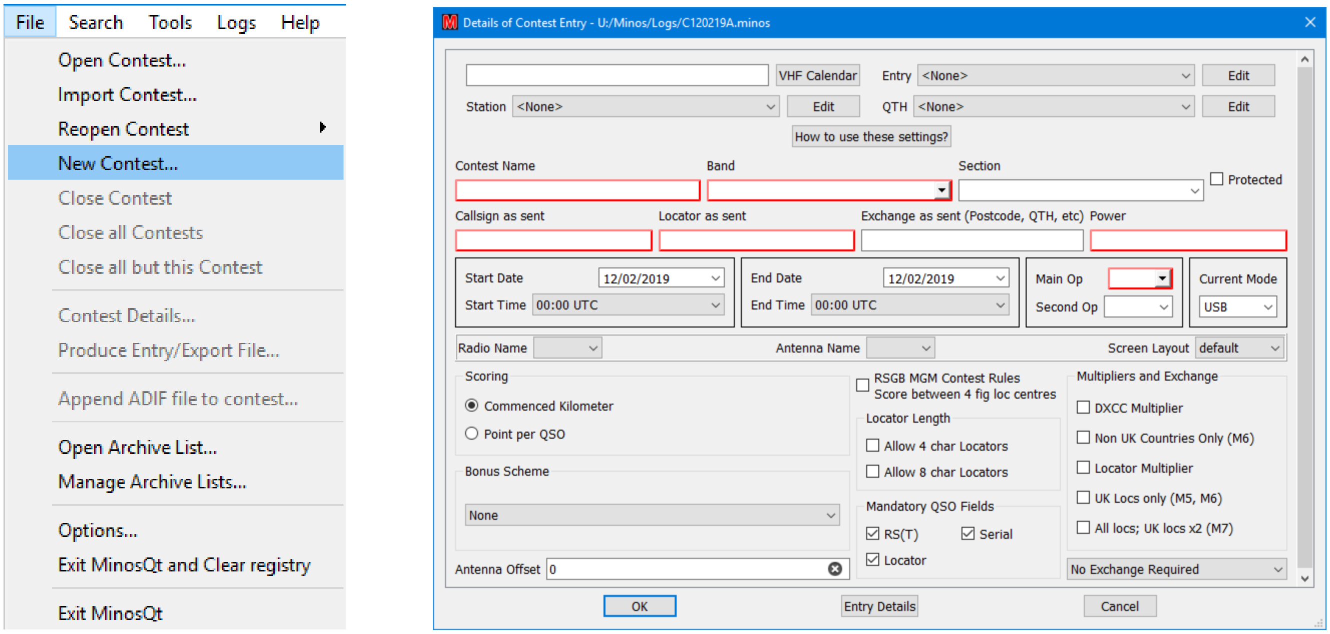

Figure 9 – File options and Contest Entry Details

This then allows the set-up of various items that will be used in future. The red outlines indicate the boxes that still require information before a contest can be validly created. When you are creating a contest the red edgings will disappear as the required information is supplied.

Click on VHF Calendar and then (because it can, and does, change) "**Download Latest Calendar**"[4]. At the time of writing, Minos ships with calendar version 19.006 6/02/2019 for the remainder of this year. Please check for updates before use, and certainly in January each subsequent year.

Clearly a new download of the calendar will be needed on **at least** a yearly basis[5].

For now, close the VHF calendar in order to edit the grouped settings[6], which can be applied to a contest all in one go. There are four basic groups of settings:

- **Entry**: All the extra bits for a real entry (entrant callsign, group name, contact details) which are stored in Entry.ini
- **Station**: Details of the rig, power, and antenna (stored in Station.ini); you will typically require one of these for each band, if only due to antennas.
- **QTH**: Where the station is located for this contest, including Maidenhead Locator used. The data is stored in QTH.ini.

[4] Any message of the form "3 of 4 files downloaded. We don't expect to load them all." can be ignored.

[5] The intention is that this facility should be available on an indefinite basis, but there is no commitment to such activity.

[6] If you have created these entries for an earlier version of the program you can copy over the INI files (to the Configuration folder). It is also useful to copy over any CSL files you may already have (to the Lists folder).

- **Contest**: Typically derived from the contest calendar, specifying contest name, band(s), and scoring. If the calendar is not used then individual boxes can be configured (see Figure 11) or modified from the calendar derived values.

To define these settings, click on Edit, then select New Setting[7]. Assign a meaningful name and fill in the requisite data. A typical collection of grouped settings might for example be called:

- **Entry**: Self, Club, Contest Group
- **Station**: 50MHz, 70MHz, 144MHz, 432MHz, Portable-144MHz
- **QTH**: Home, VHF NFD, UKAC Portable

Click on the setting name on the left to select an existing setting and then its components are shown in the right hand pane, and can be edited individually. There is **no** validation of field content at this time, although errors may be highlighted later.

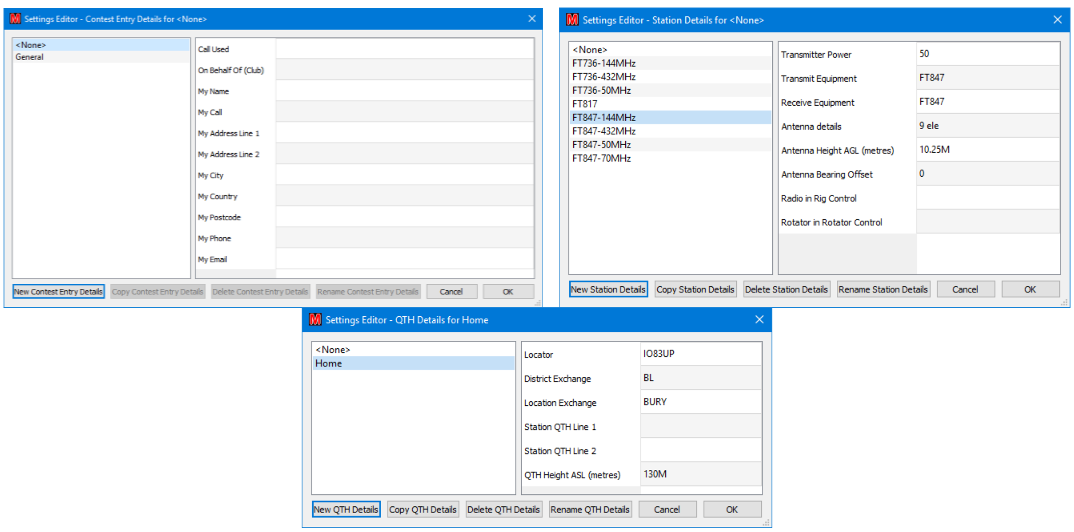

Figure 10 – Example Grouped Settings

Once the settings have been defined, use "**Entry Details**" to check that meaningful and complete data has been supplied.

During a contest "**File/ Contest Details**" can be used to check, and potentially change, these details (e.g. to add operators[8] or comments).

For now, "**Cancel**" the contest to finish set up; this will lose any other data entered into the contest (but you will do this well before the contest won't you?).

[7] Once you have created entries, an existing setting can be copied and minor changes made to cope with such things as the same rig being used on multiple bands.

[8] Operators can also be added via the pull down lists in the data entry panel.

2 Normal operation

Invoke the program from the shortcut on your desktop or the start menu[9].

Any contest files that were left open last time will remain open[10] (see section 3.4.1 for discussion of multi-band contests that would typically use this facility). Similarly, any Archive (CSL) files (see section 3.4.3) will be re-opened; these are implicitly Read-Only unless you deliberately edit them.

For normal single contest operation close anything other than the current contest, or ensure that the non-current contests are marked as "**Protected**" to prevent you accidentally modifying them. This is done by selecting "**File/Contest Details**" for the requisite contest and then ticking the "**Protected**" tick box on the right hand side just above the "**Power**" box, see Figure 13.

The protected status is preserved between invocations, and hence must be (temporarily) removed if you wish to change the content. This is done by clearing the "Protected" box, but note that it will revert to protected on the next invocation of the programme.

If a contest has been protected then all the "**Contest Details**" except for the "**Protected**" field are greyed out and any connection to RigControl or Rotator will be disabled.

If want to re-open a contest that you recently closed then "**File/Reopen Contest**" gives a list of the last 5 contests opened[11]. Opening one of the files on the list causes the entry to be removed, but it will be placed back at the head of the list when the log is closed again.

There are no known limitations (e.g. number of contacts, bands, field lengths….), but if you discover any please notify the support group (see Preface). Although it is possible to open at least 20 concurrent contests the 'tab space' at the top of the screen (on a wide, 1920x1080, screen) is full when 10 are open and it becomes complex to switch between them[12].

2.1 Creating a new contest

Use "**File/New Contest**" which will bring up the Contest Details panel as described in section 1.2.

Select the contest from VHF Calendar (normally for data entry during the contest; see section 2.2.1 for details of post-contest data entry).

Once Applications have been defined (see section 4.1) the "**Radio name**" and "**Rotator name**" pull-down lists can be used to define the default radio/rotator to be used for the contest. The connection can also be

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[9] The logger can also be invoked by directly clicking on the image (MqtLogger.exe in the \Bin folder) or by double clicking a log (.minos) file if the file association has been set up during installation.
[10] A warning message will be given if the file cannot be found.
[11] If the file that you attempt to re-open has been deleted then no error message will be given, but the entry will be removed from the list.
[12] From the operator perspective 5 is probably more than enough!

established/changed through the QSO entry panel (see Figure 67 and Figure 86) – at which point these boxes will be filled.

Select the requisite bundled settings from the drop down lists for Entry, Station, and QTH. Any bundles set to **"<None>"** will be ignored (and you will have to specifically enter the data before submitting the entry).

If the setting you want isn't there, use the "**Edit**" button to define a new setting or copy and then modify an existing setting in the manner described in section 1.2.

The "**Contest Name**" can be entered and "**Band**" selected from the pull-down list if the VHF calendar is not used.

Note that the "**Section**" selected is no longer relevant to RSGB contests; it is determined by the Contest Entry Upload Robot. However, you may want to enter the requisite value here for completeness of the log[13] or for entering a non-RSGB contest.

"**Callsign as sent**"[14] will be derived from the "**Entry**" information and "**Locator as sent**" from the QTH information; both can be overwritten as required.

If the contest is not selected from the Calendar then use the relevant fields on the "**Details of Contest Entry**" panel (see Figure 11) to set the name, date/time, scoring, multipliers etc. Any bonus scheme in use must also be selected from the pulldown list at the bottom left.

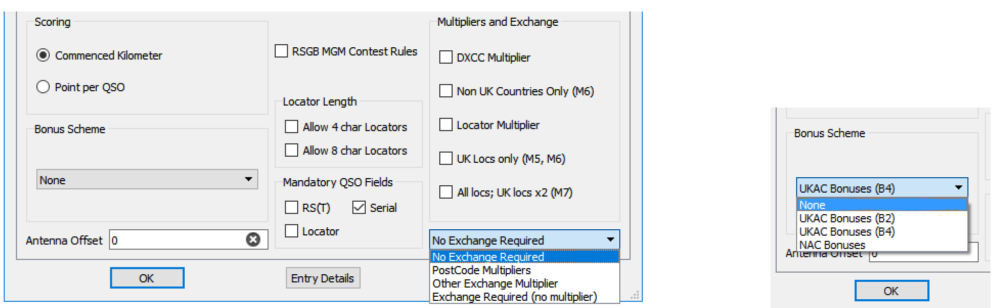

Figure 11 – Contest details and Bonus schemes

Use "**Entry Details**" to check that everything is complete and as expected. Any changes specific to the particular contest can be made via "**Entry Details**" at any point prior to exporting the log. If an error needs to be corrected it is possible to re-export the entry and re-submit (up until the closing date of the contest).

Click "**OK**" and the "**Save new contest as**" panel will generate a default name of the form **G4APJ_2017_09_02_144 MHz.minos** (the 'G' prefix will vary according to the country designated in the

[13] This can be done from the pull-down list defined for a contest from the calendar, or by entering any arbitrary text (which is completely ignored by the Logger).
[14] The "**Main Op**" field is derived from this callsign, removing prefix country information and any suffix such as /P. Once derived any changes to the "**Main Op**" field as a result of changing callsigns must be done manually.

callsign) for the file located in the "**Logs**" folder[15]. In the event that the file already exists the file will have **"_1"** (etc.) appended. It is possible to specify some other name, or browse to another folder if you want, but this may introduce undue complexity if you are unsure of what you are doing.

If you have not provided sufficient information when you click "**OK**" the "**save as**" panel will not appear and the cursor will be moved to the first field identified as requiring input. All fields still required remain outlined in red.

Clicking "**Save**" to the "**Save new contest as**" panel causes the blank Data Entry panel to be displayed ready for entering details of QSOs (see Figure 12). The contest just created is highlighted (red text on tab)[16].

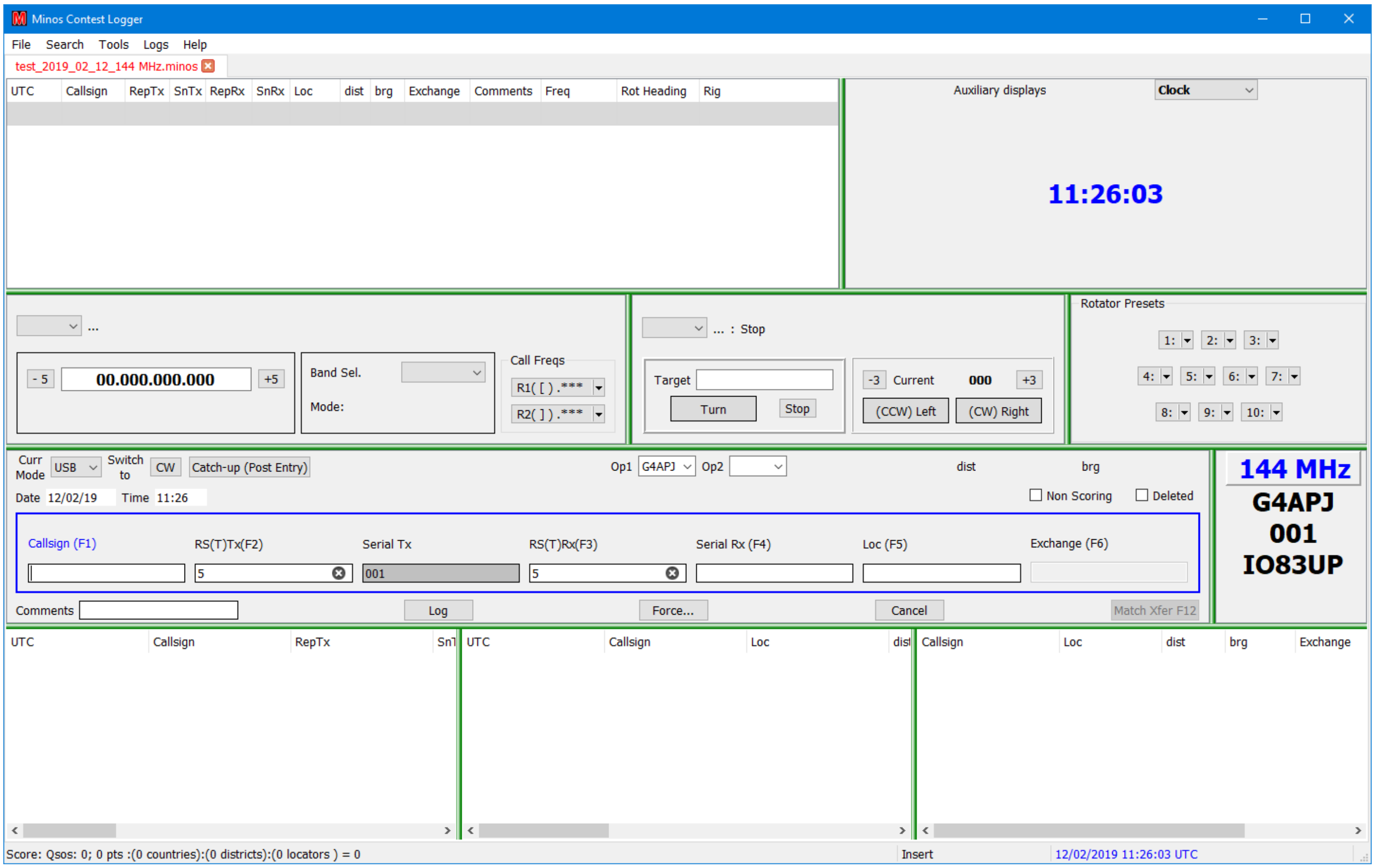

Figure 12 – Initial screen for entering QSO details

Figure 12 uses the default panel layout which includes the Rig and Rotator panels. See section 3.5 for information on how to change this default to remove these panels if they are not being used.

The width (and heights) of the sub-panels (with green edges) can be adjusted by sliding them across (up/down). Column widths can also be adjusted. The worked/un-worked filters and the fonts/panel

[15] Note that the Contest Name does not appear in the filename, regardless of how it was entered.
[16] Protected contests show green and those not selected show blue.

locations/column widths are saved across invocations (and re-installation). Defaults can be reset using the "**File/Exit MinosQt and Clear registry**" option which resets all Minos components.

2.1.1 MGM Contests

In order to support the new MGM contests (beginning in April 2018) there are a variety of options to control QSO requirements (see Figure 13). These will be filled in automatically, and correctly, if the contest is selected from the VHF calendar (which will always be the preferred method).

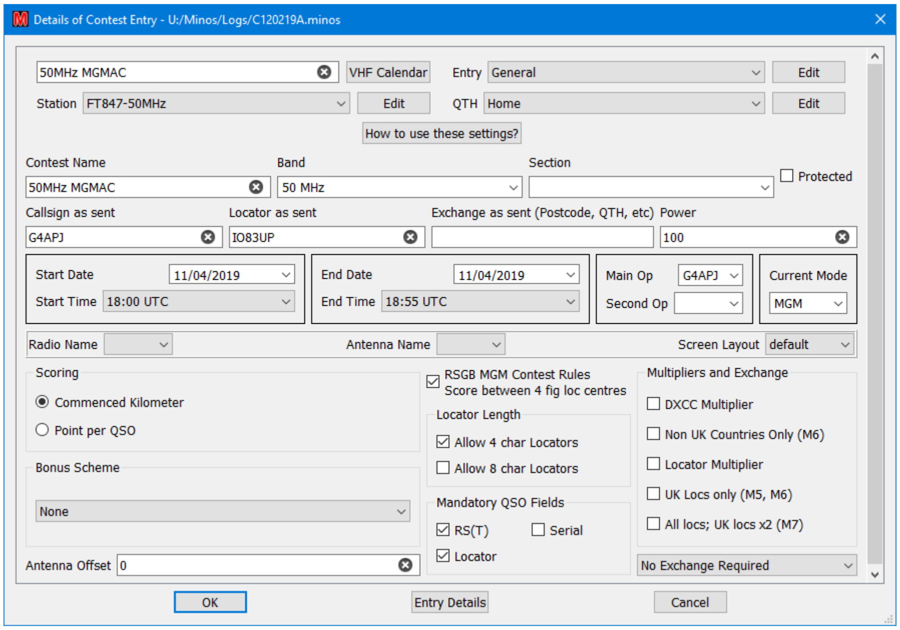

Figure 13 – MGM Contest Details

Once the "**RSGB MGM Contest Rules**" box is ticked (either automatically from the selected contest or manually) it sets things up for the scoring of the RSGB MGM contests[17].

This sets "**Locator Multiplier**", "**Allow 4 char Locators**" (scores are calculated as centre to centre of the 4 character Locators), and clears Serial in the Mandatory Fields (i.e. serial numbers are not only not required, but cannot be entered[18]).

Once you set a QSO mode to MGM (in any contest) the validation on the report is removed - it just requires it to be present – **so be careful what you type**[19].

If the MGM software being used can create ADIF files (e.g. WSJT-X ADIF Export) the log may be imported into Minos. Create the contest then select "**File/Append ADIF file to contest**" (see Figure 9) and browse for the .adi file to be included[20].

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[17] In V2.0 the options were locked to prevent changes; this restriction has been removed so be careful.
[18] Any of the "**Mandatory Fields**" not ticked is prohibited.
[19] If you switch back from MGM to e.g. USB in a mixed-mode contest you must re-enter the default report on the first contact.
[20] Defaults to the Minos log location.

Beware: the whole of the .adi file will be included so make sure the file only includes the data for the contest! Any records outside of the contest duration will be marked invalid but make cause problems with e.g. duplicate QSO marked within the contest. If the adi file is imported multiple times then many duplicate contacts will be created (as duplicates).

Concurrant logging is also now possible, and the preferred option, see section 3.6.

If you have cause to use "**Insert Before**" or "**Insert After**" (Section 2.2.5) with an MGM contest don't worry about the SerialTx being blanked out, resulting in a serial of 000.

2.2 Entering QSO details

When the Data Entry panel is displayed the cursor will initially be in the Callsign field; to return to that field click on it with the mouse or hit "F1". By default the Mode (top left of data entry panel) is set to USB, but it can be changed by means of the associated pull-down list, or will track changes made to the physical rig.

There is no required sequence of data entry, although the default order of the boxes (as selected by the "**TAB**" or "**RETURN**" key) reflects normal contest exchanges.

"**RETURN**" is normally used to move between fields[21], by selecting the next empty/invalid field[22]. For the RS(T) fields, if Autofill has been set[23], the remainder of the field will be filled with 59 or 599 as appropriate to the mode (not for MGM); any data entered will not be overwritten.

Fields may be selected 'out of order' using the mouse or the designated Function key (F1/F2/F3/F4/F5/F6 in Figure 14); a field may be partially filled (although of course must be completed before the QSO can be normally logged – see section 2.2.2 for details of logging incomplete QSOs).

[21] Clicking on the "**Log**" button has the same effect.
[22] For a new contact this will be the same sequence as if TAB were used.
[23] This is most conveniently set under "**Tools/Signal Report AutoFill**" but can also be done through "**File/Options**" (see Figure 116).

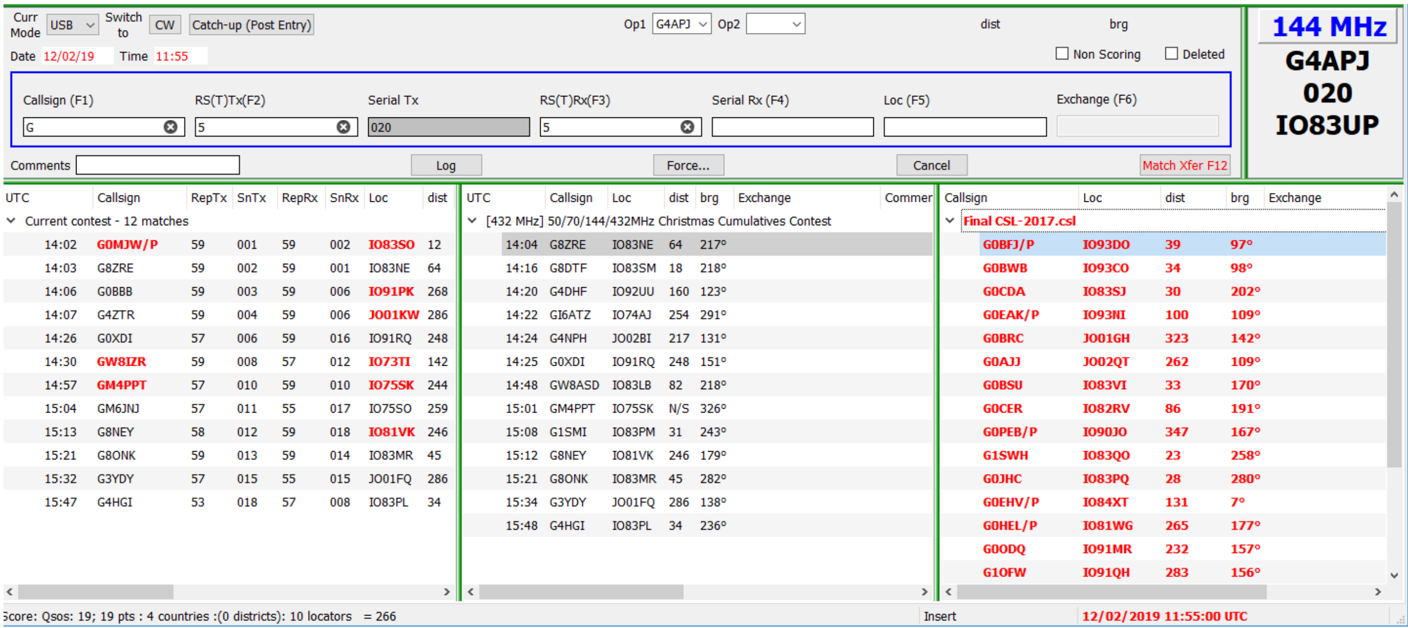


Figure 14 – Details of contact

Upper case is forced as data is entered in the Callsign, Loc, or Exchange[24] fields, and Minos displays possible matches from the current contest[25], any other contests that are open, and any open CSL file(s).

On RS(T) fields in non MGM contests, only numeric characters and a tone of A are valid; you can enter e.g. 59A for auroral contacts. In MGM contests no validation of RS(T) fields is done. On serial number fields only numeric characters are allowed, except in the received serial number; "-" shows no serial number is received (e.g. for NAC contests) .

As a Locator is entered the distance and bearing is calculated and displayed in the top right of the panel. Partial Locators give approximate figures, designated by brackets.

Note that data is not validated as it is entered, only when you attempt to log the entry. This allows for partial information to be entered (e.g. during QSB) and then be enhanced/corrected later in the contact.

If a suitable match is highlighted (with the grey bar, see Figure 14) then the contents of the selected entry can be transferred to the data entry boxes by clicking on the "Match Xfer" button or hitting the "F12" key[26] (or by double clicking the requisite entry). This *may* be faster than retyping the data, and should reduce typos (but beware, /P stations are not the only ones that may change location/locator details).

The content of individual fields may be deleted using the new "**Clear**" marker (black circle with white cross) at the end of relevant fields.

The current QSO may be abandoned, and all entry fields reset to default, by hitting the "ESC" key or clicking on the "Cancel" button. Hitting "ESC" or "Cancel" again recovers the fields, provided nothing has been typed since the original cancellation.

---

[24] If Postcode multipliers or similar are in effect.
[25] Duplicates are always displayed first.
[26] Watch out: the callsign may also get amended from that entered (e.g. /P added).

This allows the re-use of the current contact number; if the QSO has progressed far enough to make this undesirable tick the “Non-scoring” box before “Forcing” the entry to be logged.

There are a number of circumstances that might require “Non Scoring” to be used without “Force”:

- All the details have been entered, but you know that the other end missed them.
- To mark the FIRST contact with a station as a duplicate, rather than a subsequent one.
- The contact is known to be invalid – maybe with a station who will be one of your operators on the cover sheet.

Once all fields are filled with valid data, “RETURN” will log the contact[27]. This happens when the program sees that there is valid data in each required field, which under some circumstances may be before you think the contact data is complete. In this situation use cursor up to select what is now the previous contact, and edit in the required additions.

If the QSO data is incomplete[28] and you cannot log normally, then click on “Force” which will result in the dialogue described in section 2.2.2.

If your computer logging falls behind, and you miss one or more contacts, it is possible to override “Serial TX” by double clicking the field and entering the required next serial number. Minos will then check that you intend to do this and that you want to fill in the contacts later.

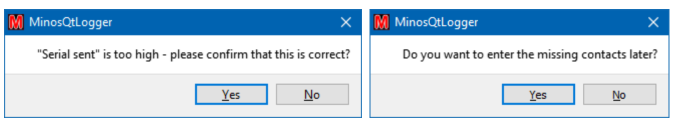


Figure 15 – Skipping contacts

Dummy entries will be created for the missing serial numbers. You can then fill in these missing contacts later (see section 2.2.4).

If you send some serial numbers twice, then double click and overtype the serial sent field to make the log reflect what you actually did.

2.2.1 Colour Coding

During normal operation various colours are displayed on the screen to indicate status or progress. Figure 16 shows an example part way through contest[29].

[27] The ‘running totals’ at the bottom of the Logger will be updated (see Figure 12 and Figure 16).

[28] For instance, some stations may not know their locator; you can work it out later from the geographical location they give you and then edit the QSO details.

[29] The screen layout has been modified from the default to remove the RigControl and Rotator application panels.

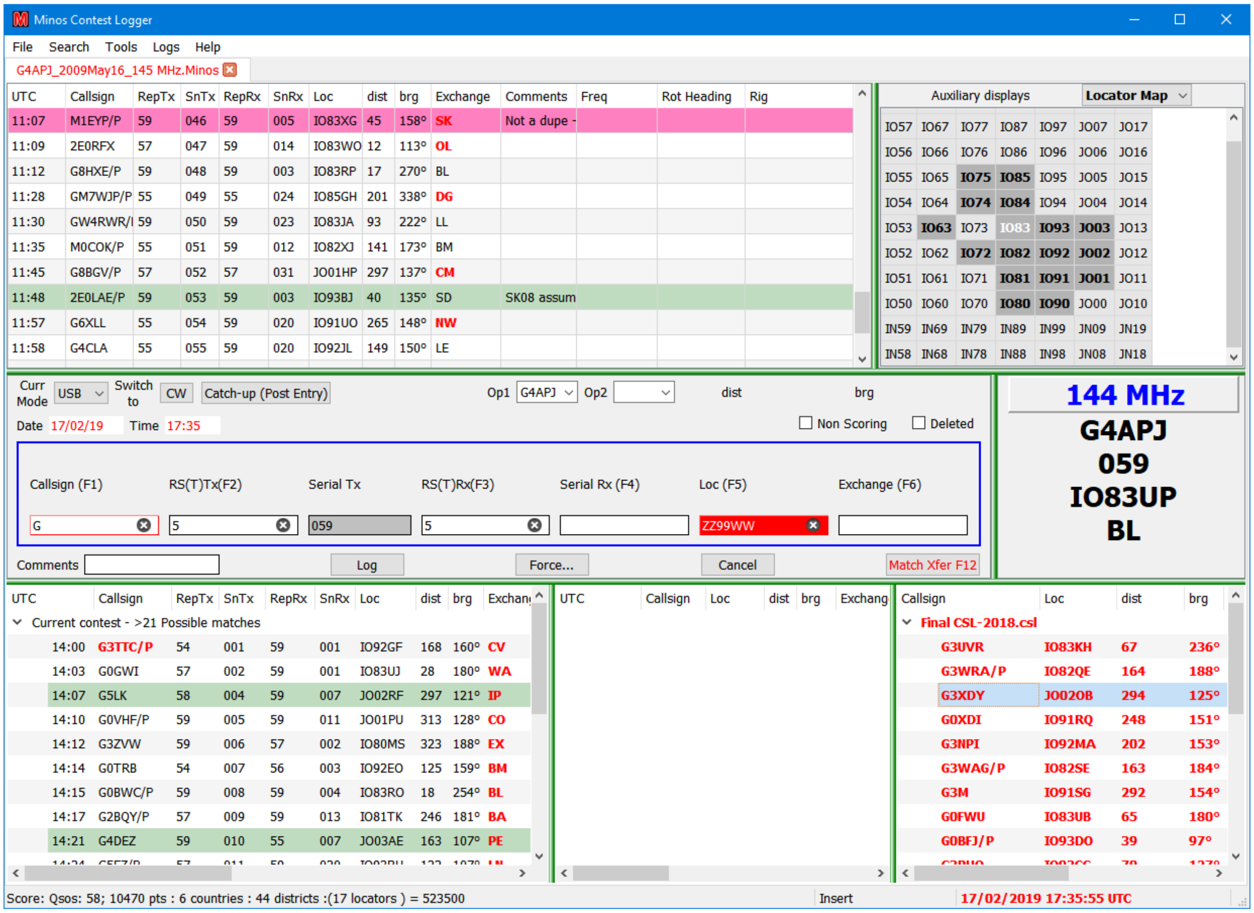


Figure 16 – Colour coding

On the main display panel:

- **Red Callsign**: New DXCC multiplier (where DXCC multiplier scoring active)
- **Red Loc**: New Maidenhead Locator (with Locator multipliers active)
- **Red Exchange**: New Postcode (or other Exchange, when multipliers active)
- **'Greenish' horizontal bar**: QSO where the details have been edited
- **'Mauve' horizontal bar**: record had to be forced; turns green once missing data entered
- **Colours in Locator stats:** Depending on the bonus scheme in use, various colours may be displayed[30]; grey indicates no bonus scheme in use. Pale indicates unworked, bold indicates worked.

Within the data entry area:

- **Red Date or Time**: current value is invalid for the designated contest duration (outside the contest time limits).
- **Blue text** above the data entry boxes: indicates the currently selected field
- **Red box** around callsign or locator indicates incomplete data.
- **Solid red fill** shows possible invalid data (including duplicate callsign)

[30] Which also show up in the other panels displaying Locator information.

Inside a match box:

- **Blue horizontal bar**: Currently selected match that will be used in a transfer operation (using F12).
- **Red Callsign/Loc/Exchange**: As per main panel

The content of the bottom right panel ('CSL match') is in red to show that panel is selected. See Figure 14 for an example of another contest panel selected.

2.2.2 Forcing a contact

If the data for a QSO is incomplete and cannot be normally logged then the "**Force**" button (or "Alt/F") can be used to store the current details pending further communication with the station or later calculation (e.g. when the other station does not know its Locator). This results in a panel as shown in Figure 17.

The current errors in the QSO are displayed, along with various options that may be selected:

- **Unfilled**: Data is missing
- **Backpacker**: If a back packer, or other station, has moved since you worked them earlier this allows the second contact to be marked as valid rather than a duplicate.
- **Insert manual score now**: (if you really want to).
- **Non Scoring**: (the default)
- **Don't Print/Export Entry**: to logically remove a contact
- **Force Country for Multiplier**: To force the use of a particular country for country multipliers.
- **No District Available**: ('auto' for non GB stations)
- **Cross band**: half score

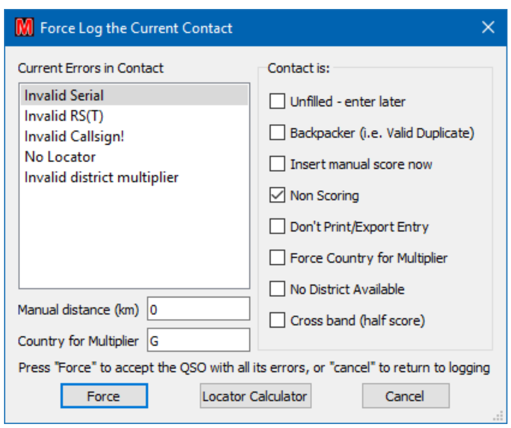


Figure 17 – Force log dialogue

The "Current Errors in Contact" may also indicate "Locator probably not within country".

Clicking "Force" (check focused, so also "RETURN") actually writes the record.

2.2.3 Modifying a previous contact

As and when more data is forthcoming for a Forced log entry the record may be edited to enhance or correct the QSO entry. Double click the requisite entry, which results in a display of the form shown in Figure 18.

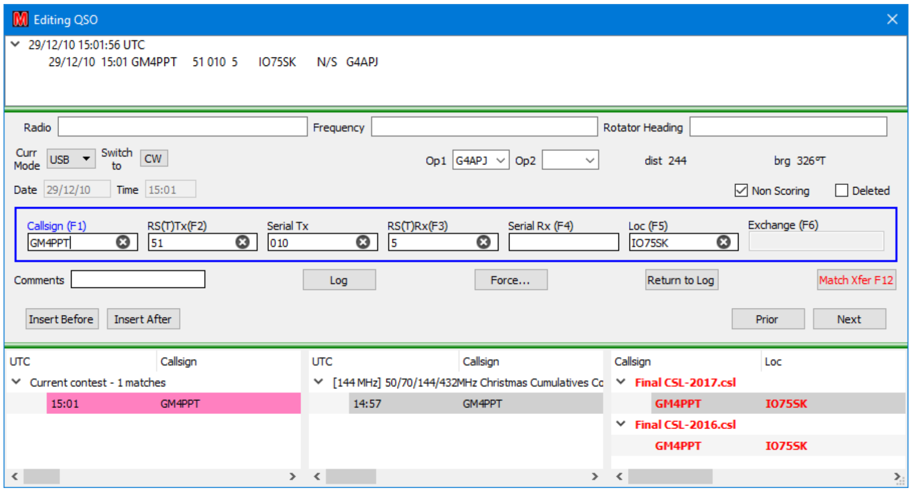


Figure 18 – Editing the details of a QSO

Once the data has been completed, ensure "Non Scoring" has been cleared (just above the Locator field) before logging the record by clicking "Log"[31]. If the data is still not complete then "Force" will invoke the force panel dialogue.

The modified entry in the main log display turns green (see section 2.2.1 for information on colour coding of records) if all information has now been supplied.

A message box (Figure 19) allows confirmation that the changes should be applied. For valid entries the distance and bearing are then calculated and the record is included in the score.

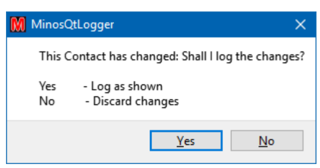


Figure 19 – Confirmation of changes

"Prior", and "Next" may also be used to navigate between entries that still require amendment. If the current selected entry has been modified when one of these buttons is selected then the confirmation of changes dialogue will appear before the action is taken. If "No" is selected then the original record will be restored and remain selected.
"Return to Log" **saves** any changes made and finishes the QSO editing activity.

Use "X" to **discard** any changes that you have made and return to the log.

[31] Note that if the serial number sent is changed it is assumed that it is a correction and no intermediate serial numbers will be offered to be created.

2.2.4 Completing unfilled contacts

If unfilled contacts have been created then the "First Unfilled QSO" button will appear within the QSO data entry panel (see Figure 20).

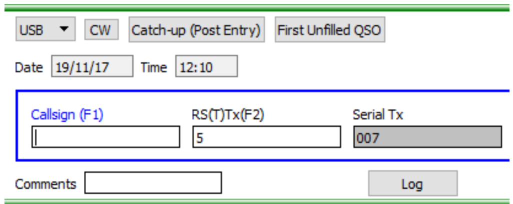


Figure 20 – First Unfilled QSO button

Click on this button to locate the unfilled QSOs. If any are found a similar dialogue to the edit QSO will appear (see Figure 22). Correct the problems, if only by marking the QSO as non-scoring, then use "Next" to move on until all have been processed.

There shouldn't be any left in the log with "UNFILLED CONTACT" in the comments, nor any incomplete information marked in 'mauve'!

Figure 21 – Search pulldown menu

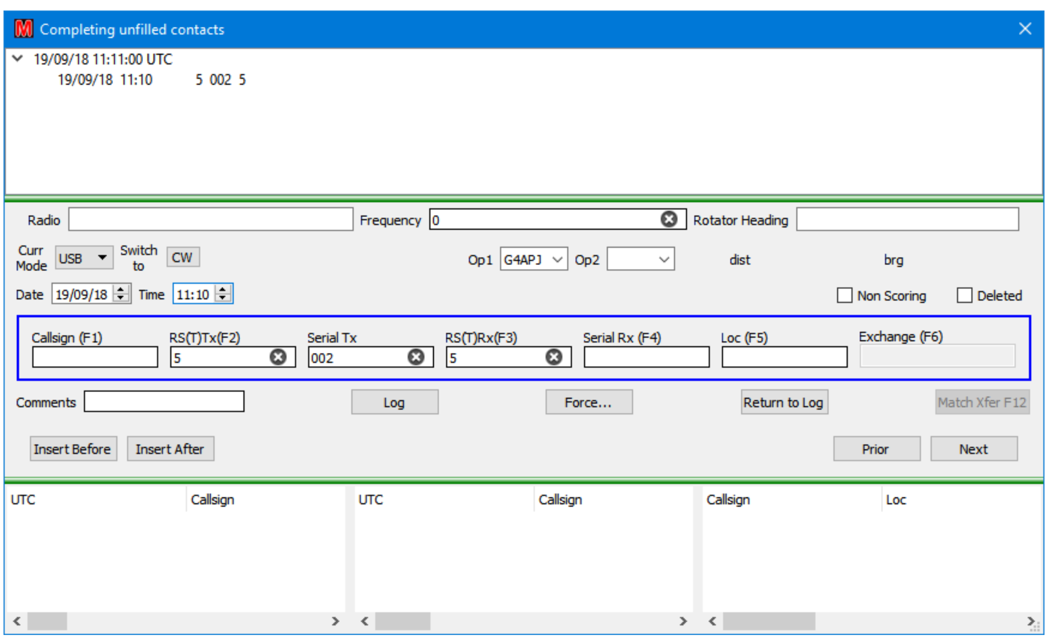


Figure 22 – Completing unfilled contacts

"**Return to Log**" **saves** any changes made and finishes the QSO editing activity.

Use "X" to **discard** any changes that you have made and return to the log.

2.2.5 Insert Before/After

When editing a QSO or filling any unfilled QSOs the "Insert Before" and "Insert after" buttons allow the creation of new QSO records in the indicated position relative to the currently selected QSO using the "Edit QSO" dialogue.

These buttons only become active after the first two records have been written, and do not allow additions before the first record or after the last.

2.3 Submitting an entry

Select "**File, Produce Entry/Export File**" and check that the 'administrative' details displayed are complete and correct (see Figure 23)[32]. Any last minute changes made at this point, e.g. overall comments, are written to the EDI file so that they get seen by the adjudicator; they do not necessarily appear on the claimed score page.

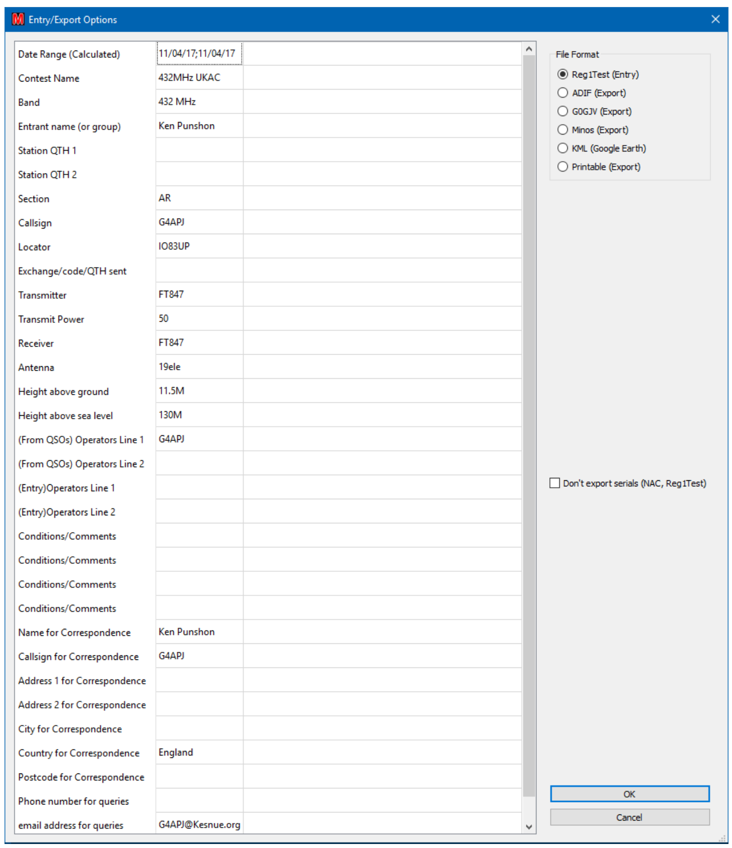


Figure 23 – Checking the Entry data

Select the required format (for contest submission this is unlikely to be anything other than Reg1Test that is accepted by the VHFCC up-loader).

[32] If changes have been made to Entry.ini or the other files since the contest was created then the requisite entry will have to be re-selected to get the data into the Entry data.

The other formats that may be used are:

- **ADIF**: to transfer data to other software packages
- **G0GJV**: (probably now redundant) to transfer to the old DOS Logger
- **Minos**: produces a 'cleaned up' log with all edits log cleaned out. It may also be used to produce a shorted abstract, e.g. for a Trophy Contest from a full IARU 24 hour event.
- **KML**(GoogleEarth): If you have GoogleEarth installed, generate one, and double click on it. It gives you all the contacts as pins on GoogleEarth.
- **Printable:** produces a plain text file for checking or printing

As can be seen in Figure 23 there is an option to suppress the export of serial numbers (NAC, Reg1Test).

Click "OK", presented with "**Save as**" dialogue.

Select requisite folder and filename (.EDI). You may want to have your own naming scheme if intending to save these files. The default location for the EDI file is the same as the log files.

Click "**Save**" to write the file.

Check the content of the resultant file. Note that in an EDI file any per-QSO comments will have been abstracted and collected under the [Remarks] header in order to comply with the EDI field definitions; this can be confusing at first sight.

Go to the entry robot (http://www.rsgbcc.org/cgi-bin/vhfenter.pl) to submit your entry (you will get a copy back from the robot by email). If you wish to practise the submission process then take a look at the new VHF Log Test Facility on:

http://www.rsgbcc.org/hf/rules/2018/rvhflogtest.shtml and
http://www.rsgbcc.org/cgi-bin/vhfenter.pl?Contest=Log%20Upload%20Tester

To exit the program, use "**File/Exit Minos**" or "**X**". If a contest is truly completed then it is advisable to close it first[33], possibly after marking it as Protected.

[33] The abstracted content of a contest may then be added to the archive list using one of the CSL utilities, or manually.

3 Additional Features

3.1 Post-event data entry

This facility is intended to allow transfer of data from a paper log. Do not use it to tidy up after the end of a contest.

Create the contest in normal fashion then click on "**Catch-up (Post Entry)**" in the Details of Contact panel (see Figure 24). The date will change to the start date of the contest (watch out in 24 hour, or longer, contest) and the cursor will move to the Time field which is set to 00:00 initially.

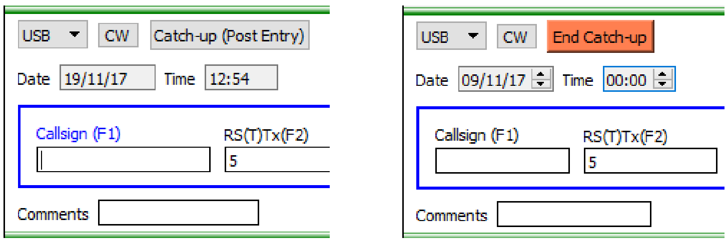


Figure 24 – Catch-up (Post Entry)

Enter the time (and date if needed) of the first contact and then enter QSO data in the normal fashion. The duplicate checking and CSL panels will operate in normal manner.

After each QSO is entered the cursor is set on the last character of the time field to allow amendment of the time to that of the contact. When finished click "**End catchup**" to return to normal operation.

Once all data entry is complete, submit the entry in normal fashion (see section 2.3).

3.2 Auxiliary displays

By default the top right of the Logger display is used to show auxiliary data in a single panel. "**Tools/Screen Layouts**" is used to control the number and position of such Auxiliary Displays (see section 3.5) and the default content.

A pull down list at the top of each panel controls what content is displayed. The panels selected are held within the contest context and are preserved when a contest is closed.

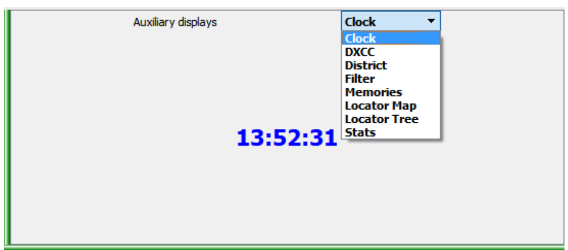


Figure 25 – Auxiliary display options

Different options may be selected to be displayed for each open contest, although any filters configured apply to all (relevant) contests. The option selected, and the filters applied, are tied to the contest and stored in the log (minos) file.

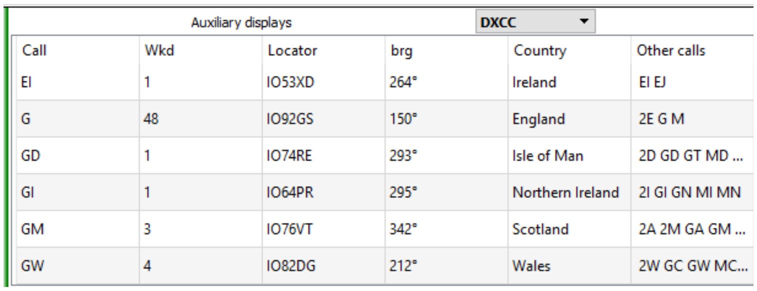
3.2.1 Clock.

The default is to display the current time in UTC. Blue means (the currently selected) contest is in progress, red means it has either finished or not yet started.

3.2.2 Statistics

There are a number of statistics displays that can be selected from the pull-down list:

- **DXCC** - displays the DXCC entities, under control of the selected filters. Most of the time only the European entities will be meaningful.

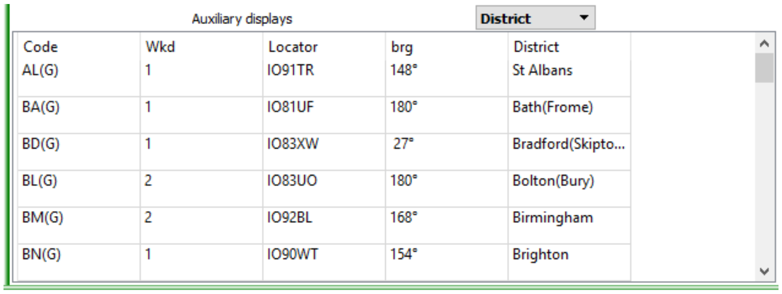


Auxiliary displays DXCC

Call	Wkd	Locator	brg	Country	Other calls
EI	1	IO53XD	264°	Ireland	EI EJ
G	48	IO92GS	150°	England	2E G M
GD	1	IO74RE	293°	Isle of Man	2D GD GT MD ...
GI	1	IO64PR	295°	Northern Ireland	2I GI GN MI MN
GM	3	IO76VT	342°	Scotland	2A 2M GA GM ...
GW	4	IO82DG	212°	Wales	2W GC GW MC...

Figure 26 – DXCC Statistics

- **District** – In Postcode exchange contests can be used to display the Districts, under control of the selected filters.



Auxiliary displays District

Code	Wkd	Locator	brg	District
AL(G)	1	IO91TR	148°	St Albans
BA(G)	1	IO81UF	180°	Bath(Frome)
BD(G)	1	IO83XW	27°	Bradford(Skipto...
BL(G)	2	IO83UO	180°	Bolton(Bury)
BM(G)	2	IO92BL	168°	Birmingham
BN(G)	1	IO90WT	154°	Brighton

Figure 27 – District Statistics

- **Filter** – controls the display of the DXCC, and Districts content **only**. Selection of the geographical regions is meaningless for the Districts display, only worked/unworked are applied. These filtering controls get saved in the minos file for each contest. The panel select works per-contest

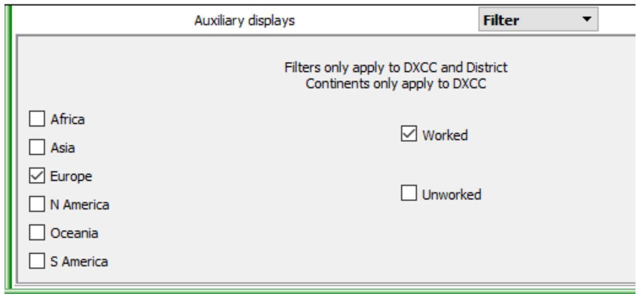


Figure 28 – Statistics Filter

- **Locator** – displays Locators using a colour coding scheme to indicate the relative bonuses[34] (if bonuses are in use). The initial display of squares will be extended, and 'sliders' added, if a square outside of this range is worked.

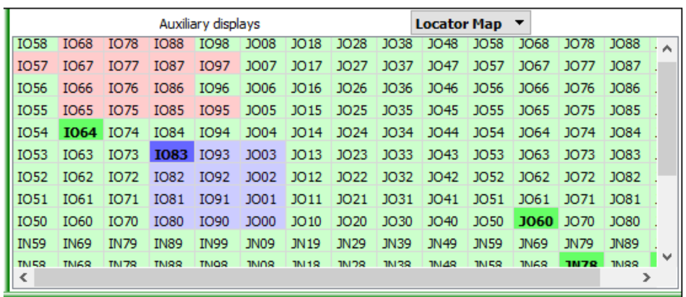


Figure 29 – Locator Statistics (Grid)

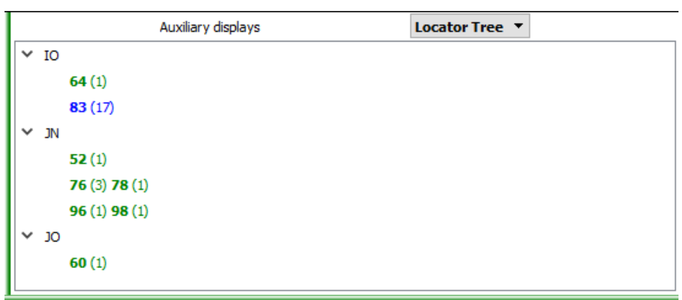


Figure 30 – Locator Statistics (Tree)

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[34] Since in the UKAC all squares now have the same bonus this is no longer really relevant. Grey indicates no bonus scheme in use. To get the colours shown, the B2 scheme would have to be in use.

- **Stats** – displays information on best contact and scoring rates. The intervals for which the DX statistics are displayed can be modified by use of the P1 and P2 boxes at the bottom of the panel.

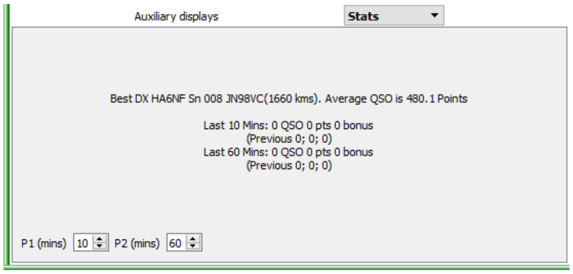

Figure 31 – QSO Statistics

3.2.3 Memories

This auxiliary panel can be used standalone as a 'scratchpad' or in conjunction with RigControl (see section 4.2.1) to provide the extended function described here. When first invoked there will be no memories present, they have to be created. The created entries are stored within the context of the currently open contest[35] and are preserved when a contest is closed.

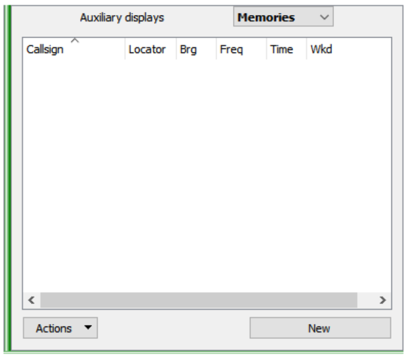

Figure 32 – Initial memories panel display

The order and size of the columns shown in Figure 32 can be changed using standard drag and drop mechanisms, and the display order can be controlled for each (ascending or descending). The column containing the callsign on the left hand side cannot be moved.

If "**New**" is clicked when the QSO entry panel has a Callsign and/or Locator present that data will be pre-loaded into the new memory (it can of course be modified), along with the current time[36] (and bearing if available). If no data is present then an empty memory is displayed (although the time is filled in). See Figure 33 for examples. Clicking "**OK**" then loads the memory into the panel.

[35] Hence are stored on the local machine, not any remote machine being used as a connection to the rig.

[36] The time information in a memory is lost if you close the contest or exit the logger.

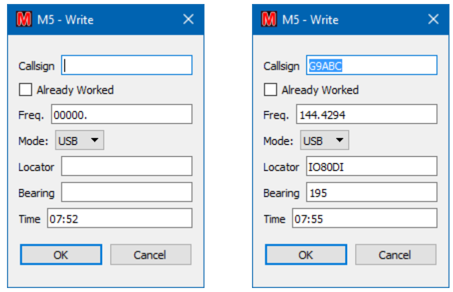

Figure 33 – Example memory content (blank and from QSO entry with RigControl active)

In order to save the memory some text is required in the callsign field; if nothing is supplied then "??" will be used, but beware when loading the contents back into the QSO entry panel.

The "**Already Worked**" box can be ticked if required which will then mark the memory entry in brackets in the same manner as any callsigns found in the log (see Figure 34) [37].

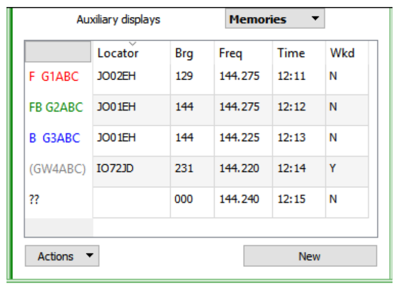

Figure 34 – Populated memories panel

If memories continue to be added then a scroll bar will appear once the available space is filled up. There is no practical limit to the number of entries, but they may be difficult to handle once scrolling starts.

The colour coding of the callsigns, and the preceding letters, indicate memories that 'match' [38] the current rig frequency and the beam heading:

- **Red F** frequency within +/- 2 kHz.
- **Blue B** bearing within +/- 15 degrees.
- **Green FB** both frequency and bearing within the limits.

Worked entries (manually designated or detected in the log) are enclosed in brackets. The '??' indicates a memory loaded with a blank (probably unknown) callsign in the QSO panel.

The "**Actions**" button provides a pull-down list of options (as shown in Figure 35) to Read, Write, Edit or Clear the selected memory or clear memories:

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[37] This might for instance be used to mark a station that you want to work on another band.

[38] The match limits are currently hard coded.

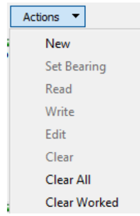


Figure 35 – Memory actions menu

Right-clicking an entry will provide the same list:

- **“New”** is an alternate way of creating a new memory.
- **“Set Bearing”** loads the rotator bearing from the memory into “Target” field of the Rotator panel.
- **“Read”** will load the callsign and locator information into the QSO entry panel, as will a single click on the callsign field of a memory.
- **“Write”** prepares to load the memory from the current QSO fields. It can be manually changed before confirming with “OK”; the current time will always be transferred.
- **“Edit”** is very similar to “Write”. It opens the selected memory to allow the contents to be modified (without copying the QSO data).
- **“Clear”** deletes all contents of the selected memory (without confirmation).
- **“Clear All”** deletes all the stored memories (requires confirmation).
- **“Clear Worked”** deletes (without confirmation) all the memories that have been marked as worked as a result of being found in the log. Any entries that have been manually marked as worked are retained).

Memories may also be loaded by right-clicking on a match panel entry (see Figure 36).

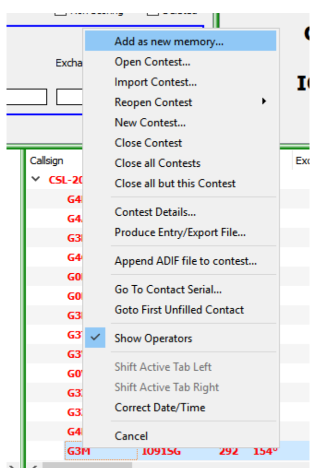


Figure 36 – Loading an entry from a match panel

A click (single or double) in the column to the left of the callsign field is equivalent to a read; a double click on any other part of the entry is equivalent to edit.

Be careful not to (double) click the callsign if editing (or clearing) an entry (e.g. if a station's frequency has changed) because the information will load to the QSO panel and the rig will QSY to the old frequency if you are using Rigcontrol.

If a locator is present in the QSO box then the calculated bearing will be added to a new memory. If there is no locator then a manual bearing may be entered when a memory is created. If a bearing is manually entered and there is a locator present then the calculated bearing overrides the manual data.

3.3 Locator calculator

The Locator Calculator can be invoked from the main screen using "**Tools/Locator Calculator**" or via a button on the "**Force**" panel. If a contest is open the current station locator will be pre-loaded as Station 1.

Depending on the option selected, you can enter a Locator, a Latitude/Longitude, or a (UK) National Grid Reference[39] into the "**Input grid ref**" box (see Figure 37) and convert between the various displays. Selecting "**Calc**" (or hitting "**Return**") will then calculate and display the other two formats. If both Station details are given it also calculates the distance and bearing from Station 1 to Station 2.

Use **Exit**, **X**, or **Cancel** to exit

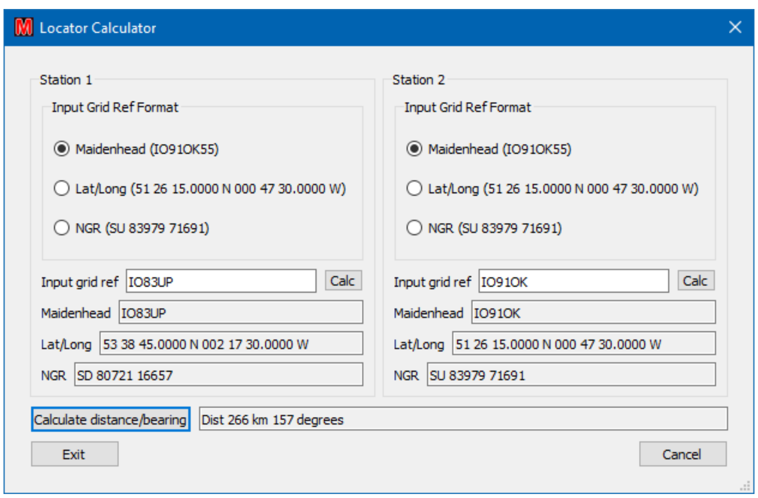


Figure 37 – Locator calculator

If the contest is set to "**allow 4 figure locators**" then the locator calculator will work out the closest point for a 4 figure locator e.g. JO88.

The distance is **always** calculated by the standard algorithm for the distance between the locator square centres.

[39] Apologies to the Irish: the Irish National Grid requires a different centre meridian to allow for being somewhat west of the rest of mainland Britain, and hence is not calculated.

Microwave locators using 8 figures such as IO80ST37 are supported. If the current contest/list has an 8 figure locator then the distance and bearing will be calculated to the accuracy of the 8 figure locators. During conversions, 8 characters always generated.

The precision used may appear excessive, but it is done in this way to enable conversion from NGR to Lat/Long and back again, and have some hope of ending up with the same result. Of course Locators are considerably less accurate.

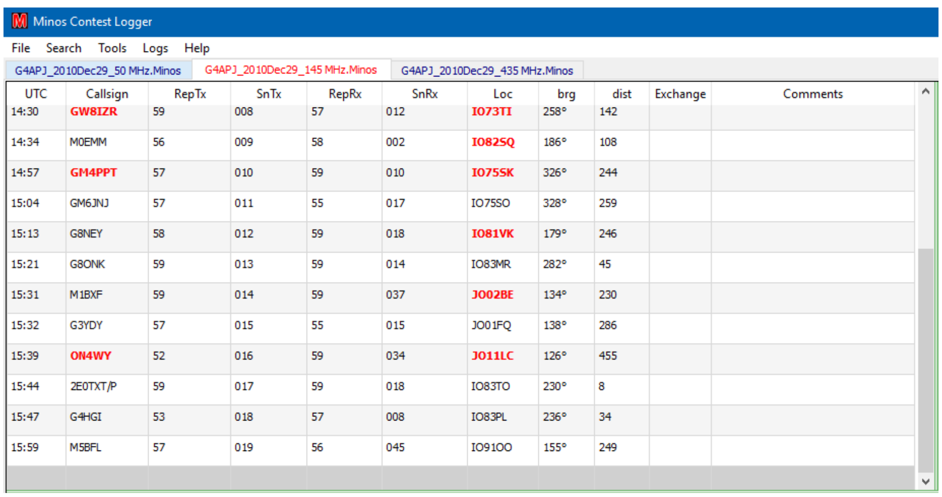
But please note that precision doesn't necessarily imply accuracy; but those locations that can be checked accurately give a similar result.

If the entered location is too far outside the UK then no NGR is displayed.

3.4 Advanced operation

3.4.1 Multiple Contests

It is possible to have multiple contests open simultaneously[40]. Each is displayed as a separate tab in the main logging screen as shown in Figure 38. It is possible to select/open multiple CSL (&LOG) files at once, but each will require acknowledgement. Contest sets (section 3.4.2) simplify the handling of multiple contests.


Minos Contest Logger

File Search Tools Logs Help

G4APJ_2010Dec29_50 MHz.Minos | G4APJ_2010Dec29_145 MHz.Minos | G4APJ_2010Dec29_435 MHz.Minos

UTC	Callsign	RepTx	SnTx	RepRx	SnRx	Loc	brg	dist	Exchange	Comments
14:30	GW8IZR	59	008	57	012	IO73TI	258°	142		
14:34	M0EMM	56	009	58	002	IO82SQ	186°	108		
14:57	GM4PPT	57	010	59	010	IO75SK	326°	244		
15:04	GM6JNJ	57	011	55	017	IO75SO	328°	259		
15:13	G8NEY	58	012	59	018	IO81VK	179°	246		
15:21	G8ONK	59	013	59	014	IO83MR	282°	45		
15:31	M1BXF	59	014	59	037	JO02BE	134°	230		
15:32	G3YDY	57	015	55	015	JO01FQ	138°	286		
15:39	ON4WY	52	016	59	034	JO11LC	126°	455		
15:44	2E0TXT/P	59	017	59	018	IO83TO	230°	8		
15:47	G4HGI	53	018	57	008	IO83PL	236°	34		
15:59	M5BFL	57	019	56	045	IO91OO	155°	249		

Figure 38 – Multiband contest

This was originally implemented to allow a single operator to handle a multi-band contest, e.g. the Christmas Cumulatives or all the microwave bands in a contest in the 432MHz-248GHz contest using a single copy of the program[41].

Separate scores, multiplier lists, etc. are maintained for each contest, but the current list of operators is shared.

[40] Whilst it is possible to have multiple contests open to one band (or one rotator) this is **not recommended** and the results are indeterminate.

[41] Be careful! It is very easy to get confused and start entering data in the wrong contest log. This can be mitigated by use of RigControl (see "**Band Select**" description in section 4.2.1).

3.4.2 Contest Sets

The above concept has now been extended to "**Contest Sets**". These are managed via "**Logs/Contest Sets/Manage Contest Sets**" (see Figure 39).

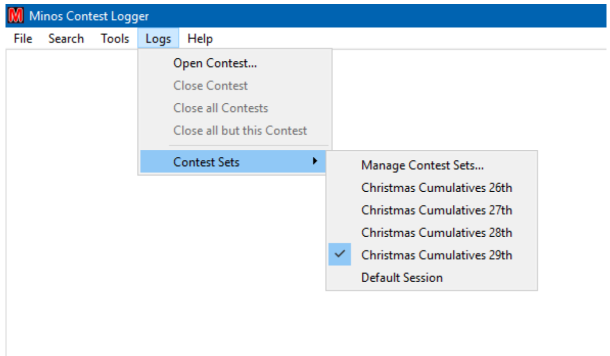


Figure 39 – Opening a Contest Set

Sets can then be created, logs added etc. When the programme is started all members of the highlighted set are opened[42]. If this is for the first time there is no requirement to acknowledge each and every "**Contest Details**" panel as there would be if each was opened in turn (or as a block).

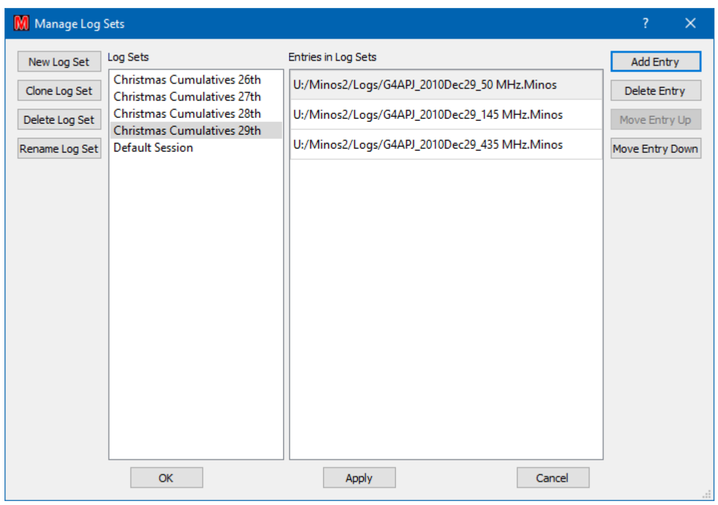


Figure 40 – Contest Set Management

Contest Sets can be considered as a single entity when opening/closing them. In the example above, selecting "**Christmas Cumulatives 28th**") would close the three logs from the 29th in one go and open all three logs for the 28th.

3.4.3 Archive Lists (CSL)

Warning: however good and accurate you think your list is, DO NOT TRUST IT and don't copy details from it that you have not heard on the air. Remember: people, and especially portable contest stations, move from

[42] So be sure to leave "**Default Session"** selected if you do not intend to be working with a set.

time to time, or sometimes just work out their locator more (or even less) accurately. Not only that, but if you ‘invent’ details that haven't been copied off air you are cheating, and if caught, you may well be disqualified from both that and other contests.

The format of a CSL file is described in section 0. It can be produced manually or the utilities detailed in Section 6 can used to create the file and maintain the content.

CSL files are opened using “**File/Open Archive list**”. Once a list is loaded the file is closed, so that any changes to the file will only be picked up the next time it is loaded. Multiple files may be used which are controlled via “**File/Manage Archive Lists**” [43], that also allows the content of a file to be edited. In the event that a CSL needs to be removed this is also done via “**File/Manage Archive Lists**”.

Archive lists apply to all open contests

3.4.4 Time adjustments

Adjustments to the date and time may be made by selecting “**Correct Date /Time**” under the Tools menu option (see Figure 41).

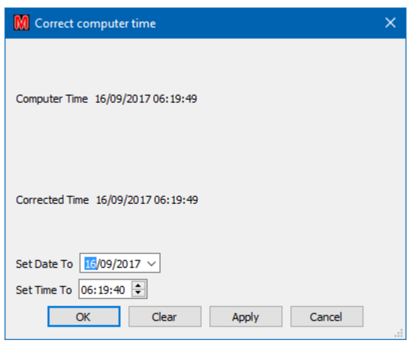


Figure 41 – Clock adjustment

3.4.5 Maintenance

The Contest Calendar should be updated at the very minimum on a yearly basis, preferably more regularly – especially in the first quarter when changes may still be in progress. This is done by clicking “**File/New Contest**”, “**VHF Calendar**”, “**Download latest Calendar**”.

3.4.6 Additional functions on screen

Right clicking anywhere outside the match panels on the contest panel[44] allows a number of functions (see Figure 42). The first two groups of these are identical to the functions available from the File menu.

[43] At least one CSL file must have been opened for this option to be active.
[44] Right clicking inside a match panel adds the option to add a memory – see Figure 36.

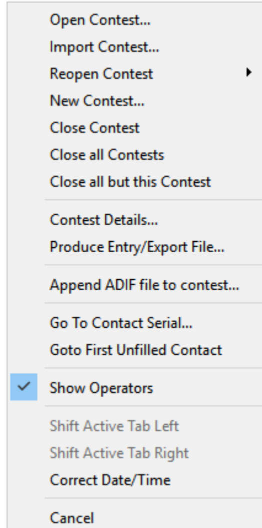


Figure 42 – Right click menu options on main screen

- **Append ADIF file to contest:** allows ADIF from another logger to be imported, as described in section 2.1.1.
- **Go To Contact Serial:** invokes the QSO edit dialogue for the selected contact number.
- **Goto First Unfilled Contact:** opens the QSO edit for the first located contact with missing information (section 2.2.4).
- **Show Operators:** Controls display of the Op1/Op2 fields within the data entry panel.
- **Shift Active Tab Left/Right:** Move the currently selected tab in the specified direction. The tabs can also be dragged.
- **Correct Date/Time:** Correct the PC date/time (see Section 3.4.4)
- **Cancel:** exits the option panel with no action

3.5 Screen layout configuration

Clicking on "**Tools/Screen Layouts**" allows selection of the screen format to use for the active contest or, by clicking on "**Configure Screen Layouts**" option, the definition of the formats to be contained in the list (Figure 43).

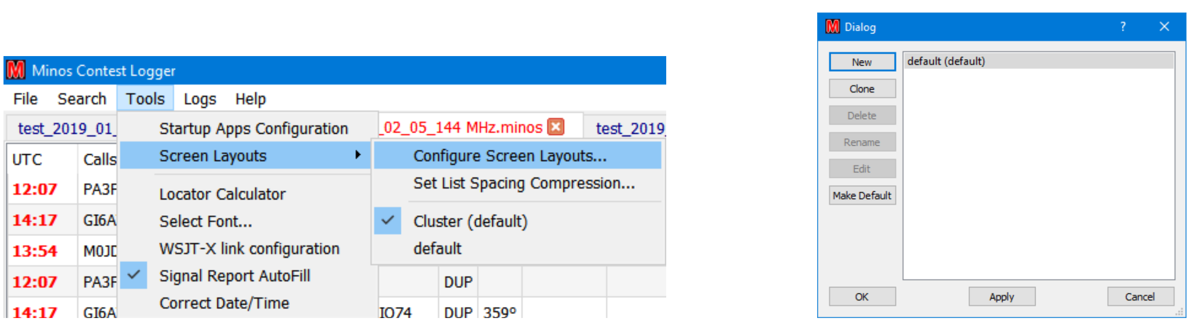


Figure 43 – Invoking the screen layout editor

A new layout is typically created by cloning the 'default' layout and giving it a name (Figure 44). A specific layout can be selected as the default for future contests by clicking "**Make Default**". The layout used for an individual contest[45] can also be changed using the "**Screen Layout**" pull-down list on the mid-right of the contest details panel (see Figure 9).

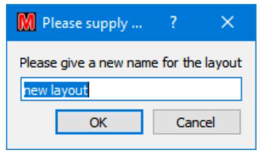


Figure 44 – Creating a newscreen layout

Clicking "OK" to a "**Clone**" operation or "**Edit**" to an existing (user defined) layout invokes a panel of the form shown in Figure 45 (the default layout) which allows the user to configure the content and placement of the Logger panels.

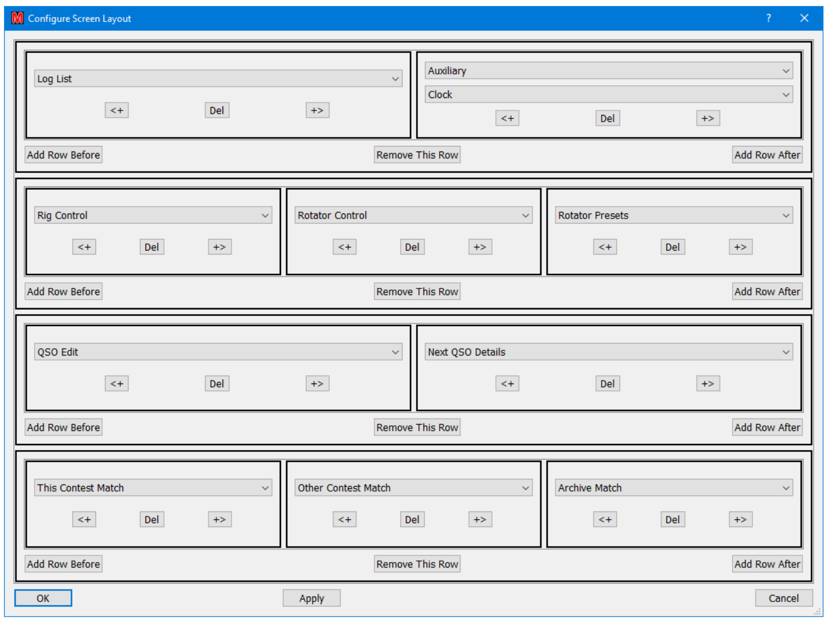
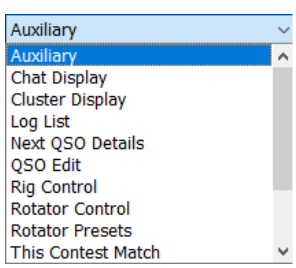
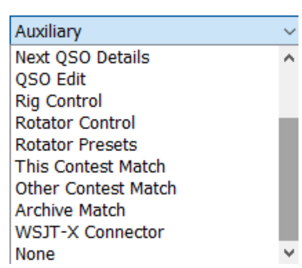


Figure 45 – Default screen layout configuration

Most of the buttons are self-explanatory, but for the sake of completeness a brief description of each is given here:

- **"<+" and "+>":** Add a box to the left or right respectively of the current box. The content of that box is then selected from the pull-down list:





[45] The layout selected for a given contest is retained within the minos log file.

Figure 46 – Panel display options

- **"Del":** Removes the panel from the display.
- **"Add Row Before"/After:** Inserts a single panel before/after the row in which it is clicked. Additional boxes can then be added to that row using "<+" and "+>".
- **"Remove This Row":** Deletes the whole row (multiple boxes if present).
- **"OK", "Apply", "Cancel"**options are available

In the event that all rows have been deleted[46] an additional option "**Add Row**" will appear (see Figure 47) to allow rows to be recreated (or click cancel and start again).

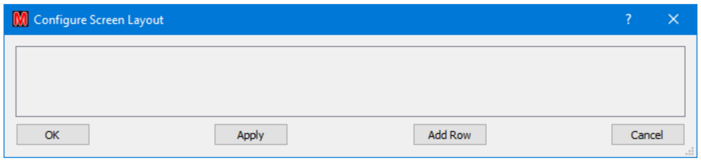


Figure 47 – Blank layout options

The Screen Layout configuration determines which panels are displayed where. To adjust the size of those panels slide the green bars (which change to red during adjustment) in the main Logger panel.

Note that (currently[47]) panels do not appear/disappear as applications are configured, hence reconfiguration is desirable (if no RigControl/Rotator then you probably want to remove that row).

There are multiple ways of achieving the same effect!

3.5.1 Set List Spacing Compression

To allow more entries to be displayed within the various frames on the screen, adjust the height of table entries within the main Logger window by selecting the "**Set List Spacing Compression**" option from "**Tools/Screen Layouts**" (Figure 49)[48].

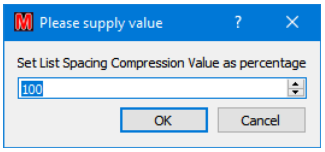


Figure 48 – Default list compression

The minimum value without causing legibility problems is likely to be around 75, but it varies with operating system and screen settings. Be careful, if it's too small then Q can appear to be O or 0, and J can look like I.

Be prepared to experiment!

[46] Or none have yet been created in a **New** layout.
[47] This *may* change at some point in the future; meanwhile use the "**Make Default**" facility.
[48] This can be combined with reducing the font size, but it is possible to make entries illegible!

3.5.2 Changing Fonts

The display font in use[49] can be changed by clicking on "**Tools/Select Font**" which invokes a window of the form shown in Figure 49 which shows the default font.

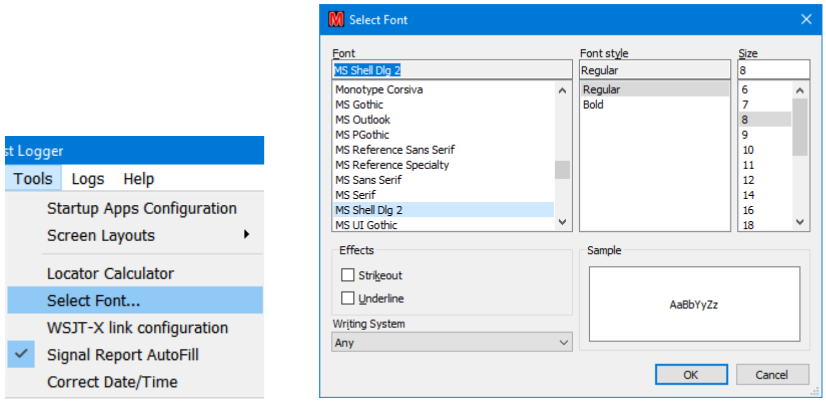


Figure 49 – Font selection

Beware, panel sizes may change and some fonts (notably 'Wingdings') can cause complete gibberish. To get out of such a problem, note that:

- "**File**" is the leftmost entry on the top banner
- "**Exit MinosQT and Clear registry**" is the last but one entry on the pull down list
- "**OK**" is the left (default) button in the confirmation.

3.5.3 Band marker colour

In the "Next QSO details" panel the band in use is highlighted in colour (Figure 50).

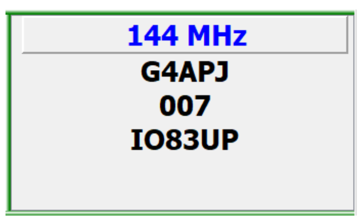


Figure 50 – Next QSO details

If you wish to change the colour used then edit **\Configuration\bandlist.xml** to change the colour at the end of the requisite line:

<Band type='VHF' flow='144' fhigh='146' unit='M' wlen='2m' UK='144 MHz' Reg1Test='145 MHz' ADIF='2m' Cabrillo='144' **Colour='blue'**/>

Available colours can be found by inspection of the other entries in the file.

Note that any changes made are lost on re-install!

[49] Within the main Logger window only.

3.6 WSJT-X and MSHV Interoperation

In order to simplify the logging of MGM contests Minos can interoperate with WSJT-X. If WSJT-X is running on the same machine as Minos (currently the only option), then it should 'just work'[50].

Changes maybe made to the Minos settings from "**Tools/WSJT-X link configuration**" (Figure 51) which allows the link to be disabled (or re-enabled) or changes can be made to the Port address (in conjunction with that of the WSJT-X configuration). The group address should be left empty.

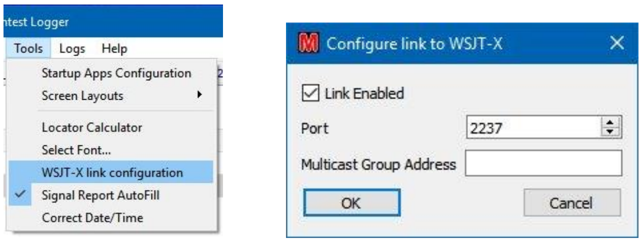


Figure 51 – Adjust settings for WSJT-X

In order to view the WSJT-X log the screen configuration must be changed (see section 3.5) to include a "**WSJT-X Connector**" panel[51].

When WSJT-X logs a contact the logging details will be replicated into Minos, into whichever contest happens to be focussed at that time! If the contest band does not match the frequency information of the QSO then a warning will be issued (Figure 52)[52].

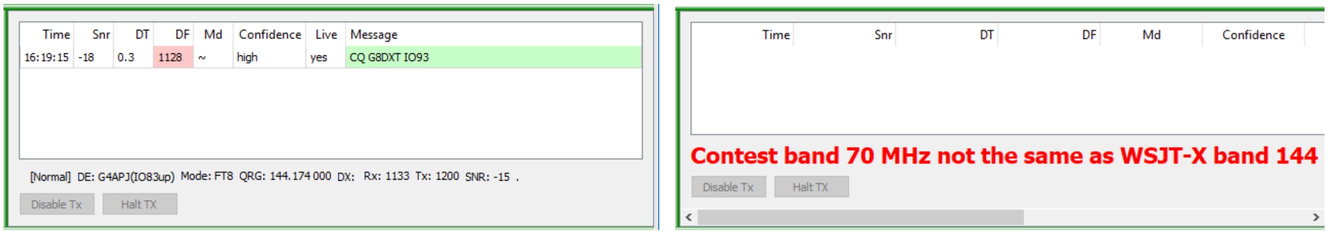


Figure 52 –WSJT-X Logging, and Band warning.

It is also possible to control some limited aspects of WSJT-X from Minos:

- Double click a CQ to call the station
- Disable Tx <<Curently disabled>>
- Halt Tx <<Curently disabled>>

[50] Remote access may or may not be possible in the future; TBI.

[51] Beware: Screen layout only allows one WSJTX panel, but doesn't give an error message – it just defaults to "None" on the additional panel(s).

[52] Actually, the warning is issued as soon as Minos detects the band mis-match which will typically, but not necessarily, be before the QSO is completed and the contact is logged.

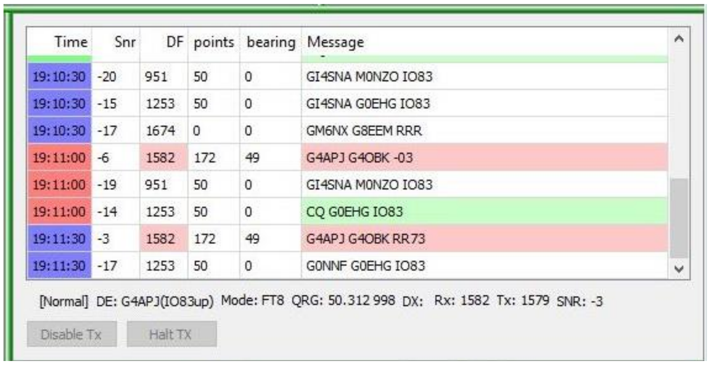


Figure 53 – WSJT-X panel in operation

The colours in the panel are derived from the message aggregator source in WSJT-X and are dependant on position:

- Time: A cycle of four colours to help keep track of the 15 second periods.
- DF: Red if 'near' own frequency.
- Message: Red if message contains own callsign, green for a CQ call.

QRG goes red when transmitting.

In MSHV you need to tick " **Enable Logged QSO ADIF Broadcast**" to activate the logging (Figure 54). Note that this only reports completed QSOs, so the sort of information from WSJT-X as shown in Figure 53 will not be available.

Minos will ignore any data seen as a result of ticking "**Enable Logged QSO Broadcast**" (there appears to be a bug in it).

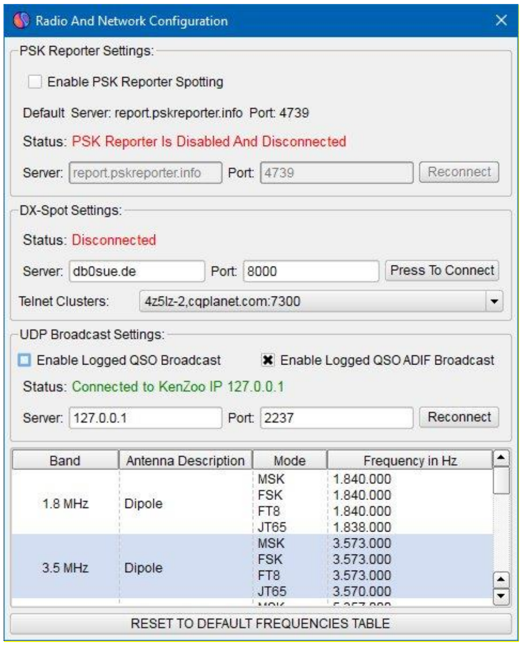


Figure 54 – MSHV configuration

4 Integrated Applications

Some of the integrated applications described in this section are not yet available, even in rudimentary form, and may not appear for some time (if ever). As and when they do appear any functionality described here may have changed.

Two applications form the infrastructure which the functional applications use:

- **MqtServer** is fundamental to the operation of the applications. It is a message broker which runs on all machines where you want the applications. It handles all the message-passing between Logger, apps, and remote servers.

- **MqtAppStarter** is the Application Starter. It is used *instead* of the Logger when you want applications but not the Logger. This will generally only be used in complex multi-machine environments (see section 4.4.3.1).

Only one copy of these infrastructure applications should be run on any given machine at any given time.

4.1 Configuration

In order to operate correctly the application configuration must be defined. The following description is for the use of the Logger to set up the applications. Usage of the Application Starter is covered in section 4.4.3.1.

This configuration process is initiated either via the "Apps" button on the Welcome Screen or from "**Tools/Startup Apps Configuration**" [53] [54]. Either method will result in small "**Minos App Configuration**" screen which should be expanded from its default size before configuring anything (see Figure 55).

To add an entry click "**New Entry**" and select the required application from the "**App Type**" pull-down list. For simple 'single user' configurations all of the information required for the applications is auto-generated[55].

Since "**Server**" is required for everything else it is automatically defined first. "**RigControl**" and "**Cluster**" are the other most likely applications for simple (single machine) environments, plus "**Rotator**" if one is present.

After applications are defined they are by default always all started. To avoid this use "**Logger Only**" in the startup splash screen or, on a more permanent basis, clear the "**Enabled**" box for the application in its definition, (remembering that Server is required to be active if **any** other application will be active).

To store the configuration Click "**Save and Close**" (this will write the information to MinosConfig.ini and exit the window) or "**Save and Start Apps**" which will then require the "**Cancel**" button to exit the display.

---

[53] Reconfiguration is only allowed if the applications are inactive – use the "**Stop Apps**" button.
[54] In complex environments it can also be done using the MqtAppStarter application on a machine that is just acting as a server and thus not running the Logger.
[55] For details of more complicated configurations see section 4.4.

By default “**Hide Server Apps**” is ticked for MqtServer to reduce the number of panels and icons displayed. Depending on configuration it may be advantageous to tick this for other applications.

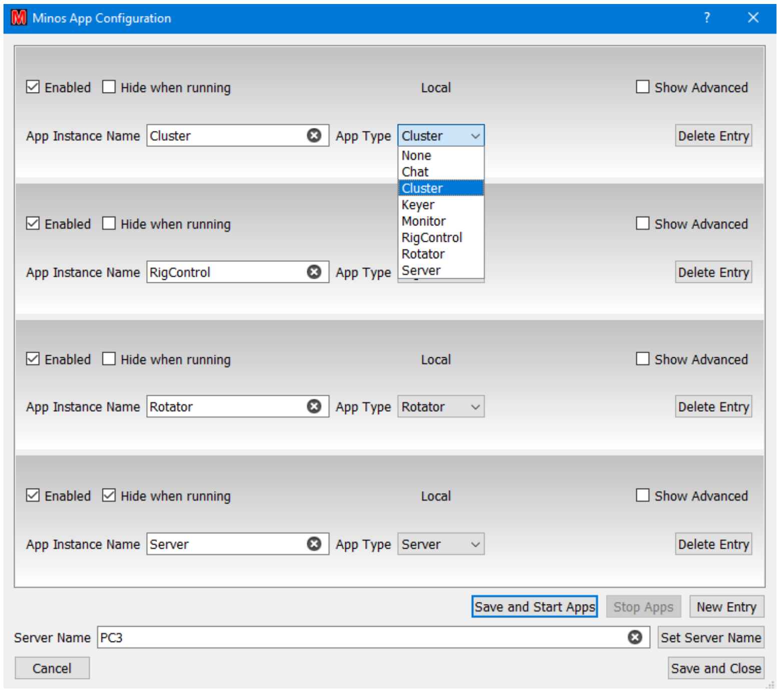


Figure 55 – Typical set of applications

“**Delete Entry**” can be used to delete erroneous information, after which “**Save and Close**” stores the changes.

When Server first runs, it requires Firewall configuration as shown in Figure 56.

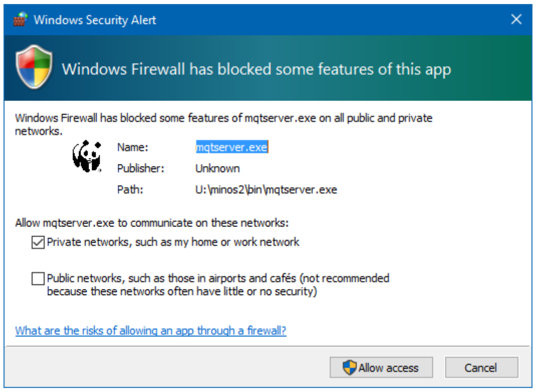


Figure 56 – Firewall message

If “**Hide when running**” has been unchecked for MqtServer then a window will appear that gives statistics on the active software components (Figure 57)[56].

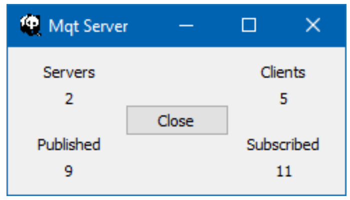


Figure 57 – Server statistics

When the Logger runs, a number of icons will also appear on the Taskbar for those applications that are not hidden. In Figure 58 the icons from left to right are:

- Main Logger
- Rig Control
- Chat (not displayed when running as a Logger panel)
- Rotator
- Monitor
- Server (not normally displayed)
- Cluster

Figure 58 – Icons on task bar

The control applications defined in this manner are then linked to individual contests using the pull-down lists from the Logger or from the Contest Details panel[57] (see Figure 59), either when creating a contest or by selecting “**File/Contest Details**”.

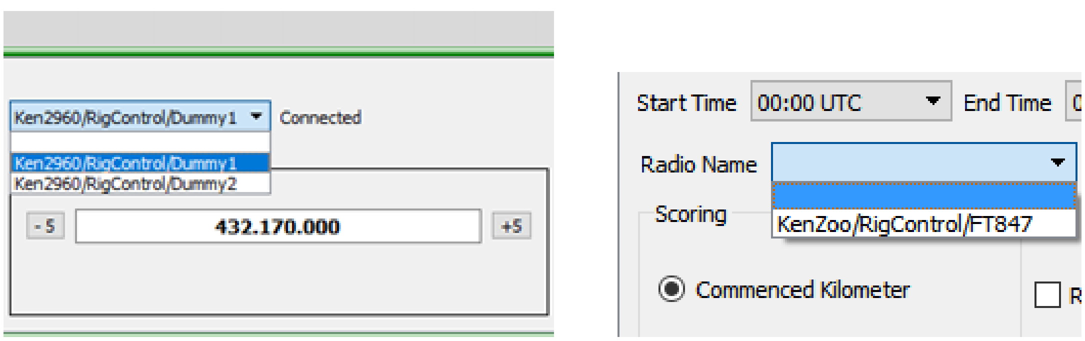


Figure 59 – Selecting a rig from Logger and Contest Details

[56] The ‘Servers’ count is more accurately ‘remote Servers’, i.e. excludes self.
[57] Both examples are for rigs, the equivalent mechanism is used for rotators.

4.2 Control Applications

These applications allow the control of rotator(s) and radio(s) using separate rotator and rig control applications communicating via (possibly USB connected) serial interfaces or, subject to finding someone who has one so it can be tested, full USB to their respective devices. The applications operate as servers for, and under the control of, the Logger.

While the programs can control a range of rotator controllers and radios using the Hamlib library, it is not possible to test all of these devices as they are not available to the developers. Some controllers may not support all the commands available or a particular controller's operation may not be fully understood.

A list of tested configurations for Rigs (and Rotators) is maintained at https://minos.groups.io/g/users/files
This location also contains a help file to give guidance with system configuration problems.
Feedback is welcomed on any information or changes that may be required to support as yet untested rotator and radio models.

The control applications can be connected to one or more contest log pages, selection of the rig in use (or rotator, if you have more than one) for a particular contest being via the drop-down lists as described above.

More complex arrangements such as different rigs for different bands are covered in section 4.4.

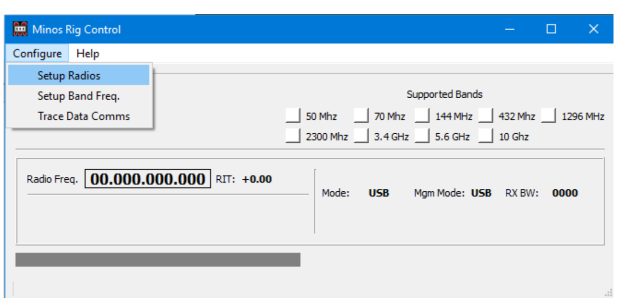
4.2.1 RigControl

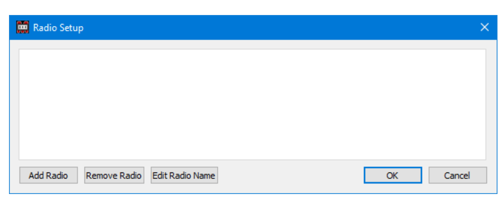
This application connects to multiple rigs[58] using CAT interfaces via serial ports (which may be simulated via a USB device), or via direct USB connection if your rig supports it. The program will also support network radios, for example FlexRadio 6xxx. When you select a radio that supports a network interface, a network address and port number must be entered. **At present this facility is completely untested!**

It provides direct control of the selected rig and extends the functionality of the memory Auxiliary panel.

Before the Logger can connect to the RigControl application one or more rigs must be defined (so there is content in the pull-down list). Transverter configurations can also be defined (see section 4.2.1.1) to adjust the frequencies seen in Minos to reflect the actual transmit frequencies, rather that the driver frequency.

Selecting "**Configure/Setup Radios**" within RigControl invokes the 'Radio Setup' dialogue (Figure 60) that allows the radio(s) to be defined.





58 More than any normal operator would want to use/can afford.

Figure 60 – Starting configuration of RigControl

Clicking "**Add Radio**" starts the definition of a rig (Figure 61).

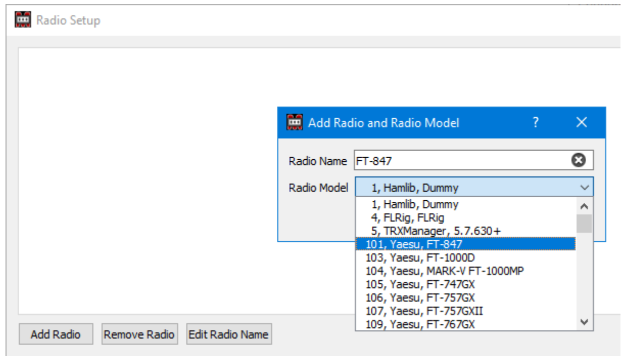


Figure 61 – Selecting the radio model

The "**Radio Name**" field is used with a contest to select (or display) which rig is currently being controlled (e.g. in Figure 67). Once defined, it may be changed using "**Edit Radio Name**" or removed (provided it is not currently active within the Logger) using the "**Remove Radio**" button.

"**Radio Model**" is a pull-down list containing an extensive range of rigs supported by Hamlib. Note that one of these is 'Dummy' which can be quite useful for testing/learning purposes.

Once the name and model are defined clicking "**OK**" continues to the setup of the interface to the device (Figure 62)

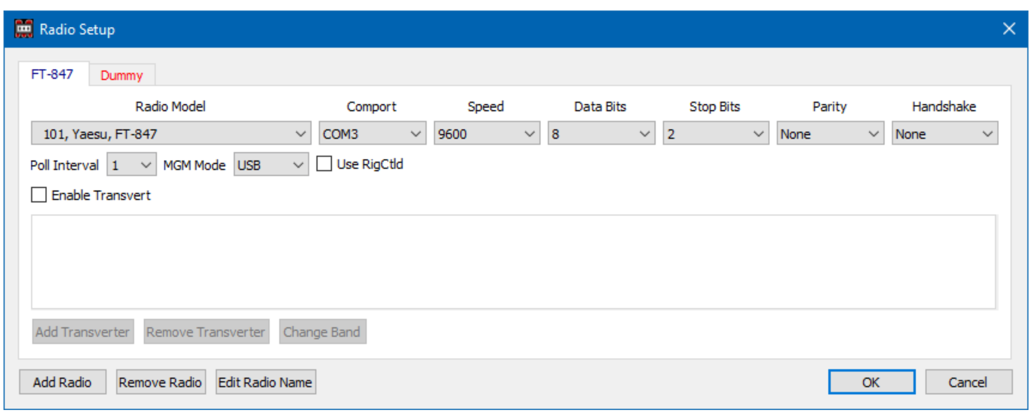


Figure 62 – Configuration of a serial connected rig

Note that the 'Dummy' tab is red, indicating that the 'rig' is in use by a contest and should not be changed. It may be possible, but the results are unpredictable.

"**Comport, Speed, Data Bits, Stop Bits, Parity**" drop-down lists configure the serial communications[59] with the CAT interface of the radio selected from the "Radio Model" drop-down list.

[59] Tip: check the settings on anything else already being used, e.g. WSJT-X.

“**Handshake**” Controls the flow control setting (e.g. XON/XOFF or CTS/RTS). Ensure this is correct for your interface (generally “None”). Some USB interfaces need CTS/RTS to be enabled to power the device.

If mis-configurations have occurred error messages such as “**Please select a Com Port**” may appear.

“**Poll Interval**” defines the interval in seconds at which RigControl interrogates the rig for information.

“**MGM Mode**” selects the rig mode to be used when MGM (data) communications are used.

“**Use RigCtld**” enables the device sharing mechanisms, see section 4.2.3.

“**Enable Transvert**” starts the transverter configuration dialogue, see section 4.2.1.1.

If your radio is an Icom then an additional “**CIV**” box will appear (Figure 63). Either leave the field blank to use the default CIV address of the radio, or enter the CIV number e.g. 0x66.

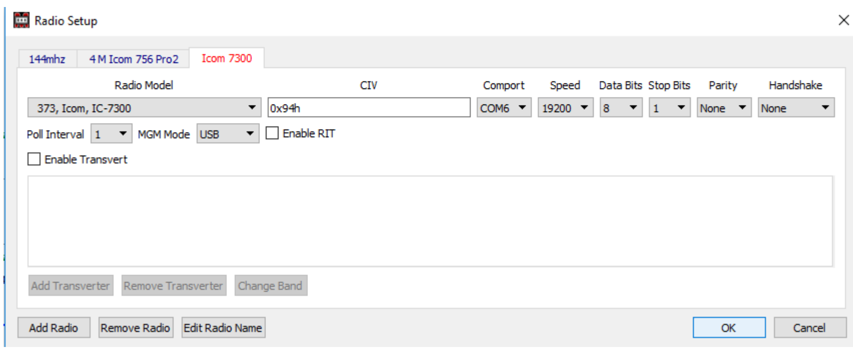


Figure 63 – Configuration of an ICOM rig

Note: With the IC7300 you need to ensure that "**CI-V Echo Back**" is "**ON**" in the settings[60].

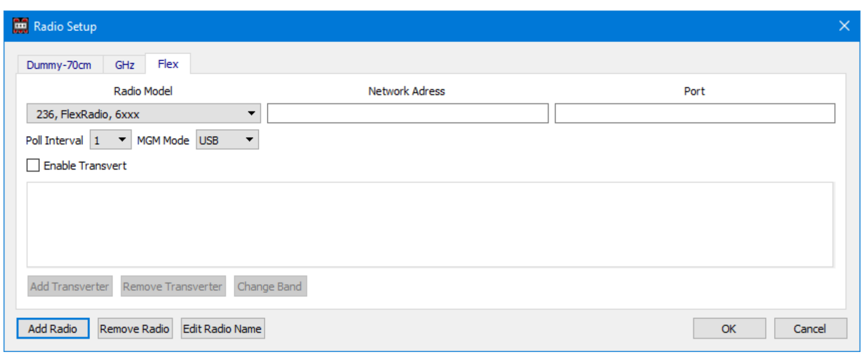


Figure 64 – Configuration of a network connected rig (FlexRadio)

A network connected radio (e.g. a FlexRadio) requires the target “**Network Address**” and “**Port**” to be entered (Figure 64).

[60] All Icom rigs need Transceive OFF and USB Port Link to remote.

If there are multiple rigs to be defined configure everything on one instance of RigControl and then restart the Logger (and thus applications). This will ensure that the definitions are visible to all instances.

"**Configure/Setup Band Freq**." (Figure 65) allows changes to the frequency the rig goes to when a contest is opened or when a "**Quick Sel**" band change is made from the Rigcontrol panel of the Logger (Figure 67). The default frequencies are chosen to avoid such things as calling channels. This option allows you to change the start frequency to your favourite start frequency on a band[61].

Beware: The 2300MHz band consists of three segments: 2300-2302 MHz, 2310-2350 MHz, and 2390-2450 MHz. Please take this into consideration when using this facility.

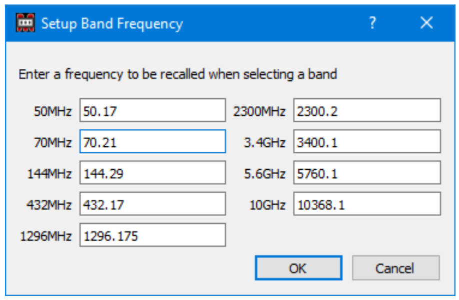


Figure 65 – Setup Band Frequency

"**Configure/TraceLog**" turns on the trace log (see section 5.3). It should be turned off (the initial default) in normal operation.

Once the configuration has been done, the radio to be connected to the Logger can be selected either via the Rigcontrol panel pulldown, which shows which radio is in use at any particular time, or the Contest Details boxes (see Figure 59).

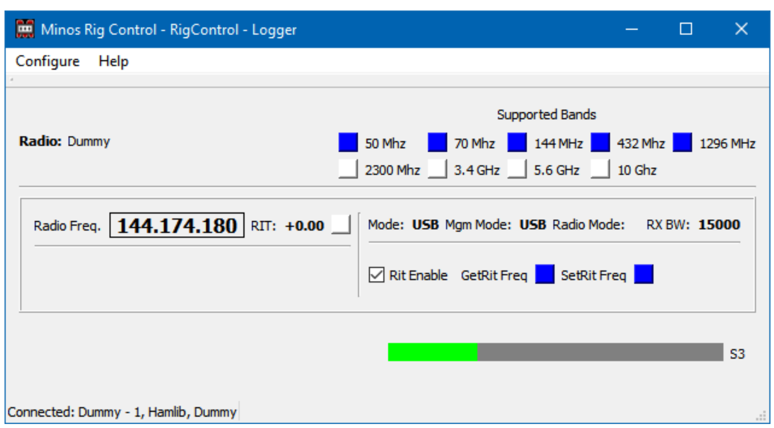


Figure 66 – RigControl operational

A status line at the bottom left of the panel (Figure 66) shows the Connected/Disconnect status[62], and which radio has been selected, with what CAT configuration.

61 It does not prevent you from specifying calling channels etc., so be careful.

62 Selecting a radio initiates connection.

Note the "**Supported Bands**" boxes towards the top right. Blue indicates that a band is supported directly whereas yellow indicates the use of a transverter.

At the bottom of the panel there is an "**S-meter**" bar, although in the example the indicated magnitude is lower than S1 and hence has a numeric value on the right.

Once configured to operate in conjunction with the Logger a contest may utilise the rig selected within the application, and the RigControl panel becomes visible in Logger (see Figure 67). As can be seen, because the application is connected to the Logger, it shows 'Logger' in the title, together with the name assigned to the instance of the app being used they are using (RigControl, by default).

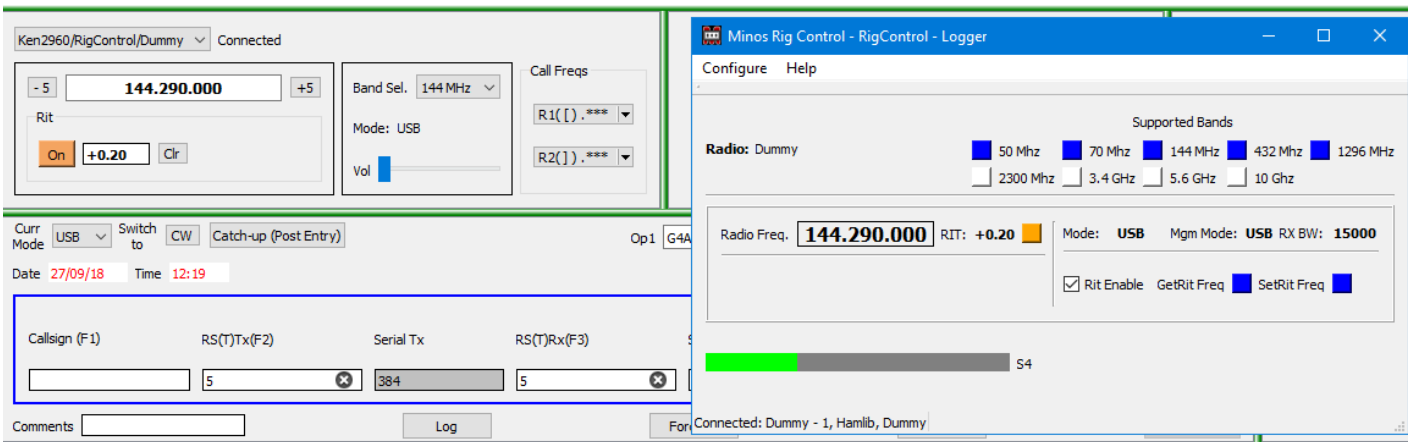


Figure 67 – RigControl connected to Logger

Within the RigControl panel (see Figure 69) on the top left is the name of the selected radio and its status which should normally be **'Connected'** for correct operation[63].

The current frequency[64] of the rig is displayed and may be changed by selecting characters inside the box and either typing in characters or using the mouse wheel to increase/decrease (band limits are enforced)[65]. If the mouse wheel is used then the changes are made immediately; if characters are type then "**Return**" causes the rig to be updated.

The border will turn red to show it is in edit mode.

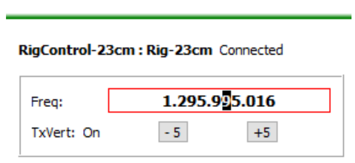


Figure 68 – Frequency change in progress

[63] If the target of a remote connection is not active when log started then it loses the connection and the target is not in the list. It reconnects automatically when the target becomes available.

[64] The calculated frequency is displayed if a transverter is in operation.

[65] Minos doesn't control the VFO selection, it always sends the commands to the 'current-VFO'.

Ctrl-f selects the KHz digit, or use the mouse to select digit, as the tuning step. With the cursor in the "Freq" box you can now use the mouse wheel or the up/down arrow to tune. Left/Right arrow change the selected digit, hence the step.

Hit "**Enter**" to select the new frequency or escape to cancel. The frequency may be 'nudged' using the -5/+5 buttons to allow calling near the frequency of a station being received[66].

If the selected radio supports RIT (through CAT) it can be enabled through the RigControl panel (see Figure 67) and it will then become visible in the RigControl panel within Logger (see Figure 12). Turning it on (Figure 67) then allows incremental tuning using the mouse wheel in the same manner as for the main frequency. A specific RIT frequency can also be entered, although that is unlikely to be very useful. There are a number of shortcut keystrokes available - listed in Figure 121.

Similarly, if the rig supplies volume control information this will be added to the display under the Band Select list/Mode display.

In multi-band contest with multiple tabs the Logger initially displays the current rig frequency on all tabs using a particular RigControl, switching to the frequencies from the "Band Frequencies" list as contest tabs are selected (or "**Band Select**" changed).

During operation any changes to that frequency are only propagated to the active tab; those not in focus do not get updated.

When an inactive tab is selected and put into focus the frequency (and mode) used the last time the tab was active is retored to the rig (whatever band that happens to be, i.e. if the 2M tab has a 6M frequency from the startup then selecting the 2M tab will cause the rig to go to a 6M frequency).

"**Band Select**" provides a quick method to get the rig to the right band for a particular tab at startup, as an alternative to physically changing the rig controls[67]. The frequency comes from the "**Band Frequencies**" list.

The current mode (USB/CW/FM/MGM) on the radio is displayed[68], and is changed as selected within the QSO data entry frame (since it may be changed in the Logger when RigControl is not in use).

The rig frequency is adjusted to compensate for changes between USB and CW.

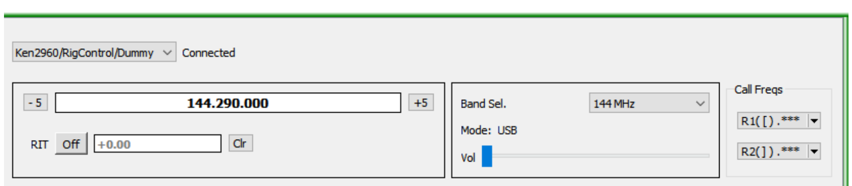


Figure 69 – RigControl panel within Logger

[66] Useful to call a station near a frequency that is in use.
[67] Restarting the Logger clears the tracking, all tabs revert to the current rig frequency at startup.
[68] At startup the radio is always switched to USB

When RigControl is in operation the memory functionality described in section 3.2.3 is fully realised, as well as the R1/R2 button functionality described here. These buttons represent the 'run frequencies' and mode selected by the operator and are configured using the same R/W/E/C dialogues as the standard memories (see Figure 70).

Extra data is also written to the log for reference purposes: "**Rig**" gets filled in with name of the selected rig that is connected to Rigcontrol and "**Freq**" is loaded with the frequency to which the rig is set[69].

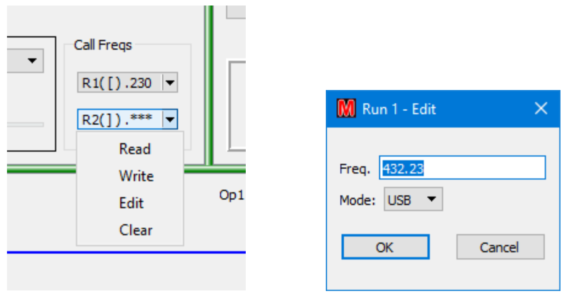


Figure 70 – Storing a run frequency

The button displays the frequency if set and "***" if nothing is stored. In Figure 69 R1 is set to (432.)290[70] and R2 is empty. The square brackets indicate the CTRL shortcut symbols that can be used. Run memories are held within, and retained with, each log.

Under some circumstances (most likely comms failures) errors may be returned from the Hamlib routines controlling the rig (Figure 71). If this occurs use "**X**" to exit RigControl, which will then restart and should re-establish communication. If it does not, then investigate the comms settings and cables.

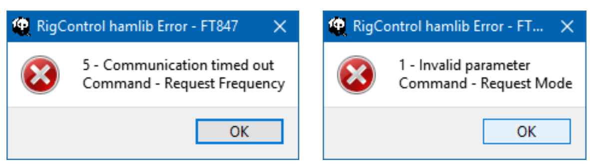


Figure 71 – Hamlib error messages

<<Looking at ways of coping with this on remote servers>>.

4.2.1.1 Transverter Configuration

If "**Enable Transvert**" is ticked the rig being defined/modified will be used with a transverter and the buttons are enabled to allow this definition (see Figure 72), including making the "**Enable Transvert Switch**" option visible.

"**Add Transverter**", "**Remove Transverter**", and "**Change Band**" are used to control the definition of possibly multiple transverters/bands.

[69] If Rotator is operational and connected to a rotator then the current bearing is written to "Rot heading"
[70] It is presumed to be **432**.290; if you have changed bands it will still be on the original band.

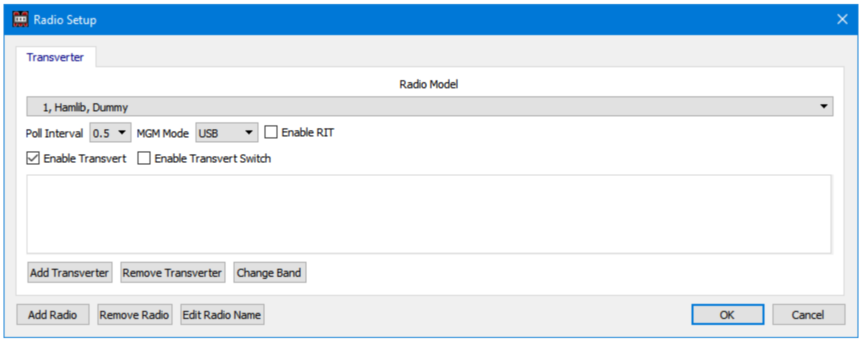


Figure 72 – Transverter configuration

When you add a transverter, a list of bands are displayed (Figure 73). The radio may support the band directly, but a transverter can still be selected for that band if for instance its RX performance is better than the radio.

Beware: The 2300MHz band consists of three segments: 2300-2302 MHz, 2310-2350 MHz, and 2390-2450 MHz. Please take this into consideration when using this facility.

If the radio supports a band for which a transverter is configured, the transverter frequency combination will be used in preference to the radio band. Configure a different radio definition (or don't configure a transverter for that band) if this is not what is wanted.

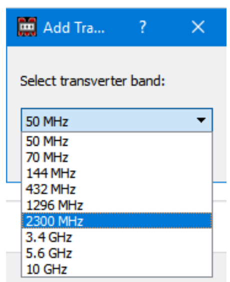


Figure 73 – Transverter band selection

Having created a transverter tab for one or more bands (Figure 74) the frequency definitions govern how it operates.

"**Radio Freq**" is the frequency the rig actually tunes to get the desired output

"**Target Freq**" is actual operating frequency

"**Offset Freq**" is the difference between them, which the transverter produces[71]

[71] Currently the focus must be moved from the entry boxes to get the calculation displayed.

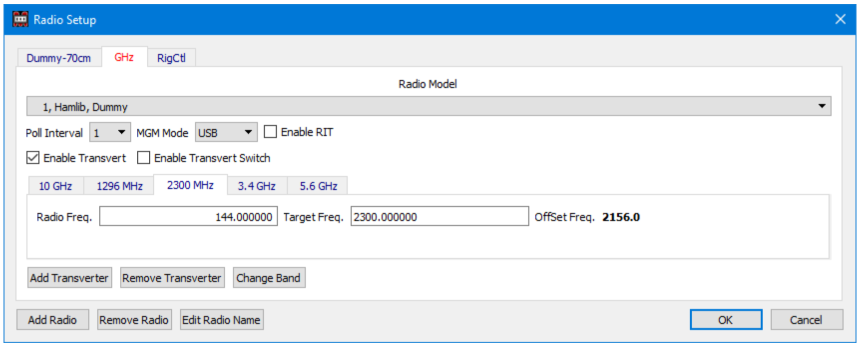


Figure 74 – Transverter definition

These frequencies are then reflected in the Rigcontrol display and passed to the rig. So, with the above definitions for 2.3 GHz if 2,300.1 MHz is entered in the Rigcontrol panel of the Logger the Rigcontrol display will appear as in Figure 75 and a frequency of 144.100 will be sent to the rig.

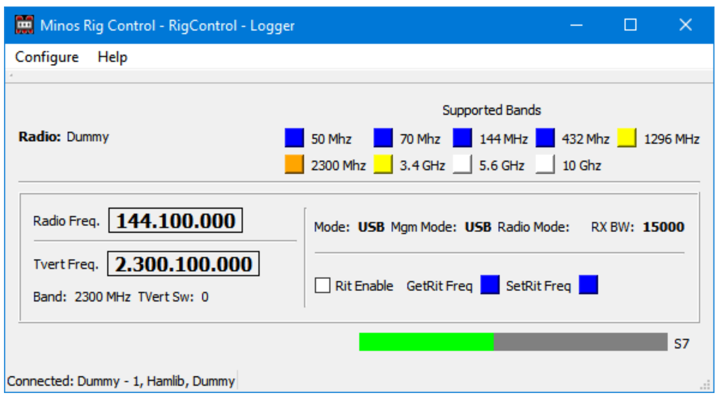


Figure 75 – Transverter in Rigcontrol display

There is the facility to drive external switching of the tranverter either via an external com port or via a network message to an external control system[72]

"**Enable Transvert Switch**" enables this facility for all defined transverter bands if ticked. It also makes the "**Enable Local Switch Control**" box and "**Transvert Switch Number**" dialogue visible.

Regardless of the path used, the message to be sent is entered in Transvert Switch Number box. It is free format, which might typically be just a simple number, and is 'wrapped' in a ':' as to start and ' LF' to terminate.

"**Enable Local Switch Control**" displays the pull down list to select the com port to be used (Figure 76). The message transmission is then via the selected com port. The 'standard' default set up is assumed (9600baud, 8bit, 1 stop bit, no parity, no flow control

The intention is that the message will be read by an Arduino or a Pi based 'switch box' (see Appendix section 7.5) .

[72] Still under development, but see section 7.5.

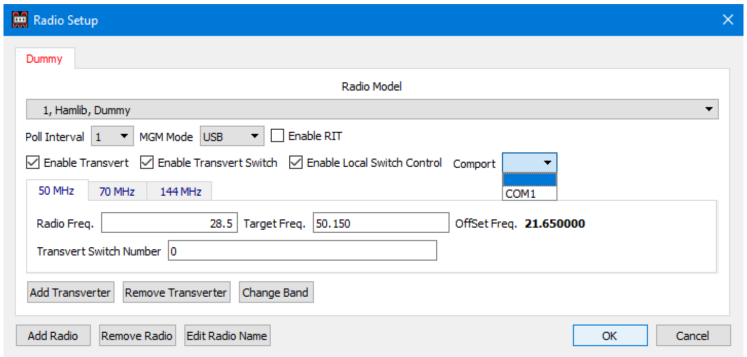


Figure 76 – Defining Local Switch Control port

The **"TxVert"** indicator within the RigControl frame of the Logger (Figure 77) shows blue when the rig is directly accessing the band, and orange when a transverter is in use (a band such as 70MHz might possible use either).

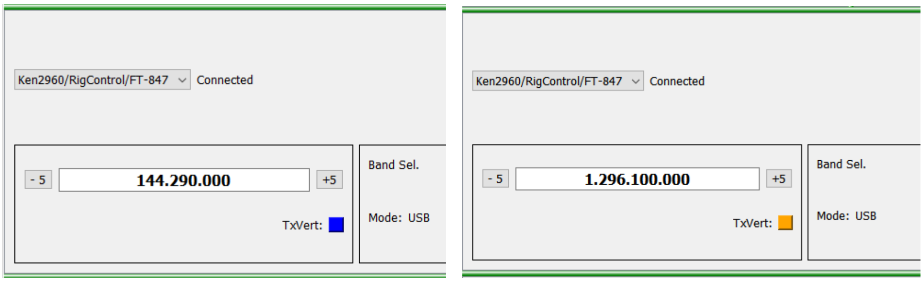


Figure 77 – Logger RigControl transverter indicator

4.2.2 Rotator

The application has many similarities to RigControl. It can connect to multiple rotators[73] (from those supported by the Hamlib library) via serial ports (which may be simulated via a USB device), or via direct USB connection if your rotator controller supports it. The application is intended to be used under control of the Logger.

Once the application has been connected to a particular contest (see above) the rotator(s) in use need to be configured so that the selected controller operates under control of the Logger. Figure 78 shows the startup with Rotator defined to a contest but with no controllers yet defined.

Selecting "**Configure/Setup Antennas**" allows the rotator(s) to be defined.

Figure 80 shows an example configuration of a rotator.

[73] Once again, more than any normal operator would want to use/can afford.

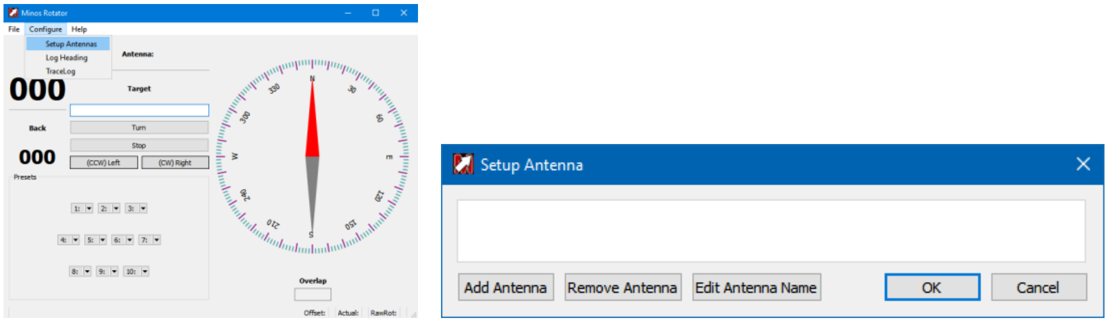


Figure 78 – Starting configuration of Rotator

Clicking "**Add Antenna**" starts the definition of a rotator (Figure 79).

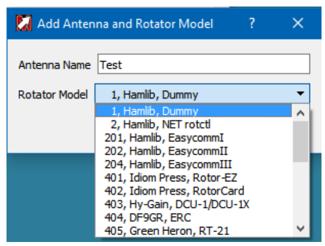


Figure 79 – Selecting the rotator model

The "**Antenna Name**" field is used to select (or display) which rotator is currently being controlled (e.g. Figure 86). Once defined, it may be changed using "**Edit Antenna Name**" or removed (provided it is not currently active within the Logger) using the "**Remove Antenna**" button.

Depending on which rotator is selected different options will appear (Figure 80).

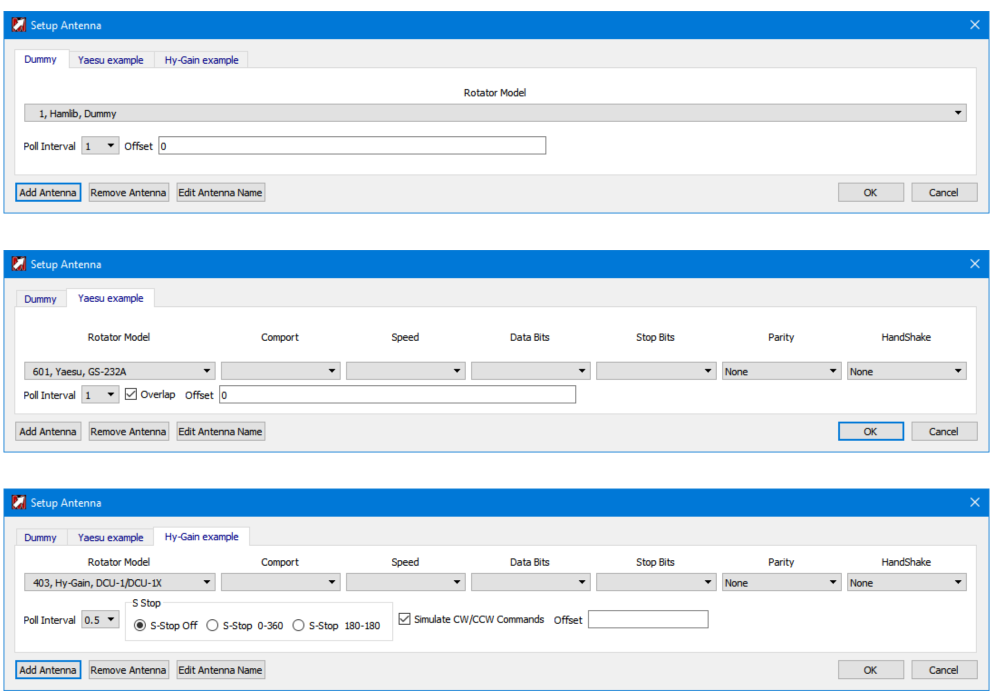


Figure 80 – Defining various rotators

If there are multiple rotators to be defined configure everything on one instance of Rotator and then restart the Logger (and thus applications). This will ensure that the definitions are visible to all instances.

The "**Antenna Name**" is the method by which the particular rotator is selected from the pull-down list.

"**Rotator Model**" is a pull-down list containing an extensive range of rotator models supported by Hamlib. Note that one of these is 'Dummy' which can be quite useful for testing/learning purposes.

"**Comport, Speed, Data Bits, Stop Bits, Parity**" drop-down lists configure the serial communications with the rotator controller selected from the "Rotator Model" drop-down list.

"**Handshake**" Controls the flow control setting (e.g. XON/XOFF of CTS/RTS). Ensure this is correct for your interface (generally 'None').

"**Poll Interval**" controls the interval between polls of the rotator; the default of '1' should be correct for most installations.

"**Overlap**" will only appear, and will be ticked by default, if the Rotator being controlled supports overlap capabilities. The capability for overlap and the amount of overlap allowed is derived from information in the Hamlib library.

Some rotator models do not support overlap and the rotator module in Hamlib may not support overlap either.

The box should only be unchecked[74] to turn off overlap when using an overlap-capable protocol with a rotator that does not support overlap. The rotator will then only turn between 0 - 360 degrees. If Overlap is unchecked then the Overlap box will not be displayed on the Rotator Display.
Beta testers are required to check out this functionality.

The "**Offset**" to support an antenna mounted offset from the rotator bearing a value between +90/-90 may be entered in the "**Offset**" box[75].

Rotator bearing is always shown in Actual on status line. With the offset (from Rotator definition) the pointer indicates the 'real' beam heading, not the rotator current setting.

With some rotators such as the Hy-Gain in Figure 80 (or if "Overlap" is unchecked on Yaesu rotators[76]) a number of settings appear to allow a rotator with a north-stop to be mounted with the mechanical stop to the south while the sensor is sending 0 – 360, instead of -180 to +180:

"**S-Stop Off**" the rotator will act as normal

---

[74] If Overlap is unchecked and an antenna is sitting in an overlap region, you may not be able to rotate the antenna. The antenna will need to be manually moved < 360 or > 0
[75] There is also an antenna offset field in contest details to allow for any skew between antennas in the same stack, or inaccuracy in the direction setting of the antenna.
[76] The Yaesu protocol is probably the most commonly emulated.

"**S-Stop 0-360**" Minos inverts the reading of 0-360 that the controller sends

"**S-Stop 180-180**" is used for rotators such as some old Stolle models, that are mounted with a south stop, but use a compass sensor that sends 180 - 180[77].

"**Simulate CW/CCW Commands**" is for rotators that don't support turn-right/left commands in their protocol. When the turn right/left buttons are pressed, a 'rotate to end stop' bearing command is used. If that causes a problem unchecking the tick box will cause the turn right/left buttons to be removed.

"**Configure/Log Heading**" allows a location and filename to specified for logging bearings when "Logging Enabled" is ticked (see Figure 81). Bearings will continue to be appended to the specified file[78] every time Minos Rotator is started until "Logging Enabled" is cleared. The "Bearing Difference" is the incremental change required before a new bearing is logged.

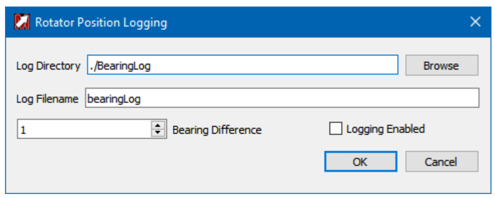


Figure 81 – Configuring Rotator Position Logging

Mon Oct 30 11:33:57 2017 Bearing is 19 Degrees
Mon Oct 30 11:33:57 2017 Bearing is 22 Degrees
Mon Oct 30 11:33:58 2017 Bearing is 25 Degrees
Mon Oct 30 11:33:59 2017 Bearing is 26 Degrees

Figure 82 – Example Bearing Log

"**Configure/Trace Data Comms**" enables the tracing of communications between Minos and the Radio[79]. It is recommended that this is only turned on for diagnostic purposes. If checked the state is saved and it will be enabled when RigControl is restarted, so remember to turn it off when done. See section 5.3 for mote information on trace logs.

Once the configuration has been done, using "**Antenna Select**" to specify the rotator in use connects the application to the Logger. The rotator in use can be changed by selecting a different entry from the pull down list; the top left of the Rotator panel in the Logger will indicate which is being used (see Figure 86). The current status of the rotator (e.g. "Turning CW") is also shown adjacent the connection information.

[77] Actually the stop is set to 181 to 179.
[78] If you want a new file, either specify a different name (saved through restart) or move/rename the existing file.
[79] Standard trace messages are now recorded at all times.

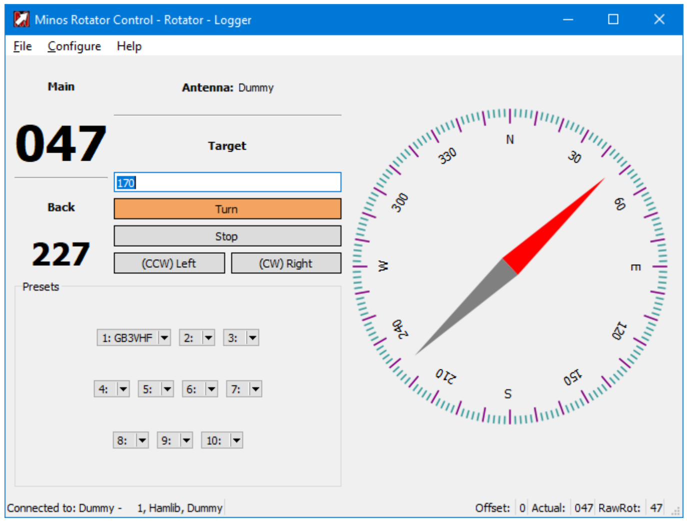


Figure 83 – Rotator control operational

A status line at the bottom left of the panel shows the Connected/Disconnect status, and which rotator has been selected.

"**Configure/Edit Presets**" allows the definition of the ten memories visible in the bottom left of the Rotator panel (see Figure 83). Each consists of a name (typically a callsign) and the relevant target bearing. These memories are effective for **all** rotators on the host machine, and are retained across restarts[80].

The presets are replicated in the Rotator Control panel of any Logger contest accessing the rotator[81].

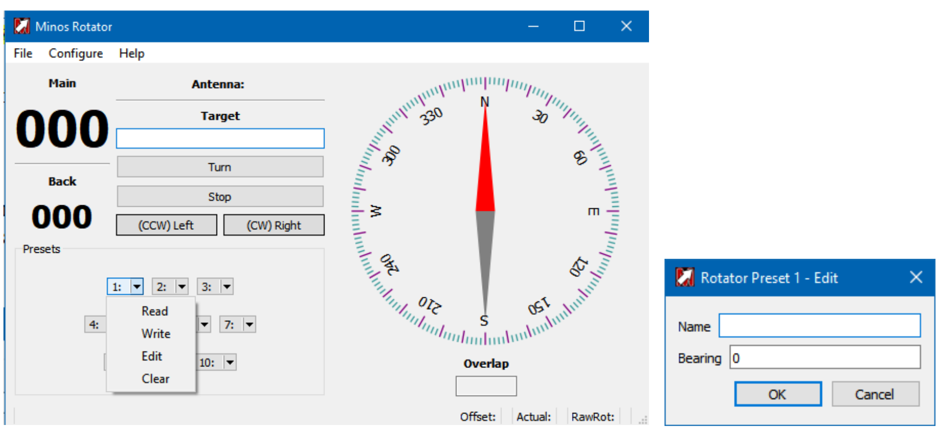


Figure 84 – Configuring Rotator control pre-sets

[80] Entries may be removed by going into "Configure/Edit Presets" and deleting the individual content.
[81] On any machine accessing the host in complex configurations.

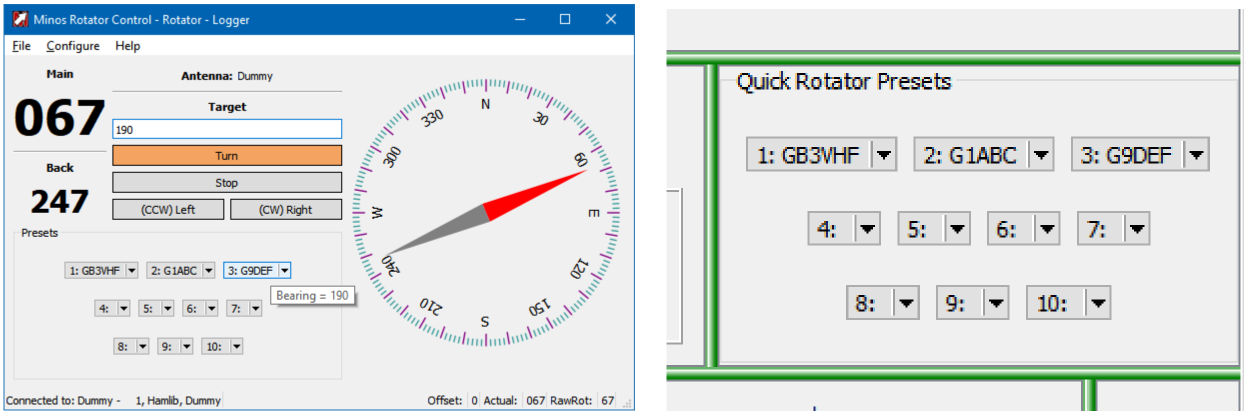


Figure 85 – Configuring pre-sets

Clicking one of these entries initiates rotations to the requisite bearing.

Once configured to operate in conjunction with the Logger a contest may utilise the rotator selected within the application, and the Rotator panel becomes visible in Logger (see Figure 86). As can be seen, because the application is connected to the Logger, it shows 'Logger' in the title, together with the name assigned to the device currently selected to be used.

With Rotator in operation there are enhancements to the memory functionality described in section 3.2.3. The memory content is extended to include bearing information. When a memory is read the bearing is inserted into the Target field for the rotator and rotation can be initiated by "**Turn**"/"**Return**"/ "**Ctrl-T**".

Extra data is also written to the log for reference purposes: "**Rot heading**" gets filled in with the current bearing[82].

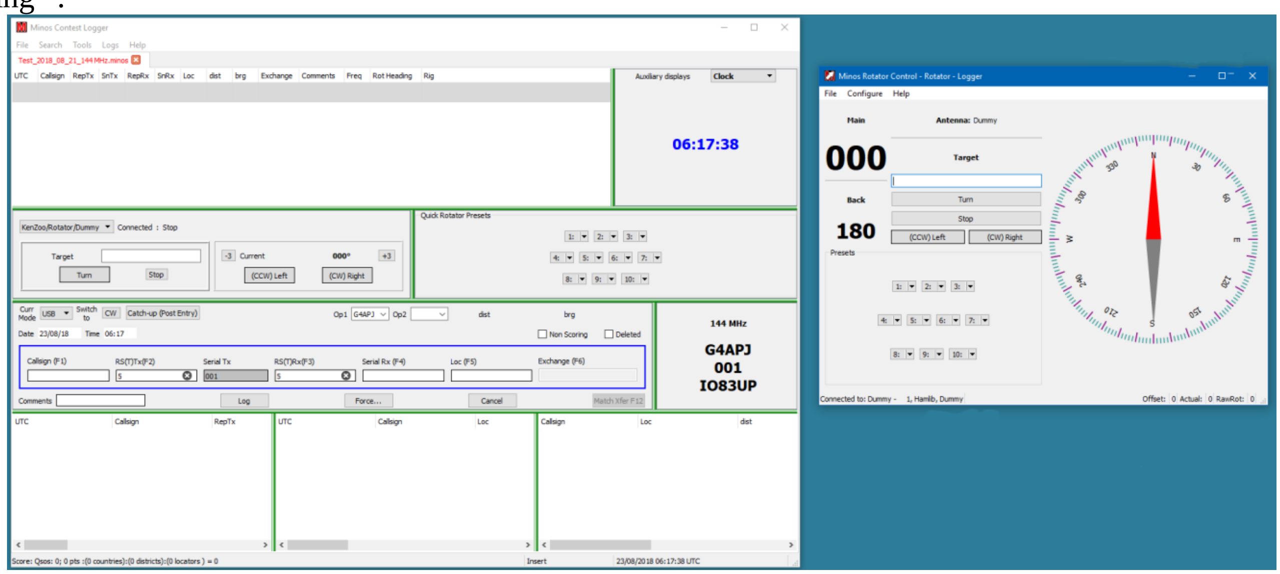


Figure 86 – Rotator connected to Logger

[82] If Rigcontrol is operational and connected to a rig then the rig name and current frequency get filled in as well.

The antenna can be controlled from the Rotator application or from the buttons in the Logger. If actions are initiated from the Rotator application the panel in Logger shows what is happening.

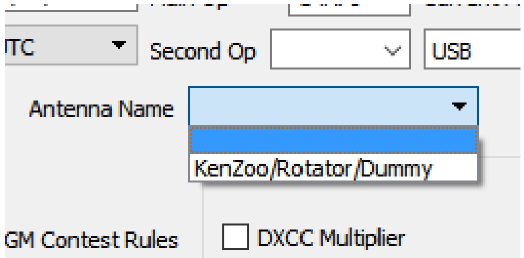


Figure 87 – Selecting the Rotator Application Instance

Within the Rotator panel in Logger (see Figure 88) the text at the top left shows the rotator in use (‘Dummy2’ in this example) and the status (e.g. ‘Turning to bearing’).

The left hand box shows the target bearing (‘170’) and the “Turn” button changes colour to show that an operation is in progress (it clears when complete). The rotation to the requested bearing will occur via the shortest valid movement using overlap.

The right hand box shows the “**Current Heading**”[83] for the rotator in use as well as the control buttons “**(CCW) Left**” and **“(CW) Right**” [84] and the ‘Nudge buttons’ (-3/+3) which nudge the rotator 3 degrees every time the button is pressed[85] (in the direction of the adjacent box, i.e. + is CW).

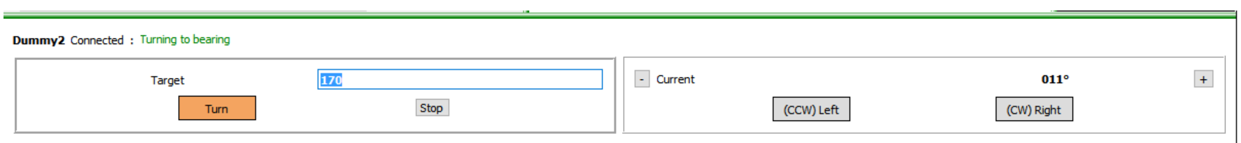


Figure 88 – Rotator panel within Logger

The “**Target**” (heading) may be entered manually, or is calculated from any QRA locator entered in the data entry panel in the Logger. This locator itself may be derived from an entered callsign found in the archive and transferred using F12 (or equivalent).

If a target bearing has been manually entered then rotation may be started by hitting RETURN, else click the Turn button or Ctrl-t to start rotation. Rotation may be stopped by clicking the “Stop” button on the left or by using the shortcut (Ctrl-s).

While rotating the bearing is checked for end stop, if reached a stop will be automatically issued.

[83] Red indicates overlap > 360 degrees and blue for overlap < 0 degrees.
[84] Left and Right are from the perspective of physical buttons on the rotator.
[85] If a button is selected when rotator is moving, a Stop command is issued before the selected command is issued.

Logging a contact while the rotator is turning logs the 'incomplete' bearing (i.e. the bearing at the moment of logging).

The Rotator application (see Figure 83) displays the bearings (Main and Back), together with a 'compass bearing' schematic of the current bearing heading (actual, calculated using any offset defined). A status line at the bottom left of the panel shows the Connected/Disconnect status and the name assigned to the rotator that has been selected.

There are control buttons to "**Turn**", "**Stop**", **"(CCW) Left**" and **"(CW) Right**".

Some rotator controllers do not support a Rotate Left (CW - ClockWise) or Rotate Right (CCW - Counter ClockWise) Command. To allow the CW and CCW buttons to function, the command rotate to minAzimuth (typically 0 degrees) is sent for CCW command and the command rotate to maxAzimuth (typically 360 degrees) is sent for CW. The CW and CCW buttons are latching, selecting the buttons again will stop rotation. Rotation will also stop when the rotator reaches the endstop if a stop command is not sent first.

If a control button is selected while the rotator is moving, a stop command is issued before the selected command is issued.

If you change any value in the Current Selected antenna, you will be prompted to reload the values. You can choose "No" to skip and the values will be loaded when the program is restarted.

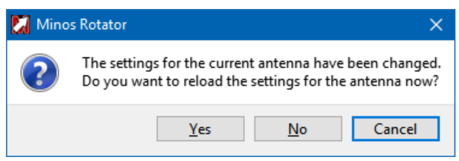


Figure 89 – Rotator settings changed

4.2.3 Device Sharing

A common requirement is to be able to share control of the rig(s), and possibly rotator(s), that Minos is using with other programs such as WSJT-X.

Hamlib provides a mechanism for doing this; it is called rigctld (or rotctld for rotators[86]). These network servers or 'daemons' control the actual radio and use a TCP/IP based call/response protocol to handle requests from multiple client programs.

In principle Minos could treat a rig as local but control it using remote access to rigctld on the target server, but this approach has been rejected. Despite the fact that rigctld and its companion utility programme rigctl implement no security measures it can be a bit tortuous to successfully configure such remote access in a Windows environment[87].

[86] Currently not, maybe never, supported in Minos.
[87] This is still being investigated.

It is much simpler to handle the rigctld access locally and 'advertise' the result(s) through the Minos server communications.

To enable this operation, click the **"Use RigCtld"** box in the radio definition (Figure 90, left). When the rig is selected from a contest the rigctld daemon will be started with the requisite parameters, and connection established through it to control the rig. Once the connection is operational the **"RigCtld Active"** indicator turns orange in the RigControl Window (Figure 90, right).

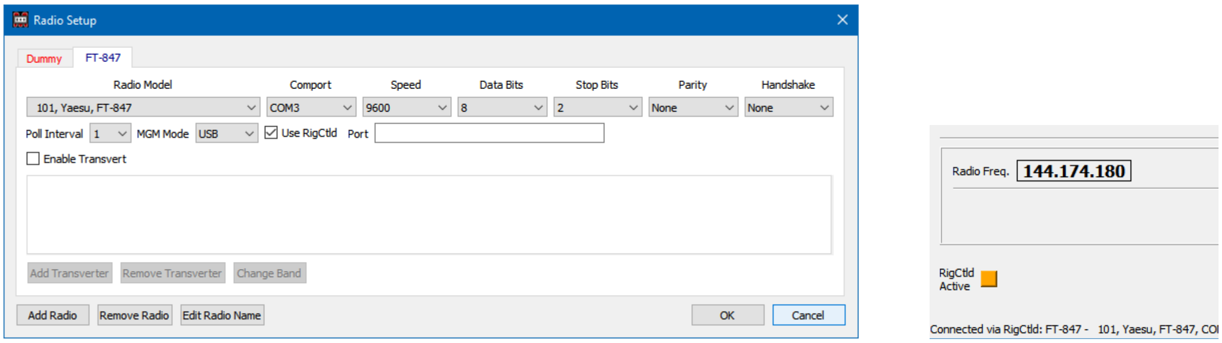


Figure 90 – Defining rigctld connection to Rigcontrol

The rigctld connection is established on the local loopback address (127.0.0.1) using port 4532 by default[88] (if **"Port"** is left blank in the radio setup). Other applications accessing the rig through rigctld must use these parameters (see WSJT-X example below).

Any such applications must be started **after** Minos has create the rigctld daemon, and terminated before Minos[89] is disconnected from the rig (hence terminating rigctld). This restriction can be avoided by starting rigctld externally (typical from a BAT file, section 4.2.3.1) prior to starting Minos[90], which also allows the user to select the location and specific version of rigctld will be used.

By default RigControl initiates the daemon using **rigctld.exe** from the /Bin folder – make sure a copy of **rigctld.exe** has been copied there (i.e. rename **rigctld-wsjtx** if that is being used**)**.

A different location may be selected by editing **/Configuration/MinosRigControlConfig.ini** and inserting text of the form[91]:

[RigCtld]
RigCtldPath=D:/Radio/Minos/hamlib-w32-3.3/bin/

---

[88] If multiple devices are in use it is recommended that even port numbers be used in sequence from this default (one instance of rigctld, and hence port, is required per concurrent connection to a rig), but it doesn't really matter as long as there are no clashes (and the right ones are connected).

[89] This is also the preferred sequence to synchronise the various views of what frequency the rig should be using!

[90] The **"Use RigCtld"** box must still be ticked within the rig definition. Note that the "RigCtld Active" indicator will not light under these circumstances.

[91] Note the use of forward slash "/" as the seperator.

Once again, **rigctld-wsjtx** would have to be renamed.

For WSJT-X to share a rig the same information is supplied via the "**File/Settings**" dialogue, although the layout is somewhat different (Figure 91); other applications will have their own methods of defining the parameters.

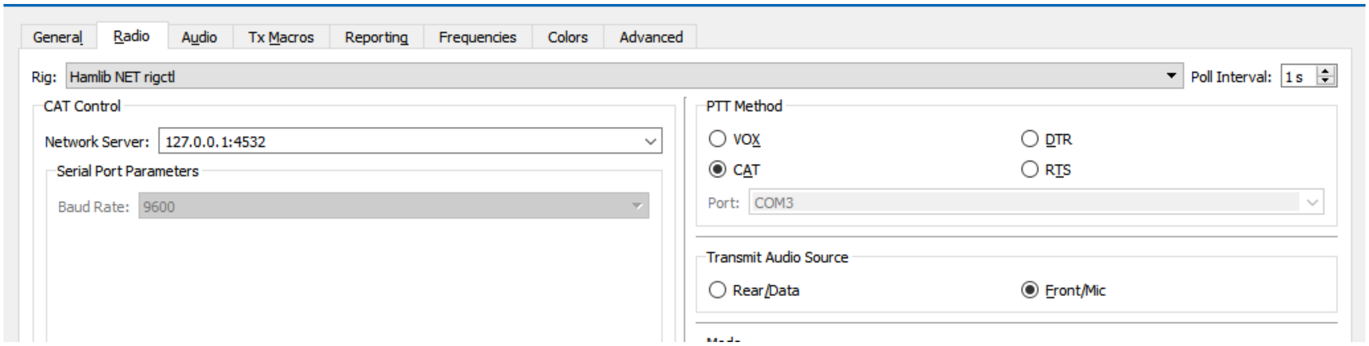


Figure 91 – Defining the rigctld connection to WSJT-X

4.2.3.1 Rigctld commands

There are several different versions of the software. Minos is built on the premise of using Hamlib V3.3 (not earlier versions), however WSJT-X includes a special, 'heavily modified' version of rigctld thought to have been created because Hamlib versions prior to 3.3 were in some way deemed unsatisfactory.

No problems have been identified with using either version with either programme. The only identified difference is that WSJT-X maintains the CAT connection all the time whereas the Hamlib 3.3 version drops the connection once all connections have terminated (thus allowing a direct connection from another programme, but raising the risk of contention).

If you have WSJT-X on your system it is probably better to use that version (and hence avoid installing Hamlib).

In you wish use the Hamlib version then to install it get the installation programme from
https://sourceforge.net/projects/hamlib/files/hamlib/3.3/hamlib-w32-3.3.exe/download

Run hamlib-w32-3.3.exe to install into <target folder>; only the DLL and program options are required to use the software although the documentation *may* be of interest to some users, particularly if multiple rigs are to be shared.

Before the daemon can be used it must be started. This is most easily done from a .BAT file, perhaps at system startup, containing a single command line of the form:

<WSJTX location>\bin\rigctld-**wsjtx** -m 101 -r com3 -s 9600 -T 127.0.0.1 -t 4532

This example connects to the FT-847[92] using COM3 at 9600 baud and makes the service available on the loopback address, port 4532, which is the easiest way to use it with Minos.

The equivalent command for the Hamlib version merely changes the folder from which the program is invoked:

<target folder>\hamlib-w32-3.3\bin\rigctld -m 101 -r com3 -s 9600 -T 127.0.0.1 -t 4532

For an ICom rig the CIV number (typically 66 hex) has to be expressed in decimal, e.g.:

< folder location>\rigctld -m346 -rCOM6 -s19200 **--set-conf=civaddr=102** --set-conf=data_bits=8 --set-conf=stop_bits=1 --set-conf=serial_parity=None --set-conf=rts_state=ON --set-conf=dtr_state=ON

In this example (for an ICom 746PRO) some handshake lines are set to enable the USB interface with power.

Rotator sharing is only done via manually configuring rotctld (Figure 92) and has not been tested. Volunteers wanted!

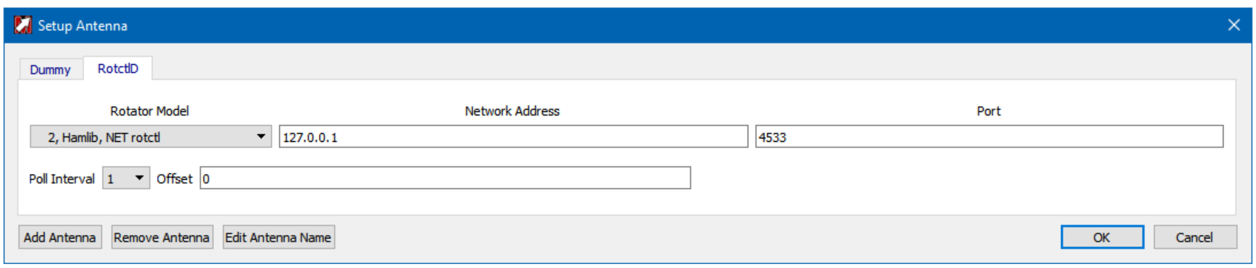


Figure 92 – Defining rotctlD connection to Rotator

The equivalent command to start the rotator daemon[93] for the above Antenna definition to use would be:

<target folder>\hamlib-w32-3.3\bin**rot**ctld.exe -m 1 -T 127.0.0.1 -t 4533

Note the different port number! If multiple devices are in use it is recommended that even port numbers be used for rigs (one instance of rigctld is required per physical connection to a rig) and odd numbers for rotators, but it doesn't really matter as long as there are no clashes (and the right ones are connected).

[92] The numbers for the device type (-m) are the Hamlib values, visible in the pulldown list of the "Radio Model" in radio setup (see Figure 61).

[93] WSJT-X does NOT include a version of rotctld.

4.3 Information Applications

4.3.1 Chat

Chat is similar to an IRC application allowing text messages between stations. It can now be run in either (or both) of two configurations: as a separate window or, with suitable Screen Layout definition, as a panel in the Logger.

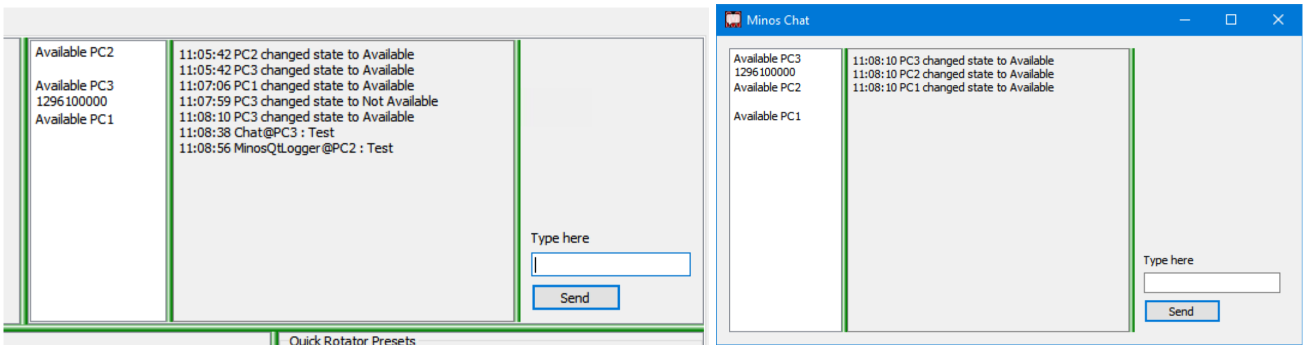


Figure 93 – Examples of Chat usage (Logger Panel and Standalone app)

The application is self configuring as machines come Available (chat application is visible) or go NotAvailable (chat is not visible, machine may or may not be active). To send a message (to all available stations) enter it in "**Type here**" and click "**Send**" (or hit "**Return**").

4.3.2 Monitor

<<Probably at some point also changing to be another contest panel >>

Monitor allows display of the logs on both remote machines and the local machine. The application is self configuring as Loggers come online/disappear and contests are opened/closed. The contests on a particular machine may be expanded/hidden using the chevrons on the left.

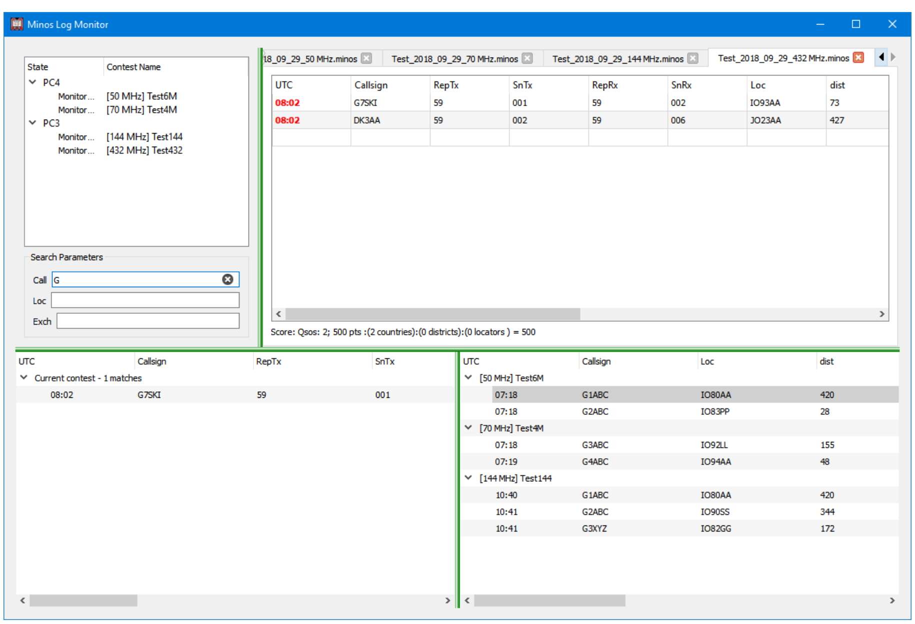


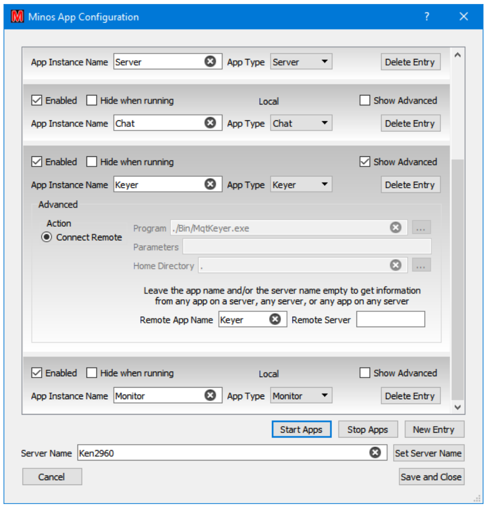
Figure 94 – Example of Monitor usage

To start monitoring a contest double click the contest entry in the top left hand panel. A new tab will be open, displaying data from the monitored log. Click "X" on the tab to terminate monitoring of that contest log.

There is a search facility beneath the log selection panel, the results of which are displayed into the lower panels (current contest to the left and other contests to the right). The search, and hence its results, can be cleared by deleting the search content or by pressing ESCape.

4.3.3 Keyer

<<TBA>>
Can only be defined to operate remotely.



4.3.4 Cluster

This facility monitors a selected DX cluster node[94] and passes spots into any open contest tabs that have a Cluster Client frame configured. The spots displayed in each contest can be controlled by various filter mechanisms on a per-frame (contest) basis.

Before Cluster data can be displayed the Cluster Server must be operational.

4.3.4.1 Cluster Server

The Cluster Server Application is initiated in the same manner as any other Application, see section 4.1. It can be defined as operating remotely so that one or more Cluster Servers can provide data to multiple client machines, each with multiple contests.

Once the Cluster Server is running, invoke the Setup screen from the Configuration pulldown list (see Figure 95). This allows the cluster server to be configured to get the spots that are then filtered/displayed in the Cluster Window(s).

[94] Using any filters defined in **…\Configuration\Cluster\cluster_start.txt** if it is used during Minos startup.

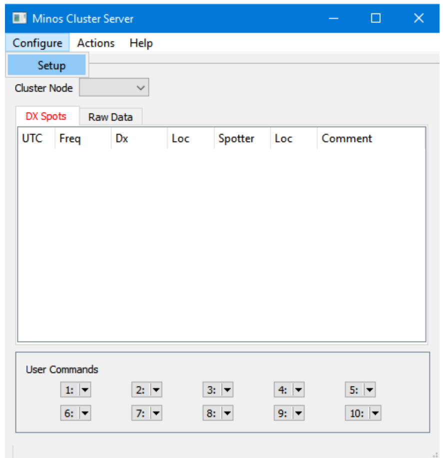


Figure 95 – Initial Cluster Server panel display and Setup

The General tab (Figure 96) allows adjustment of the Time to Live for spots, as well as controlling the Logon/Logoff command file activation (see section 4.3.4.2).

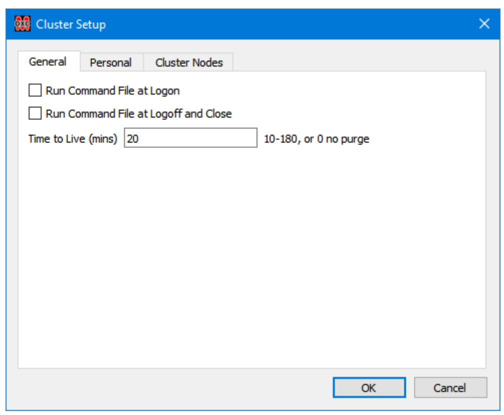


Figure 96 - Adjusting the timeout for spots

The time to live must be between 10 and 180 minutes; the default is 20 minutes. Spots will be removed from the Cluster Client panel(s), including filtered call/locator entries, after the period specified.

Spots are not saved in the clients or server. It may be useful to have a predefined button (show/dx) to recall the last x spots if a restart of the program has occurred[95].

The "**Personal**" data[96] (Figure 97, left) must be supplied before a successful connection can be made to a DXcluster server. The DXcluster server to be used is selected from the "**Cluster Node**" pulldown list in the main screen. The contents of this list is derived from the "**Cluster Nodes**" tab (Figure 97, right) which

[95] This can also be done through the DXcluster startup file; at least one DXcluster node (GB7RAU) is configured to do this automatically.
[96] The 'Locator' is not derived from the contest details and must be changed for different locations.

contains a suitable selection of nodes, but allows specific nodes to be added or deleted; password information can also be supplied if required.

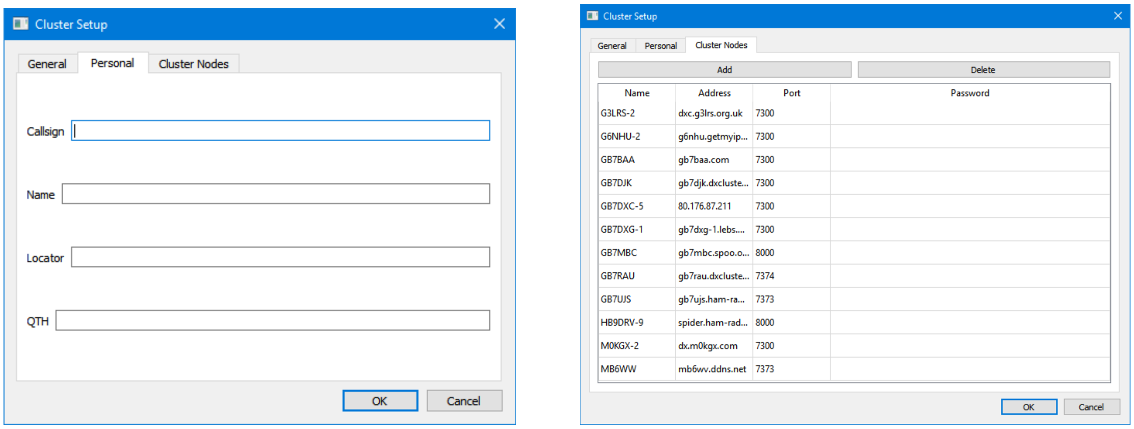


Figure 97 –Cluster Server configuration

Once the “**Personal**” data is fully defined, return to the main Cluster Server screen and select which of the defined DX clusters will be used as the source of data[97].

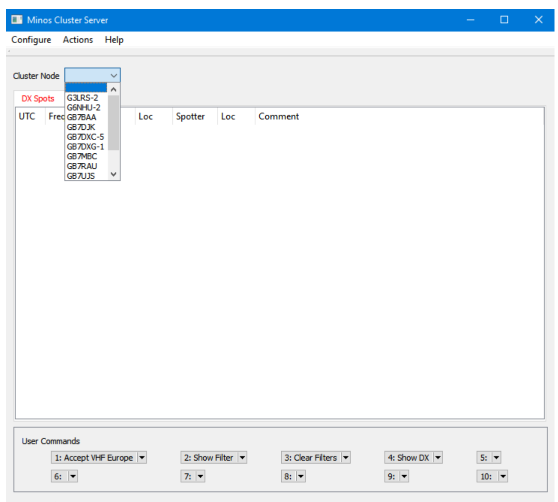


Figure 98 – Selecting the source DX cluster

Attempting to connect to a DX cluster before configuring **all** personal data will result in an error message:

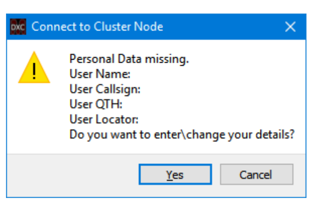


Figure 99 – Personal data missing for connect to DXcluster

[97] Note that at least some DXclusters only allow a single login under a given callsign so a different callsign may have to be used if multiple servers are invoked.

Clicking “**Yes**” initiates the “**Cluster Setup/Personal**” screen to input the information. Once done, deselect then reselect the DX cluster node to retry the connect. Check the “Raw Data” tab to see that the connect has been successful.

Once the server has successfully connected to the DXcluster spots should start appearing in the server display for all (monitored) bands from 50MHz upwards (see Figure 100).

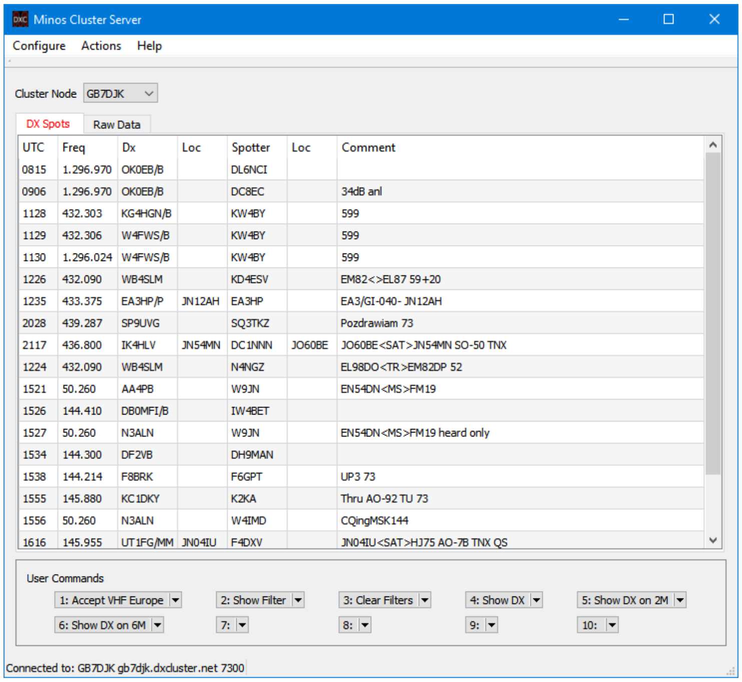


Figure 100 – Initial Cluster Server panel display

4.3.4.2 DXspider Commands

When configuring the Cluster Server (Figure 96) there are two tick boxes that control operation of commands to the DXcluster node when a connection is made or terminated.

If the “**Run Command File at Logon**” box is ticked then commands contained in **…\Configuration\Cluster\cluster_start.txt** are sent to the DXcluster when the connection is made[98]. By default this file contains a command to set filters to only pass spots on VHF in zones 14,15,15 (West, Central, and Eastern Europe), although it is not activated[99].

Conversley, the “**Run Command File at Logoff and Close**” tick box will, if set, invoke **cluster_end.txt** before the connection is terminated, thus enabling filters (or other information) to be reset to their default values.

[98] This is in addition to any DXcluster defined startup file, that may be there from years ago. Use **show/startup** to check.
[99] The Cluster Server discards any HF spots received although they will still be displayed on its ‘Raw Data’ tab.

Selecting the “**Raw Data”** tab allows the operator to view the dialogue with the DX cluster. This is most likely to be of use when initiating “**User Commands**” (defined in the lower part of the Cluster Server panel – see Figure 100), including for diagnosis if no spots occur (there may not be any of course!)[100].

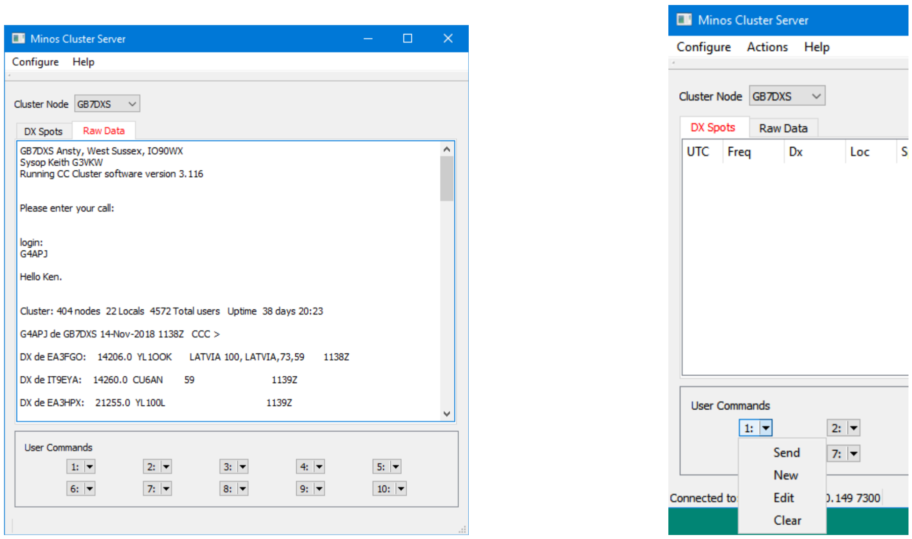


Figure 101 – Raw Data and User Commands

Up to ten “**User Commands**” may be defined using the pulldown options on each button (Figure 101)[101]:

- **Send:** Send the command (if one is defined) to the DXcluster.
- **New:** Create a new user command.
- **Edit:** Allows changes to be made to a command (if no command exists then no action)
- **Clear:** Clears the content after a positive response to the confirmation message.

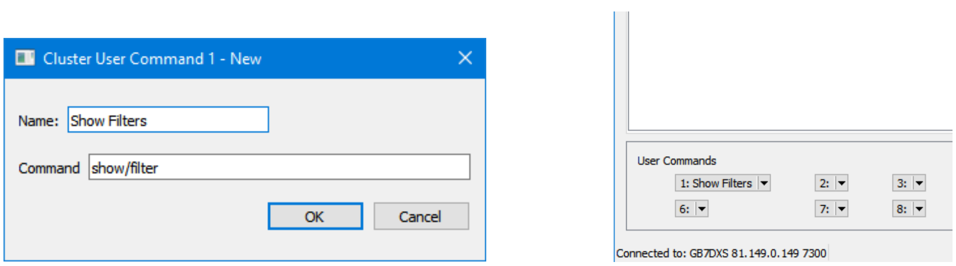


Figure 102 – Defining a User Command

Once defined, these commands are stored in **…\Configuration\Cluster\ClusterCommands.ini** which provides another mechanism for defining/editing their content, and to ensure that they appear each time the Cluster Server is started.

[100] **Beware:** command syntax *may* vary by DXcluster; DXspider based nodes are assumed.
[101] A useful document for the adventurous can be found at http://www.dxcluster.org/main/filtering_en.html

The **"Actions/Clear All Spots"** facility deletes all spots in the server; it does not affect the client displays.

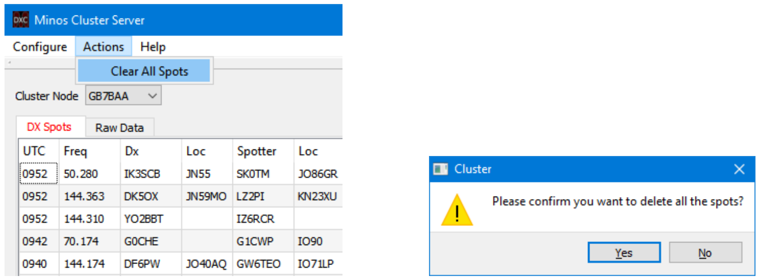


Figure 103 – Clearing spots.

Once the Cluster Server has been configured and is fully operational it **may** be advantageous to use Application Setup to change it to hidden, although the "DX Spots" tab does provide a useful summary across all bands if there is space on the screen.

4.3.4.3 Cluster Display

Once the Cluster Server has been configured (section 4.3.4.1) the Cluster Client (display) frame is added using the "**Configure Screen Layouts**" capability (see section 3.5)[102].

One cluster frame is allowed in each open contest tab.

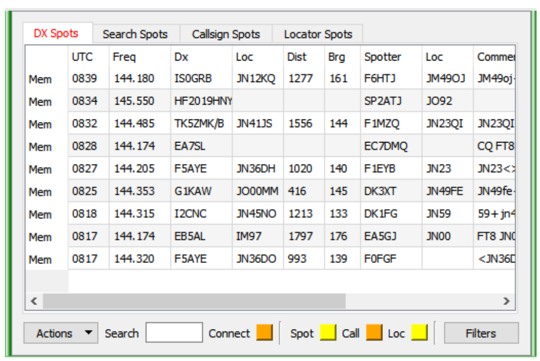


Figure 104 – Initial Cluster Client panel display

The "**Connect**" indicator turns orange when communication has been established with a Cluster Server (local or remote) that has established communication with its designated DXcluster.

"Search Spots" (Figure 105) allows the Dx and Loc fields within the panel content to be searched for strings of interest. Enter the required data into the "Search" field and hit return. The display will switch to the **"Search Spots"** tab to display any results found[103].

[102] The layout including "**Cluster"** should be made the default if the facility is used with most contests.
[103] Currently only partial callsigns and four character Locators are processed.

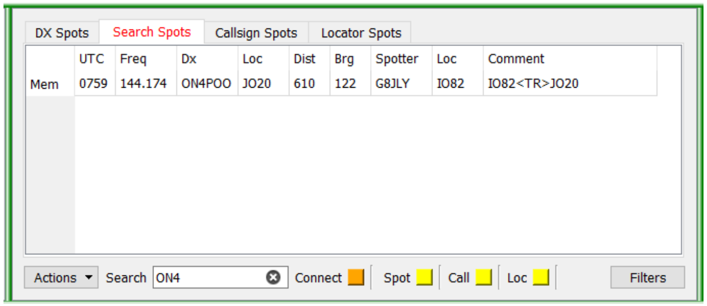


Figure 105 – Search facility

The content of the display can be controlled by clicking the "**Filters**" button and adjusting the Band (/Mode) of interest or the callsigns/locators to be watched.

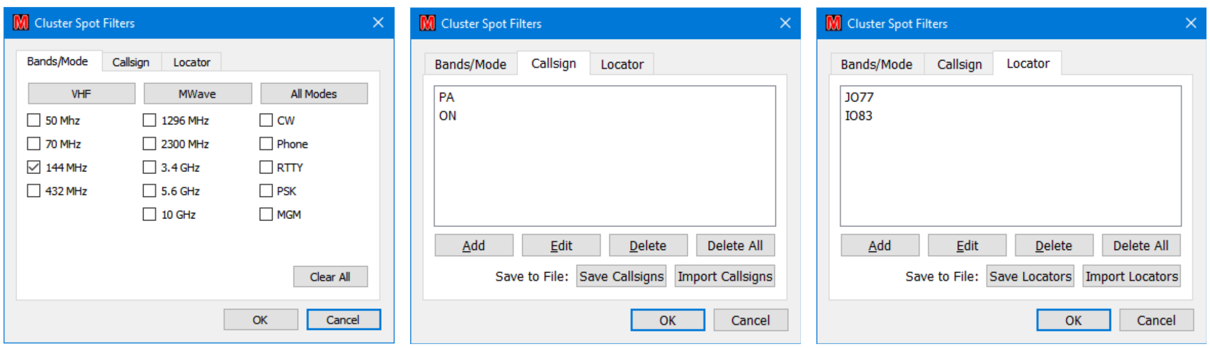


Figure 106 – Setting up filters

By default each frame is set to receive spots just for the band configured for the contest within which it is defined. The Callsign and Locator tabs allow predefined callsigns and (four character) locators to be watched[104]. These Callsign and Locator filters only apply to spots that satisfy the Band (/Mode) filters.

The Save/Import Callsigns/Locators buttons can be used to export/load favourite callsigns or locators via file(s) (in a default location of **…\Configuration\Cluster\callsignFilterLists**). The lists are maintained within each contest context so this facility would typically only be used when a contest is created.

Once content is displayed in the Cluster frame a single click may be used to transfer information to other panels (Figure 108):

- **Mem:** Sends the spot to memory panel and 'tags' the spot in blue. A spot may only be sent to the memory panel once, unless the Logger is restarted.
- **Callsign:** Load the logger QSO panel with the spot details. Note that in many cases this may only be a 4 character Locator and hence will be flagged as invalid in non MGM contests.
- **Freq:** If Rigcontrol is connected to a radio then it is set to the frequency of the selected spot.
- **Brg:** If Rotator is connected then rotation will be initiated to the bearing of the selected spot (provided it is not blank).

[104] DX locators are a bit hit and miss, it depends upon the spotter sending them in the comments.

The "**Actions**" pulldown list also provides these functions, as well as allowing individual or all spots to be cleared (after a confirmation message). To use this feature it is probably best to click on the time to avoid activating the 'field functions' listed above.

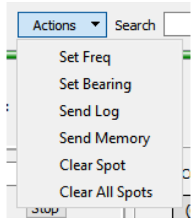


Figure 107 – Actions menu on spots

The callsign turns red if the station has been worked, the locator will turn red if the square has been worked on the current band. Callsign and locator matching is disabled for monitored bands other than the current contest band.

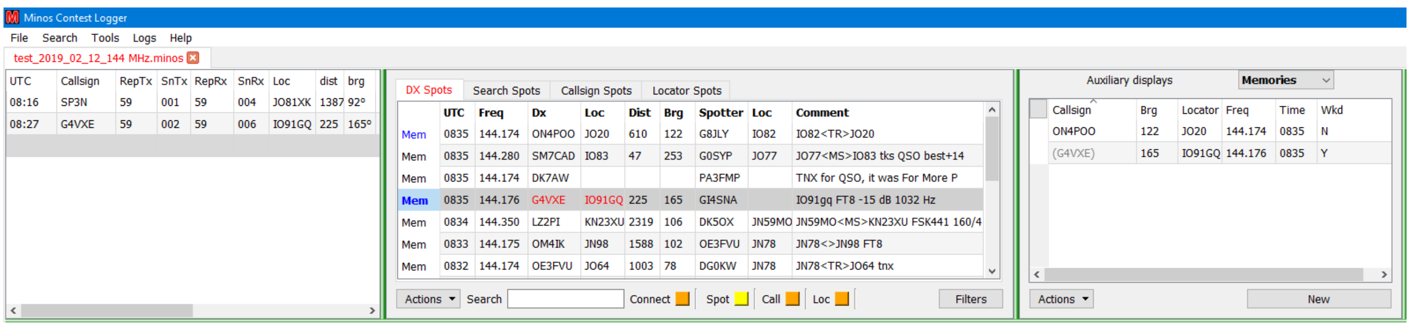


Figure 108 – Colour coding in spots panel

The indicators "**Spot**", "**Call**", and "**Loc**" at the bottom of Figure 109 are normally yellow, but will turn orange if a new entry appears in one of those views when that tab is not selected[105]. Selecting the tab will clear the indicator.

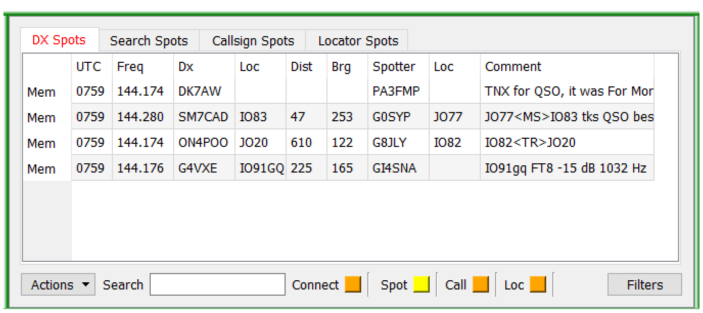


Figure 109 – New matching spots available

If a new spot arrives whilst the Callsign or Locator Spots tab is selected the Spot indicator will turn orange (Figure 110):

105 Figure 104 shows that a match has occurred on a callsign filter.

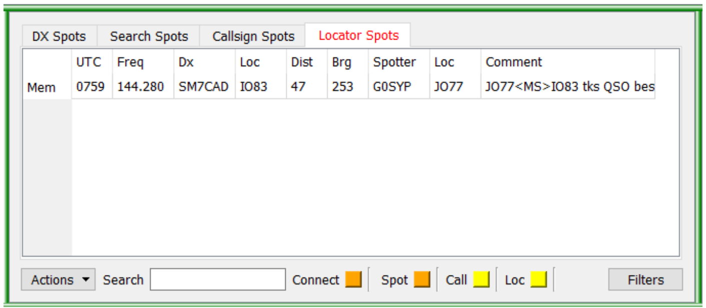


Figure 110 – Indication of new spot arrived

4.3.5 Band Map

Another future facility that is planned is Band Map. It requires Rig Control to be working first and hence will take a bit longer.

Configuration
Usage

4.4 Advanced Configuration

Advanced configuration is required for several possible activities:

- Development and testing
- Multi-PC environments necessitated by limitations on space, ports, or PC performance
- Multi-operators

Advanced configuration is invoked by clicking "**Show Advanced**" on one or more applications within App Configuration (see Figure 111)[106].

Complex/multi-machine environments involve other components and careful planning is required, particularly of the names used.

The defaults are quite sufficient for single machine operation with multiple Rigs or Rotators attached, with manual selection of the one being used at any particular time.

4.4.1 Development/debug

For development/debugging it is a simple matter to change the "**Program**" invoked for a particular function. Use the selection button on the right to browse to the development/test version of the application; the "**Parameters**" passed to it and the "**Home Directory**" used may also be modified.

It may be necessary to change the "**App Instance Name**"[107] ("**Rotator-debug**" in Figure 111) in the "**Apps Configuration**" in order to link the contest being used for the testing with a particular (debug/development) application.

Some test scenarios may require the use of a remote machine, and thus may be considered in the same manner as complex/multi-machine environments

[106] Any applications configured for remote access will always have "**Show Advanced**" ticked.
[107] The App Instance Name must be unique on any one machine.

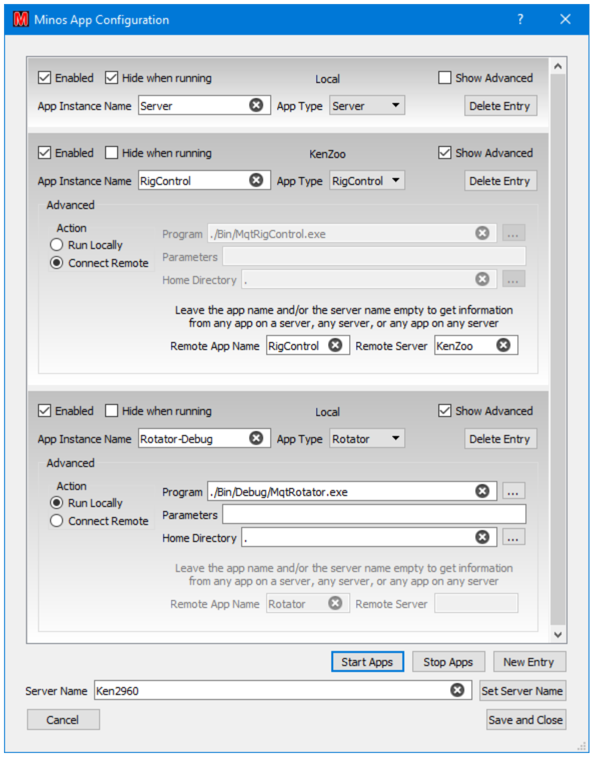


Figure 111 – Advanced configuration

4.4.2 Multiple rigs

The rigs defined through "**Configure/Setup Radios**" are common to all instances of RigControl on any given PC[108]. This means that a given rig can be accessed via multiple routes such as **PC/RigControl1/Rig1** and **PC/RigControl2/Rig1**.

A number of factors may force the use of a multi-computer environment:

- Number of available ports on any one PC
- A single PC is not powerful enough to handle everything
- The number of concurrent contests, hence:
  - Operators
  - Available monitors
  - Contest logs open
  - Active instances of Minos (only one per machine)
- Physical separation

For information on configuring multiple machines see section 4.4.3

Within each contest that uses a specific rig, change the Contest Details "**Radio Name**" definition (or use the pulldown list in the Logger) to select the requisite rig; the result is stored within the contest, but can be changed at any time.

[108] The same would be true for Rotator if so configured.

Configuration information is shared between all instances of RigControl on a single machine. If the wrong entry from the "**Select Radio**" pull-down list was inadvertently selected it would cause problems that might be difficult to diagnose (especially under contest pressures) [109].

The same technique would apply if multiple rotators were in use (but changing the Rotator App names obviously), and of course both techniques can be used together.

4.4.3 Multi-machine environments

In multi-machine environments all the machines have to be in the same LAN broadcast domain; no intervening routers.

The server name for a given PC defaults to the machine name, but can be changed by setting the new name in the "Server Name" box at the bottom of the App Configuration panel and clicking "**Set Server Name**".

The default instance of RigControl controls the locally connected rigs, and acts as the target for remote access to the local systems.

In order to have different rigs on different machines, multiple instances of RigControl must be defined; to avoid confusion these instances should have meaningful names. Probably the easiest is to append the server name[110] to which the instance connects, e.g. Rotator-PC2, or the functional area, e.g. Rigcontrol-VHF. Similarly, the target band could be appended, e.g. RigControl-2M and RigControl-70cm as shown in Figure 112, although this implies that each target PC only handles one band (possibly through multiple machines).

The choice will depend on the requirements/topology of the target environment, but make sure the planning is done before the contest, the result is documented and tested, and the final list is adhered to[111]!

---

[109] There is a degree of protection due to contention for the comms port(s).
[110] Which need not be the same as the PC name, if changed through the Apps Config panel.
[111] You can, if you want, leave the "**Remote App Name**" and/or "**Remote Server**" fields blank within a 'remote' Rigcontrol definition. This will result in **all** instances becoming visible and possibly resulting in confusion, so once again; plan it!

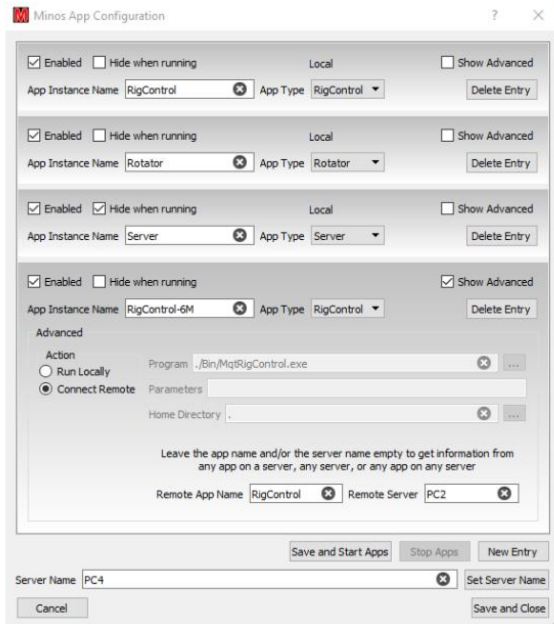


Figure 112 – Configuring two rigs, using remote RigControl

It might be useful to stagger the configuration process; get one working before doing the next.

"**App Instance Name**" is the name by which the connection is known on the local machine. "**Remote App Name**" is the name on the target machine, typically the default. Consistent naming helps – especially in more complex configurations (see section 4.4.4).

If the app is set to "**Connect Remote**" then "**Remote App Name**" and "**Remote Server**" must match the definitions on the remote machine. It is probably desirable to change the names on both machines to indicate that the app is remote/being used by a remote machine.

Obviously the name of the server on which the instance runs must match the name defined in that machine's server definition.

Beware: there is nothing to stop multiple contests sharing a rig or rotator; any such access must be coordinated!

4.4.3.1 AppStarter

If running a machine with no Logger (e.g. a Pi, acting just as a server device) then the Application Starter is used *instead* of the Logger to control the applications.

If the server machine is a Windows PC then the Application Starter can be started using the command

>start /d C:\Minos2 C:\Minos2\Bin\MqtAppStarter.exe

(within the system startup file) to ensure the correct default directory is used. AppStarter itself does not display much information (Figure 113) but the configuration dialogue is the same as the Logger EXCEPT that appstarter has an "autostart" checkbox.

Figure 113 – AppStarter

If the server machine is a Raspberry Pi then Aplication Starter is run from the login script, invoked by having the machine auto-login. It then runs headless, with any access through SSH.

4.4.4 Example complex configuration

This configuration is intended to show the methods and capabilities rather than reality; it is overly complex!

Analyse requirements for a particular situation (cable lengths?)

Figure 114 shows an example is a sort of 'super contest group' combining VHF NFD with the 432MHz-248GHz contest, and a possible way such a hypothetical situation might be configured:

- PC1 Runs the Logger for 6M and 4M with a rig for each band, both connected to and controlled by PC2. It has a single rotator with two Antennas, one for each band, serving the two contests.

- PC2 Runs the Logger for 2M, which can use either of two rigs - one of which is controlled by PC4. It has two rotators, one of which is controlled by PC4. It also runs the 6M & 4M rigs for PC1.

- PC3 Runs the Logger for 70cm 'and up' (23, 12, 9, 6, 3), with a separate rig for 70cm (controlled by PC4) and locally controlled rig/transverters for the other bands. It has two rotators; the one for 70cm is controlled by PC4, the locally controlled rotator handles all the antennas for the GHz bands.

- PC4 runs the AppStarter to control the 2M rig/rotator from PC2 and the 70cm rig/rotator from PC3.

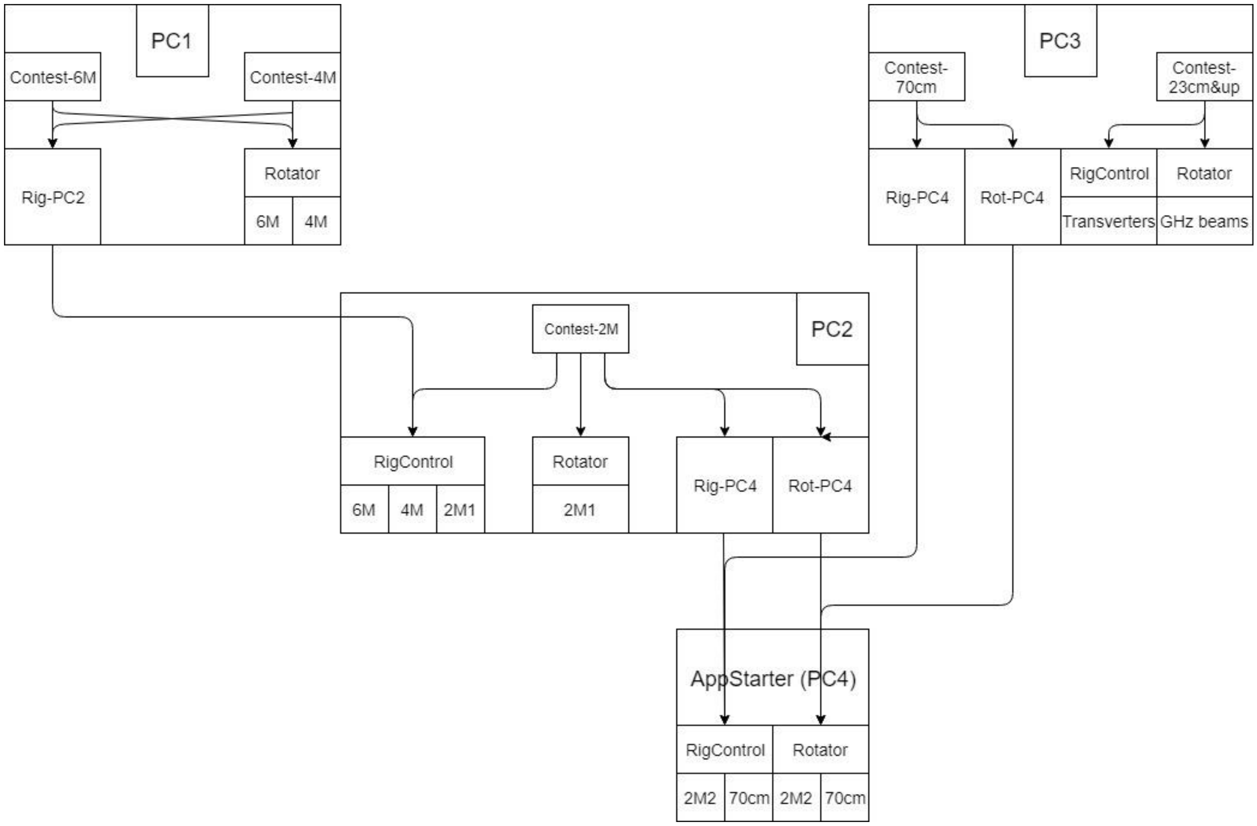


Figure 114 – Example complex configuration.

An example naming plan is shown in Figure 115

	Log	**Appname**	**Connects to**
PC1	6M 4M	RigControl-PC2	PC2/RigControl/6M PC2/RigControl/4M
		Rotator	6M & 4M aerials
PC2	2M	RigControl	6M, 4M, 2M1 rigs
		RigControl-PC4	PC4/RigControl/2M2
		Rotator	2M1 aerial
		Rotator-PC4	PC4/Rotator/2M2
PC3	70cm	RigControl-PC4	PC4/RigControl/70cm
	23cm-3cm	RigControl	23cm-3cm rigs (transverters)
	70cm	Rotator-PC4	PC4/RigControl/70cm
	23cm-3cm	Rotator	23cm-3cm aerials
PC4	(AppStarter)	RigControl	2M2 & 70cm rigs
		Rotator	2M2 & 70cm aerials

Figure 115 – Naming plan.

5 Internals

This section is still very much work in progress.
Minos is written using the open source Qt framework. If you want to get involved, then ask on the GJVList reflector. Details of how to build the software are documented in the main source repository.

When you do anything that makes a log change Minos appends a record to the file with the changes; one of the fields is a QSO sort key so that records can be correlated. The exact details are not documented here; anyone who really wants to 'go inside' it can look at the code (or ask G0GJV).

There should be no requirement to edit the file; indeed, it is highly recommended that you do **not** do so, in case of corrupting it.

5.1 Folders and files

The overall folder structure is:

- **Bin:** programs and DLLs (MqtLogger.exe is the main Logger programme)
- **Configuration:** configuration and INI files
- **Docs:** Documentation
- **Help:** <<Placeholder>> for help file(s)
- **Lists:** the default location for CSL file(s)
- **Logs:** the default location for storage of the .minos log files. They are not particularly interesting reading in raw format; certainly ***don't even consider*** editing them!
- **Tracelog:** diagnostic traces are written to this folder

5.1.1 Configuration files

- **Bandlist.xml:** Describes the amateur bands and the various formats possible. This file should not be edited!
- **Cty.dat:** The DXCC countries and prefixes list. Documentation of the CTY.DAT format for CT9 can be found on the K1EA website (http://www.k1ea.com/). CTY.DAT should be used **unchanged**; any modifications required should be made to CTY.SYN. This file is updated at the same time as downloading the calendar[112].
- **Cty.syn:** As countries keep adding to their valid prefixes this file may need to be modified to keep up to date. Each line starts with a primary DXCC prefix, followed by synonym prefixes, separated by commas. You can use # on the start of a line to signify comments.
- **District.ctl:** Postcode information.
- **DISTRICT.SYN:** Synonyms for single character postcodes.
- **Prefix.syn:** Allows postcodes to be checked for correct country and also duplicate checking if a station changes country in the middle of a contest.
- **ScreenConfigs.json:** Screen layout information
- **vhfcontests.*:** Contest calendar(s) downloaded from the RSGB web page.

[112] The URL is http://www.country-files.com/ .

5.1.2 INI files

The content is fairly self-explanatory, but if you want detailed information then look at the Open Source code. Some of the INI files are only created when program first run or as various applications are configured; others (not documented here, see main text) are created by the user for special purposes. **In general there should be no need to touch any of these files directly;** use the provided interface to make changes!

"File/Options" provides a summary of many of the Logger configuration options. Several of these (e.g. WSJTX configuration) are configured through 'command panels' rather than direct alteration in the display.

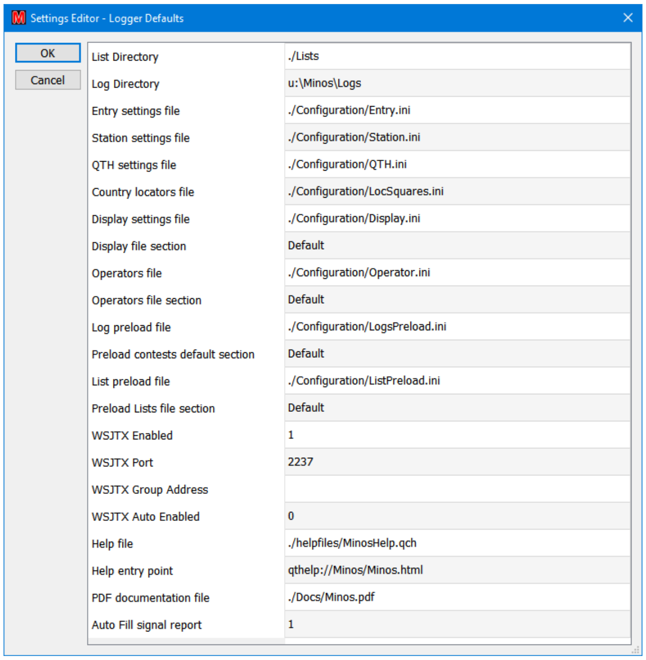


Figure 116 – Options Panel

\Configuration

- **AppConfig.ini:** the template for application definition.
- **Display.ini:** holds the selected statistics options and screen layout information.
- **Entry.ini, QTH.ini, Station.ini, Contest.ini**: contain the various bundles of information used to define a contest entry (as described in section 1.2).
- **ListPreload.ini**: contains the list of CSL files to be read on startup.
- **LocSquares.ini:** maps locator squares to countries to help validation**.**
- **LogsPreload.ini**: details the Contests and Contest Sets to be loaded on startup.
- **MinosConfig.ini**: defines the configuration of the applications (see Section 0).
- **MinosLogger.ini**: holds a pointer to the logs location and the autofill setting.
- **MinosRigControlConfig.ini:** tracelog and MGM mode control.
- **MinosRotatorConfig.ini:** the memory pre-sets from Rotator.
- **Operators.ini:**
- **UKACBonus.ini:** definition of the bonuses to be used for various squares within UKAC.

\Configuration\Antenna

- **AvailAntenna.ini:** which antennas are available, and the settings to access the rotators.
- **RotatorCurrentAntenna.ini:** the currently selected rotator

\Configuration\Radio

- **AvailRadio.ini:** which rigs are available, and the settings to access them.
- **FreqPresets.ini:** the list of initial frequencies for each band
- **RigControlCurrentRadio.ini**

\Configuration\Radio\Transverter

- **<RadioName>TransVertRadio.ini :** the definition of the bands and settings for the transverter(s)

\Configuration\Cluster

- **ClusterSettings.ini :** Personal name etc from Cluster Server configuration
- **ClusterSites.ini:** A list of possible DXcluster sites (this is where password information should be provided if it is required.

5.1.3 CSL format

Archive lists (.CSL) are now proper CSV files, and the fields may be quoted, plus there is now a possible extra field in the list for "number of contacts" with the call/locator/exchange combination.

```
# Any line with a hash as the first character
# is considered a comment and is ignored
# Format is
# <Callsign>,<Locator>,<Exchange>,<Count>
#
2E0BMO,IO83PO,WN,
2E0CNJ,IO83TO,,
2E0DOD,IO83XI,,
2E0DOD/M,IO83XI,,
2E0DOD/P,IO83XG,,
2E0LAE/P,IO93BJ,SD,
2E0NEY,IO81VK,,
2E0RFX,IO83WO,OL,
2E0UAC,IO92FJ,,
2E0UOG,IO83PN,WN,
2E0XLG/P,IO83WV,,
2E0XLG/P,IO84WC,BD
…
```

The 'count' field is the 'number of contacts' which is used in various ways by the archive list maintenance tools.

5.2 Matching Rules

Matches are evaluated with an embedded space character representing any character (single character wild card). Matching is soft - if no exact matches are found, the program will progressively soften the match criteria - first, remove any /x (suffix, e.g. /P) from the callsign, then ignore the locator and QTH fields, and finally ignore the callsign prefix and number (e.g. G0GJV/P will be looked for as GJV). At each stage, if any matches are found the softening stops. The match display varies - for the current contest it is the same as the main display; for other contests it loses irrelevant information - date/time, reports and serial numbers, giving more room for more important information. As the match process progresses the header of the match list will indicate progress. Typing any character in these fields will kill the current match process and restart it.

In addition to the above match process the callsign is subject to a rigorous duplicate check. A duplicate (if present) will always appear first in the match list.

5.3 Trace Logs

Trace logs can be very useful for diagnosing problems. In the event of finding a problem please try to supply a detailed list of steps leading up to the fault, and the appropriate trace log.

Rigcontrol and Rotator log trace messages all the time apart from those passing over the comms link to the radio(s) or rotator(s).

Trace logs are pruned by age - 10 days

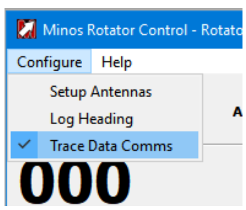


Figure 117 – Enabling extended trace log for Rotator

Tracelogs can be generated for diagnostic purposes. This can be enabled from the Configure menu where the TraceLog line can be enabled or disabled by checking the menu item. It is suggested that this should be disabled in normal operation.

Example tracelog with current antenna settings[113].

09:28:07.731 *** Antenna Updated ***
09:28:07.731 Rotator Name = Test Antenna
09:28:07.731 Rotator Model = 603 Yaesu,GS-232B
09:28:07.731 Rotator Number = 603
09:28:07.731 Rotator Comport = COM8

113 These settings can also be obtained via "Help/About Rotator Config" in the Rotator application. RigControl has the same capability.

09:28:07.731 Baudrate = 19200
09:28:07.731 Databits = 8
09:28:07.731 Stop bits = 1
09:28:07.731 Handshake = 0
09:28:07.731 Antenna Offset = 0
09:28:07.731 Current Max Azimuth = 450
09:28:07.731 Current Min Azimuth = 0
09:28:07.731 South Stop Flag = 0
09:28:07.731 Overrun flag = 1
09:28:07.731 Rotator Max Baudrate = 9600
09:28:07.731 Rotator Min Baud rate = 150

5.4 Registry entries

All registry content is collected under [HKEY_CURRENT_USER\Software\Minos2Qt]

Don't even consider editing it, even if you think you understand it. It can be helpful to reset defaults when re-installing the programme for test purposes. This may be done by using the option on the file pull-down list.

6 CSL Management Utilities

6.1 Archive Utilities (G4AUC)

These four utilities can be used to create and populate archives (CSL files), check for possible errors in archives and logs, and merge archives. They keep track of the number of times (and the dates when[114]) a station has been worked. All use standard browse mechanisms to locate (or create) the files being processed.

Download "setup_ArchiveUtilities_2.4.0.6.exe" to a convenient location on your computer from https://sourceforge.net/projects/minos/files/ and then run it.

It uses the standard install dialogue to install the utilities to a default location of "**C:/Program Files (x86)/Archive Utilities 2.4.0.6**"

The .NET Framework 4 (full version) is required, but should already be installed on Windows 10.

After successful installation, the following facilities will be available (via desktop icons if you have chosen to create them):

- **Archive Maker**
  Optionally creates, and then populates, a .CSL file with the content of a contest entry (.EDI) file.

  - "**Create A New Archive (.CSL file) and add .EDI file(s) to it**" allows you to select a location for the Archive file (typically in the /Lists folder) and enter a suitable name for the file before saving (creating) it. A blank file is created; if this is all you want to do then click on "**Cancel**" instead of selecting any .EDI files.

  - "**Add .EDI file (or files) to an Existing Archive**" Open lets you select an existing Archive to which the .EDI file(s) will be added.

  In both cases, a new window opens to select the **".EDI"** files. Browse to, and select, the required file(s) then click "**Open**". The main programme window then shows all the contacts that are added to the file.

- **Archive Checker**
  The selected archive is scanned and entries logged to the main programme window. Possible matches (where the Callsigns appear to be very similar, or Locators the same[115]) are displayed.

  Ticking the "**Show Similar Locators**" box will additionally display contacts where the Locator is similar rather than exact match.

- **Contest Reporter**

[114] There must be some limit to the number of dates retained; it is not known at present what that limit is.
[115] Note that this is often caused by one of the entries having Postcode information.

This program is intended to suggest where there may be typographic errors in your .EDI file (such as typing G9BAC instead of G9ABC). Remember that there may already be erroneous entries in the archive from earlier mis-typed QSOs.

The .EDI file for checking is selected, and then the .CSL file which will be used as a reference. The selected log is scanned and entries logged to the main programme window.

The programme distinguishes between Callsigns not found in the archive and those already present. For the latter, possible matches (where the Callsigns appear to be very similar, or Locators the same) are displayed.

Ticking the "**Show Similar Locators**" box will additionally display contacts where the Locator is similar rather than exact match.

Remember people do change QTH from time to time and change callsign, particularly from Foundation to Intermediate to Full etc. Good locations are also used by multiple stations, sometimes even at the same time!

Whether a problem exists is entirely up to your judgement.

- **Merge Archives**
  The content of the first selected .CSL is added to the second selected CSL.

  The main programme window then shows all the Entries from the first file that has been added to the second.

All of these utilities have "**Close the Archive Maker Application**" buttons (or "**X**" can be used) and (extensive) "Help".

6.2 CSL Manager (G4APJ)

CSLmanager is a simple portable application[116] to add contacts from one or more EDI (Minos) or LOG (old G0GJV Logger) files to the designated CSL file.

It handles (pun intended) the names of remote operators, and merges Postcode information into the archive.

"**CSLmanager (V2.2.1).zip**" can also be downloaded from https://sourceforge.net/projects/minos/files/

The Zip file includes release notes, and usage information.

[116] It may be necessary to register an OCX file if your system does not already use it; instructions are given.

7 Appendix

7.1 Hints, tips, and ‘tricks’

Reader contributions are welcomed here!

As with all software it is a good idea to take regular backups, but it is particularly important to save the configuration information before and after reconfiguring Applications – especially if you are doing an in-place upgrade and/or using a beta release. The internal format of the configuration files has changed several times, and there is no guarantee that it will not do so again!

It is also a good idea to periodically purge the TraceLog folder since multiple logs accumulate in there.

- If Rigcontrol panel disappears it means there is a problem connecting; could be target not present OR duplicate servername!If it appears to hang, check that there is not an error message displayed in a window that is hidden behind everything else on the screen.
- If you are getting weird bearings (and distances) check that you are using 4 char locators (if you are).
- If 'error' is displayed in the RigControl box when using remote access go check the rig – it has probably disconnected.
- When working NAC stations that don't give serials it is a fair few hoops to jump through to get Minos to accept the QSO with a serial less than 1, i.e. 000 as suggested by RSGB. When you click log the Dialog box comes up. Make sure the insert manual score now is ticked then click force. After the contest it is useful to add a comment in the comment box for each such QSO saying “No serial number received”.
- The TS2000X supports 23cms, but Hamlib does not contain a suitable definition for that band. This results in a message 'No 1296 MHZ Band found for this radio'. To work round this, add a transverter for 1296 in the TS2000X radio config, setting both Radio and Transmit frequencies to 1296. It may be necessary to restart the Logger.
- If you accidentally lose the contents of the cluster client frame then you can look on the cluster server DX spots tab which retains them indefinitely (?)

7.2 Suggestions

Improvements may be suggested by anyone, and if they make sense to G0GJV and, possibly after some rethinking of the request, fit in with the general philosophy for the program they may be included. Several improvements have been made at the request of users.

However, since the project is now Open Source you also make the change you want and submit it for possible inclusion in the main project.

Current possibilities include:

- There are around 25 to 30 feature requests logged
- Foreign language support for French

7.3 Future directions

- The remainder of the Applications defined in section 4.3.

7.4 Known problems (as of 05 Mar 19)

This section should be considered the equivalent of 'release notes'.

- There are a few *known* 'glitches' in more complex configurations, but nothing likely to affect simple single band contest operation; please notify any problems you find.
- At least one of these concerns connection of a second contest when first defined – merely restart the Logger to get round the issue (that works for a couple of other things as well).

7.5 Minos Transverter Switch

Minos Rigcontrol can be configured to send a switching message when a transverter is selected from a radio. The ASCII string message is sent out of the rigcontrol computer comport at the time of selection. A 'clear switches' message is sent when no transverter is selected.

A micro-controller such as an Arduino Uno can be utilised with a suitable relay board. The Arduino USB port is connected the computer USB.

Minos can also use a com port defined in the radio configuration tab in Minos (see Figure 76). The serial port is hard coded as 9600 Baud, 8 data, 1 stop, no parity.

Messages are sent of the form <':'><Message Body><'\n'>.

<Message Body> can be any ascii sequence of characters when a transverter is selected. The values are entered in to the Switch Number box (Figure 76), available for each transverter. If this is left blank, no message will be sent.

An example interface that will probably suits most users' requirements is shown below. This software will switch relays 1 to 8. If '1' is entered in to the transverter switch number box, relay 1 will be turned on when that transverter is selected. If more than one number is entered – for example '158', then relay 1, 5 and 8 will be selected. So, it is possible to turn on more than one relay at a transverter selection, if that is required. Use '0' to clear all relays.

The 5 volt relay board is an example that is available on the internet for a reasonable price, boards with less relays can also be found if 8 aren't required. Arduino Unos are also available for a reasonable price. This document does not take you through loading the software, for which there are many tutorials on the internet to guide you through the process.

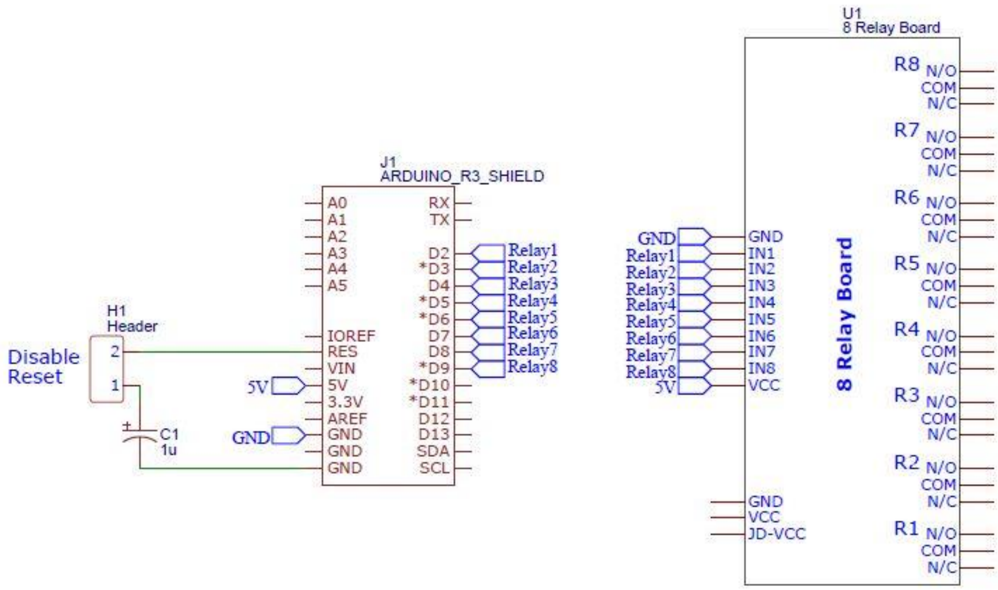


Figure 118 – Minos Transverter Switch Circuit

Figure 118 shows how to connect the IO ports to the relay board. Power for the relay board and Arduino can be taken from the computer USB port. The capacitor C1 is important, as it prevents the board performing a reset when the comport is initialised on the computer. **C1 needs to be out of circuit when uploading the switch control software to the UNO, and in circuit when the software is running.**

Inter-connections between the two boards can be achieved with the male-female Dupont jumper leads that can be purchased on the internet.

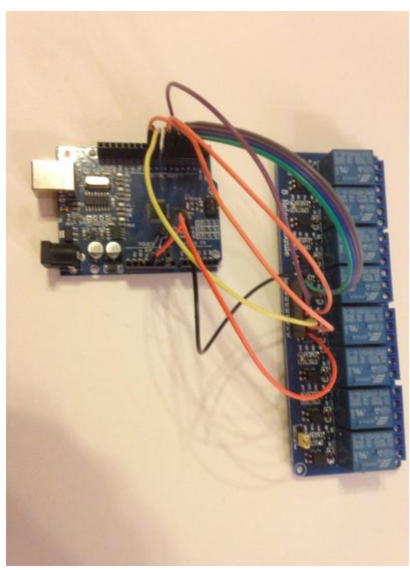


Figure 119 – Arduino and Relay Board

The Arduino Sketch can be downloaded from https://gitlab.com/dgbbal/Minos_Arduino_Transverter_Switch

7.6 Shortcut Keys

There are numerous shortcut key combinations available. They are all case-insensitive.

Rotator within Logger			
Function	**Shortcut**	**Function**	**Shortcut**
Rotate Left (CCW)	Ctrl-L	Rotate Right (CW)	Ctrl-R
Rotate to Bearing - Turn	Ctrl-T, Enter	Stop	Ctrl-S
Preset 1	Ctrl-alt-1	Preset 6	Ctrl-alt-6
Preset 2	Ctrl-alt-2	Preset 7	Ctrl-alt-7
Preset 3	Ctrl-alt-3	Preset 8	Ctrl-alt-8
Preset 4	Ctrl-alt-4	Preset 9	Ctrl-alt-9
Preset 5	Ctrl-alt-5	Preset 10	Ctrl-alt-0

Figure 120 – Rotator Shortcuts

RigControl within Logger			
Function	**Shortcut**	**Function**	**Shortcut**
Select KHz in Frequency Box	Ctrl-f		
Select Run Frequency 1	Ctrl- shift- [	Select Run Frequency 2	Ctrl- shift-]
Turn RIT On/Off	Ctrl-o	Select RIT 100Hz digit (if on)	Ctrl-i
Clear RIT frequency	Ctrl-k		

Figure 121 – Rig Control Shortcuts

[End of Document]